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1A

Contrasting ciliate species richness and abundance in soils derived from *Tectona grandis* and *Canella winterana*: testing the plant effect. DIMARIS ACOSTA-MERCADO and DENIS H. LYNN, University of Guelph, Guelph, Ontario, Canada.

We still have a rudimentary understanding about the mechanism by which plant roots may stimulate soil microbial interactions. A biochemical model involving plant-derived biochemical fractions, such as exudates, has been used to explain this ‘‘rhizosphere effect’’ on bacteria. However, the variable response of other soil microbial groups, such as protozoa, to the rhizosphere suggests that other factors could be involved in shaping their communities. Thus, two experiments were designed to (a) obtain a better understanding of the mechanism by which ciliate species richness and abundance differ among plant species and (b) to determine whether this mechanism is maintained via stimulatory and/or inhibiting factors associated with particular plant species. Bacterial and chemical slurries were reciprocally exchanged between two plant species known to differ in terms of ciliate species richness and abundance (i.e., *Canella winterana* and plantation *Tectona grandis*). The ANOVA showed that the bacteria plus nutrients, and the nutrients-only treatment have no significant effect on the overall ciliate species richness and abundance when compared to the control treatment. However, the use of only colpodean species to increase the taxonomic resolution of treatment effects showed that bacterial slurries have a significant effect on colpodean ciliate species richness. These results suggest that for particular rhizosphere ciliates, biological properties, such as bacterial diversity or abundance, may have a strong influence on their diversity and possibly abundance. These results are consistent with a model of soil bacteria-mediated mutualism between plants and protozoa.

2A

Short-term dynamics and long-term recolonization of protozoa in soil. SINA M. ADL* and DAVID C. COLEMAN**, *Department of Biology, Dalhousie University, Halifax, Canada, **Institute of Ecology, University of Georgia, Athens, Georgia, USA.

Most studies of Protozoa in the soil are based on the ‘‘most probable number’’ (MPN) estimates from cultured sub-samples. This approach has been criticized in recent years by protistologists. In order to work around these criticisms, we have tried to develop a set of procedures that rely on direct counts, without culturing. We show that the method is more sensitive and requires less effort than the MPN approach. Our species extractions focus on ‘‘active species at the time of sampling’’, and we tried to estimate the variation between days, as species adjust to soil moisture and temperature changes over several days. These variations are compared to species composition and abundance fluctuations observed between seasons, in decomposing leaf litter bags. We also obtained abundance and community structure data based on samples from a 25-year agro-ecosystem chronosequence under no-tillage management. We found that direct count methods, without prior culturing of samples, were more sensitive in detecting changes over several days, over several months, and in decadal succession from field samples. This provides a great advantage over culture-based methods that generally fail to distinguish between encysted and re-activated species.

3A

Two putative peroxidases of bacterial origin in the hydrogenosomes of the anaerobic protozoan *Trichomonas vaginalis*. SIMONE AHR and KATRIN HENZE, Institute of Botany III, University of Düsseldorf, Düsseldorf 40225, Germany.

Trichomonas vaginalis is an anaerobic protozoan parasite of the human urogenital tract. It lacks functional mitochondria but contains hydrogenosomes, anaerobic forms of mitochondria that generate ATP from the fermentative degradation of pyruvate. To determine the similarities and differences between hydrogenosomes and mitochondria, we are characterizing the proteome of purified hydrogenosomes from *T. vaginalis* by 2D electrophoresis and ESI-Q-TOF MS/MS. In the process, we have identified two proteins of bacterial origin that may function as peroxidases and this may be part of the answer to the unsolved question of how hydrogenosomes in *T. vaginalis* are protected against the detrimental effects of oxygen: rubrerythrin is a non-heme iron protein found predominantly in anaerobic bacteria that can reduce H₂O₂ and was shown to complement catalase-deficient *E. coli* mutants. Among eukaryotes we have found only one further rubrerythrin homologue in the anaerobic, amitochondriate amoeba *Entamoeba histolytica*. Peroxiredoxins are ubiquitous peroxidases using thiols to reduce alkyl hydroperoxides. *T. vaginalis* hydrogenosomal thiol peroxidase (Tpx) shows 65% homology to bacterial 20-kDa Tpx. These peroxidases contain three highly conserved Cys residues, Cys⁶², Cys⁸², and Cys⁹⁵, and belong to a peroxiredoxin subfamily that previously contained eubacterial sequences only. The finding of two putative bacterial-type peroxidases in *T. vaginalis* hydrogenosomes suggests that proteins involved in peroxide detoxification in these anaerobic organelles may have been acquired laterally from eubacteria.

4A

Experimental studies of the relative effects of leaf litter from four hardwood trees on protozoan community productivity. O. ROGER ANDERSON, Lamont-Doherty Earth Observatory of Columbia University, Palisades, NY 10964, USA.

Leaf litter is an important source of organic matter as nutrients for aquatic and terrestrial microbial communities. Biogenic decomposition of leaf litter contributes substantially to atmospheric CO₂, possibly producing global warming, and comprises an important part of the terrestrial carbon cycle. Small-scale variability in terrestrial protozoan communities may be attributed to localized variations in leaf matter. This study examined the effect of aqueous leaf extracts from four trees (beech, maple, red oak, and white oak) on the abundance and diversity of protozoa in laboratory aquatic and soil cultures. Beech leaf extract was the most productive followed in the descending order by white oak, red oak, and maple in both aquatic and soil microcosm cultures. Dilution experiments indicated that concentration-related inhibitory substance(s) in the maple and red oak leaves contribute to lower productivity. Removal of tannins by treatment with activated charcoal (Norit®) improved productivity especially for red oak leaf extracts, suggesting that tannins may contribute to the inhibition of protozoan standing stock. To determine if the lower productivity was mainly due to the inhibition of protozoan excystment, aliquots of thoroughly dried soil were inoculated into maple and red oak leaf extracts. A comparable growth of protozoa occurred in these treatments as in aliquots from moist soil indicating that inhibition of excystment is not a likely explanation for reduced productivity of the protozoan communities.

5A

Phagotrophy in chlorarachniophyte algae: implications for eukaryotic genome evolution. JOHN M. ARCHIBALD, Genome Atlantic and the Canadian Institute for Advanced Research, Program in Evolutionary Biology, Department of Biochemistry and Molecular Biology, Dalhousie University, Nova Scotia, Canada.

The chlorarachniophytes are a group of marine amoeboflagellate algae that acquired photosynthesis by secondary endosymbiosis. Together with cryptomonad algae, the chlorarachniophytes are unusual in that they have retained the nucleus of their eukaryotic endosymbiont. The host component of chlorarachniophyte cells is derived from a member of the Cercozoa while the chlorarachnion endosymbiont is green algal in origin. In addition to being able to photosynthesize, chlorarachniophytes have been shown to ingest other algae and bacteria, although the exact nature of engulfment, and the frequency with which this occurs in nature is unknown. Here, I present a light micrographic study of phagotrophy in three chlorarachniophytes, *Chlorarachnion reptans*, *Lotharella* sp. 240 and *Gymnochlora stellata*. These organisms readily ingest a wide variety of algae, including cryptomonads, haptophytes and green algae. The mechanism of capture and ingestion involves adhesion to, fusion with, and digestion within the pseudopodia formed by chlorarachniophyte cells. The implications of phagotrophy on the chlorarachniophyte nuclear genome are discussed.

6A

Two adjacent genes encoding copper-induced-type metallothioneins in *Tetrahymena thermophila* may have resulted from a relatively recent gene duplication. ARTA BAKSHANDEH, EILEEN P. HAMILTON and EDUARDO ORIAS, MCD Biology Department, University of California, Santa Barbara, CA 93106, USA.

Tetrahymena thermophila is a valuable model organism for the study of metallothioneins, which play important roles in the metabolism of essential metal ions and in heavy-metal detoxification. Starting from a *T. thermophila* expressed sequence tag (EST) originally annotated as a Zn-Cu-Cd metallothionein (GenBank Acc #BM394105), we determined the genomic sequence corresponding to this EST and its flanking region. The sequence contains two adjacent genes encoding metallothioneins of the copper-induced type: the *MTT2* gene (submitted to GenBank by Boldrin et al. as Cu-inducible while this work was in progress) and a second, novel gene (*MTT4*), located ~1.4 kb downstream from and with same orientation as *MTT2*. Our entire sequenced segment is contained in the *T. thermophila* whole-genome sequence recently completed at TIGR. Both *MTT4* and *MTT2* are intronless and encode two nearly identical proteins (two differences out of 108 amino acids). The 5' and 3'-flanking regions of both the metallothionein genes show significant sequence similarity as well, suggesting recent gene duplication. All glutamines use the CAA codon, reminiscent of codon usage observed in highly expressed *Tetrahymena* genes. *MTT4* and *MTT2* map to a 1.5-Mb macronuclear chromosome from the right arm of micronuclear chromosome 2.

Supported by NIH grant RR02931. Sequence data were obtained from The Institute for Genomic Research *Tetrahymena* website, www.tigr.org/tdb/e2k1/ttg/

7A

Genetically mapping a *Tetrahymena* cloned and sequenced gene: *ASII*. REBECCA A. BARBEAU*, EDUARDO ORIAS*,

KATHLEEN M. KARRER** and EILEEN HAMILTON*, *MCD Biology Department, University of California Santa Barbara, Santa Barbara, CA 93106, USA, **Department of Biological Sciences, Marquette University, Milwaukee, WI 53201, USA.

The *Tetrahymena thermophila ASII* gene encodes a leucine zipper protein that is upregulated during post-zygotic development (Udani and Karrer 2002). AT-rich sequence flanking chromosome breakage junctions in *T. thermophila* frequently contain insertion/deletion polymorphisms between inbred strains B and C3 that are amenable to genetic mapping. *ASII* contains one intron, which likewise is AT-rich and presumably not under stringent selection. PCR-amplification with intron-flanking primers revealed a B-C3 PCR product size polymorphism. The segregation pattern of B and C3 alleles of *ASII* in meiotic segregant panels showed that it maps to the linkage group JO6 on micronuclear chromosome 3R. The segregation pattern in a terminal assortant panel showed that *ASII* does not co-assort with any previously mapped polymorphism. Comparison of the *ASII* sequence with the genome sequence database at TIGR showed that this locus is carried on a 0.98-Mb sequence scaffold. Therefore mapping the *ASII* polymorphism has identified one additional genetically mapped and sequence-tagged macronuclear chromosome and the corresponding sequence scaffold. DNA polymorphism-tagged MAC chromosomes and sequence scaffolds should accelerate the co-assortment mapping of mutant genes with interesting phenotypes to known sequence scaffolds, which will facilitate positional cloning and candidate gene identification.

Supported by NIH grant RR02931. Preliminary sequence data were obtained from The Institute for Genomic Research website.

8A

Ionic currents and cytoplasmic streaming in *Amoeba proteus*. SUSAN R. BARRY and ERICA D. CORSON, Department of Biological Sciences, Mount Holyoke College, South Hadley, MA, USA.

Amoeba proteus may drive a current through themselves, and this current may determine the direction of cytoplasmic streaming. Cytoplasm streams from the more contracted tail or uroid region to the extending pseudopods. The positions of the uroid and pseudopods change as the amoeba changes direction. The membrane potential of the amoeba may change with location in the cell. This voltage gradient may regulate the magnitude and direction of cytoplasmic streaming. Bingley and Thompson (1962), recorded from amoebae using intracellular microelectrodes and reported that the membrane potential of the uroid region was more negative than that of the pseudopods. In contrast, Nuccitelli et al. (1977), using a vibrating probe, observed that current entered the uroid region and exited through the pseudopods. They concluded that the uroid region was more depolarized than the pseudopods. We revisited this issue by recording membrane potential simultaneously from two regions of the amoeba and by recording ionic currents using the two-electrode voltage clamp. Our data support the findings of Nuccitelli et al. Cytoplasm may stream from more depolarized to more hyperpolarized regions. Moreover, overshooting depolarizations of the cell may produce global cytoplasmic contractions and rounding of the amoeba.

9A

A Sec7-related Protein in *Tetrahymena*. AARON J. BELL, CHARLES F. GUERRA and PETER SATIR, Anatomy/Structural Biology, Albert Einstein College of Medicine, Bronx, NY, USA.

Sec7 was first identified in mutants of *Saccharomyces cerevisiae* that accumulated Golgi stacks and cisternae when incubated at 37 °C in low glucose medium. Since then, Sec7-related proteins have been found in a variety of organisms including *Homo sapiens*, *Arabidopsis* and *Paramecium*. In their role in vesicular transport, many of the Sec7-related proteins have been shown to function as guanine-nucleotide exchange factors (GEFs) for small G-proteins such as ARF. *Paramecium sec7* (*psec7*) has been cloned and shown to be up-regulated upon deciliation, implicating a role of pSec7 in ciliogenesis and/or ciliary function. The presence of a Sec7-related protein in *Tetrahymena thermophila* has been shown by immunoblotting and immunofluorescence using pSec7 antibodies against *T. thermophila* proteins. These experiments confirm the localization of tSec7 to the cilia and ciliary membranes of *Tetrahymena*. These results suggest that tSec7, like pSec7, is a ciliary-membrane-associated protein that could participate in signal transduction processes. The genomic sequence of *tsec7* is now known. RT-PCR has revealed the presence of 9 introns within *tsec7*, the largest of which is 327 bps in length. Predicted start and stop codons within the genomic sequence will be verified with mass spectroscopic analysis of the protein. Alignment of pSec7 and tSec7 amino acid sequences indicates a 32% identity and a 52% homology between the two proteins. In addition, motifs found in pSec7 (i.e. the Sec7 domain, truncated IQ motifs and PH domains) are also found in tSec7.

10A

How many novel eukaryotic “kingdoms”? Pitfalls and limitations of environmental DNA surveys. CÉDRIC BERNEY, JOSÉ FAHRNI and JAN PAWLOWSKI, Department of Zoology and Animal Biology, University of Geneva, Geneva, Switzerland.

Over the past few years, the use of cultivation-independent techniques to detect eukaryotic diversity has proven to be a powerful approach. Based on small-subunit ribosomal RNA (SSU rRNA) gene analyses, these studies have revealed the existence of an unexpected variety of new phylotypes. Some of them do not seem to be related to any molecularly described lineage, and have been proposed to represent novel eukaryotic kingdoms. In order to critically review the evolutionary importance of this novel high-level eukaryotic diversity and to test the potential technical and analytical pitfalls and limitations of eukaryotic environmental DNA surveys (EES), we analysed 484 environmental SSU rRNA gene sequences, including 81 new sequences from sediments of the river Seymaz (Geneva, Switzerland). Based on a detailed screening of an exhaustive alignment of SSU rRNA gene sequences and the phylogenetic re-analysis of previously published sequences using Bayesian methods, our results suggest that the number of novel higher-level taxa revealed by previous EES was over-estimated. Three main sources of errors are responsible for this situation, namely (1) the presence of undetected chimeric sequences; (2) the misplacement of several fast evolving sequences; and (3) the incomplete sampling of described, but yet unsequenced eukaryotes. EES undoubtedly contribute to unravel many novel eukaryotic lineages, but there is no clear evidence for a spectacular increase of the diversity at a megaevolutionary level. After our re-analysis, we found only five candidate lineages of possible novel high-level eukaryotic taxa. To ascertain their taxonomic status, however, the organisms themselves have to be identified now.

11A

A molecular perspective on the phylogeny of amoeboid protists. CÉDRIC BERNEY and JAN PAWLOWSKI, Department of

Zoology and Animal Biology, University of Geneva, Geneva, Switzerland.

The origin of amoeboid protists is one of the greatest enigmas of eukaryotic phylogeny. Although it has long been accepted that amoeboid protists are polyphyletic, the lack of molecular data for many amoeboid taxa impeded their placement in the tree of life. We have sequenced the small-subunit ribosomal RNA and actin genes from a broad taxonomic sampling of amoeboid protists, including three lineages of heliozoans (Actinophryida, Desmothoracida, and Taxopodida) and one lineage of radiolarians (Phaeodarea) for which no molecular data were available yet. Phylogenetic analyses of our data show that (1) all lobose amoebae sequenced to date belong to the recently defined phylum Amoebozoa and (2) Heliozoa and Radiolaria both represent polyphyletic assemblages, as suggested by previous ultrastructural studies. However, although axopodia evolved at least five times independently during the evolution of eukaryotes, our study also reveals that most protists possessing filopodia, reticulopodia, and/or axopodia are closely related, and constitute a new supergroup of amoeboid protists, the Rhizaria.

12A

Saccamminid foraminifers from anoxic sediments: implications for Proterozoic ecosystem dynamics. JOAN M. BERNHARD*, ANDREA HABURA** and SAMUEL S. BOWSER**, *Department Geology & Geophysics, Woods Hole Oceanographic Institution, Woods Hole, MA 02543, USA, **Wadsworth Center, New York State Department of Health, POB 509, Albany, NY 12201-0509, USA.

Eukaryote origin and earliest diversification occurred in the Proterozoic when Earth's atmosphere was undoubtedly different from that of today. Atmospheric oxygen levels were increasing from the primordial anoxic atmosphere due, for example, to cyanobacterial oxygenic photosynthesis. Late Proterozoic (~0.6–0.9 Gya) deep-ocean oxygen concentrations are less certain, but geochemical evidence suggests anoxia and hydrogen-sulfide enrichment. It can, therefore, be postulated that initial eukaryotic diversification occurred in oxygen-depleted, sulfide-enriched environments. Foraminifera are aerobes and, thus, not expected in anoxic settings. Recently, however, we found a saccamminid allogromian in a deep-water anoxic, sulfidic setting. Samples were collected from Santa Barbara Basin (California) when bottom-water oxygen was undetectable and sediments smelled strongly of hydrogen sulfide. Foraminiferal SSU rDNA sequences recovered from sediments included one from a previously uncharacterized saccamminid. Ultrastructural analysis indicated the presence of intact Golgi, mitochondria, and prokaryotic endobionts. Saccamminid occurrence in environmental conditions known to exist during the Proterozoic supports the possibility of their origin early in eukaryotic evolution. Extant saccamminids could have competed well in the prokaryote-dominated Proterozoic benthic ecosystem given their diet includes bacteria, bacterial biofilms and unicellular algae. Thus, Proterozoic foraminifers may have been top carnivores.

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13A

Genotypic identification and species names in *Acanthamoeba*: correlation and conflict. GREGORY C. BOOTON, DARYL J. KELLY and PAUL A. FUERST, Department of Evolution,

Ecology, and Organismal Biology, The Ohio State University, Columbus, Ohio, USA.

Many named isolates of *Acanthamoeba*, e.g. isolates designated *Acanthamoeba polyphaga*, are used widely in studies of pathogenicity, and other biological characters. However, molecular analyses show that isolates designated *A. polyphaga* are polyphyletic. Isolates may be very dissimilar at the molecular level, but share a species name, leading investigators to believe that they should share biological characteristics. Here, we present a summary of the correlation in *Acanthamoeba* between genotypes and the named species of this genus. Analyses of sequences of the nuclear small subunit ribosomal RNA gene (ssu rDNA) have identified at least fifteen genotypes of the free-living amoeba genus *Acanthamoeba*. Genotype determination has been defined by a sequence dissimilarity of five percent or greater in pairwise sequence comparisons. The ssu rDNA genotypes correlate well with many of the named species of *Acanthamoeba*. However, in some cases a species name exists as an invalid synonym of other species names within monophyletic genotypic clades. In addition, numerous species names are associated with the large and widely distributed *Acanthamoeba* genotype T4. Using molecular examination, some of these species designations are shown to be polyphyletic. The genotypes identified by nuclear ssu rDNA are supported by the addition of data from the mitochondrial ssu rDNA, which also provide some resolution of clades within genotype T4. Some lineages within the T4 genotype are associated with named species. Robust identification of *Acanthamoeba* isolates, including molecular data such as genotypic designation, is required so that data from studies on physiology or pathology may be correctly interpreted.

14A

Ciliates as a tool to assess planktonic patchiness: the case of *Myrionecta rubra*. CELIA BULIT, Depto. El Hombre y su Ambiente, Universidad Autónoma Metropolitana-Xochimilco, Calzada del Hueso 1100, 04960 México DF, México.

Protists, and especially planktonic ciliates, exploit space by forming distinct patches at different scales. Powerful new statistical techniques for the analysis of structure in spatial and temporal distributions are being developed. One of these, geostatistics, has been used to assess patchiness of the autotrophic ciliate *Myrionecta rubra* in a tropical coastal lagoon. Patchiness varies with scale and over time: *M. rubra* forms patches of different size, and abundance exhibits a temporal pulse-like pattern. While large-scale patches (~2000 m) are likely to be related to fresh- and marine water influx and seasonal variations in nutrient inputs, mesoscale patches (100–1000 m) may originate by dissipation of larger patches, migration and growth. Fine-scale patches (1–100 m) may be formed and advected by physical-behavioural aggregation of cells.

15A

On *Micromastigotes* and the evolution of the hypermastigont condition in the Parabasalida. STEPHEN L. CAMERON* and PETER J. O'DONOGHUE**, *Department of Integrative Biology, Brigham Young University, Utah, USA, **Department of Microbiology & Parasitology, University of Queensland, Queensland, Australia.

This study redescribes the genus *Micromastigotes*, Hollande and Caruette-Valentin, 1971, a parabasalid flagellate symbiotic in termites, on the basis of light and electron microscopy. The genus *Micromastigotes* is characterised by possessing flagella bands which spiral around the anterior portion of the cell, the location

of the nucleus and Golgi bodies at the base of the anterior flagellated area, and an axostyle, which runs the length of the cell. Individual flagella are derived from the central axis of the cell exiting perpendicularly, each is offset from the preceding flagellum by 12–18° thus forming a spiral band. Ultrastructurally, the flagella bases form a structure resembling a spiral staircase. Peltoaxostylar and preaxostylar fibres arise from the anterior most kinetosomes and the striated lamina forms a weakly undulating sheet directed posteriorly from each flagellum. Dense lamina and parabasal fibres are absent. *Micromastigotes* was originally classified as part of the Spirotrichonymphina and there are some similarities to other genera in this group, but *Micromastigotes* lacks a flagella gutter, a U-shaped band at the base of the flagella composed of the striated and dense lamina, which is diagnostic of the spirotrichonymphines. The spiral pattern in *Micromastigotes* is not consistent with previous schemes for the development of a polymastigont condition in spirotrichonymphines suggesting that *Micromastigotes* represents an independent derivation of a polymastigont condition from a trichomonad-like ancestor.

16A

Possible artificial introductions of freshwater peritrich ciliates and their implications for assessment of protistan biodiversity. JOHN C. CLAMP, Department of Biology, North Carolina Central University, Durham, NC, USA.

The planktonic, sessiline peritrichs *Epicarchesium pectinatum* (Zacharias 1897) and *Pelagovorticella natans* (Fauré-Fremiet 1924) were collected from different tributaries of the St. Lucie River, Florida. *Epistylis procumbens* Zacharias, 1897, another planktonic sessiline, was collected near the mouth of the Kissimmee River, Florida. *Epicarchesium pectinatum* and *P. natans* were not seen again in the St. Lucie drainage, probably because prolonged drought had increased the salinity of habitats drastically. *Epistylis procumbens* was absent from other, nearby sections of the Kissimmee River and nearby lakes and streams. Efforts to collect the three species from other bodies of water in Florida and from lakes and ponds in central New York met with no success; however, *P. natans* was found in two of the many localities sampled in North Carolina, an artificial pond and a natural lake. *Pelagovorticella natans* was found on only one of several visits to the latter locality. The three species occurred in habitats that were either within or near populated areas and had been extensively modified. This and the disjunct, ephemeral distributions of the peritrichs suggest that they may have been introduced artificially. Artificial introduction of ciliates or other protists is poorly documented but probable, considering their small size and consequent ease of transportation, and peritrichs should be especially suspect because many of them attach to aquatic plants. In the end, research on biodiversity of free-living protists must include the possibility of artificial introductions and develop methodologies to separate them from natural occurrences or else be reduced to a meaningless exercise.

17A

Paramecium aurelia revisited. ANNETTE W. COLEMAN, Biomed, Brown University, Providence, RI, USA.

A major running battle was fought in the 1940s, 1950s and 1960s in the continuing war between advocates of biological species versus taxonomic (morphological) species. The subject was the taxonomic treatment of *Paramecium aurelia*. At that time, proponents of the newly accredited concept of a biological species (Mayr 1948) demanded that the 16 sexually isolated syngens of *Paramecium aurelia* be named as separate species. T. M.

Sonneborn, who was largely responsible for the mating studies revealing the presence of multiple syngens (isolated mating groups) within the morphologically uniform *P. aurelia*, felt that species descriptions requiring biochemistry or cell culture for identification were impractical. Finally, in 1975, he capitulated, describing *P. primaurelia*, *P. biaurelia*, etc., on isozyme and mating affinity criteria. *P. aurelia* is not the only taxonomic species encompassing multiple cryptic species. The appropriate taxonomic solution is still moot, but we have obtained DNA sequence data now that reveal the relationships among the *P. aurelia* syngens, and that can be used to compare with similar cases, both in ciliates and in the microalgae, as well as plants and animals.

18A

Massive differential expansion of the *Trichomonas vaginalis* adaptin genomic complement. JOEL B. DACKS^{*,**}, JANE CARLTON^{***}, ROBERT P. HIRT^{**} and T. MARTIN EMBLEY^{**}, *Department of Zoology, Natural History Museum, London, Cromwell Road, London, UK, **School of Biology, King George VI Building, The University of Newcastle, Newcastle upon Tyne NE1 7RU, UK, ***The Institute for Genomic Research (TIGR), 9712 Medical Center Drive, Rockville, MD, USA.

Membrane transport is crucial to eukaryotic cell function and its evolution is likely to be critical for eukaryogenesis. Tetrameric adaptin complexes form part of the transport machinery, selecting cargo for transport through the late secretory and endocytic systems. With a few protistan exceptions, the majority of adaptin studies have been performed on animal and yeast models. Trypanosomatids and yeast have a minimal complement of adaptin genes, while humans and *Arabidopsis* show a moderate expansion. This expansion is functional, with adaptin subunit isoforms substituting combinatorially into their complex to provide novel localization. To address questions of evolution, parasitology and comparative cell biology, we examined the adaptin complexes of the sexually transmitted parasite *Trichomonas vaginalis*. In coordination with the *T. vaginalis* genome project at The Institute for Genomic Research, we identified all adaptin genes in the current 3.3X genome coverage data (<http://www.tigr.org/tdb/e2kl/tvg/>). Surprisingly, *T. vaginalis* has massively and differentially expanded its adaptin arsenal, with six times as many genes as other unicellular organisms examined and three times as many multicellular organisms. Eight *Trichomonas* adaptins have been confirmed by cDNA, demonstrating concurrent expression of the same gene. Phylogenetic analysis shows that the duplications giving rise to the various adaptin subfamilies occurred prior to the last common ancestor of diverse protists. Our data provide important insight into the evolution of the adaptin complex and the complexity of the *Trichomonas* late secretory/endocytic system. The differential expansion may illustrate the relative functional importance of the adaptin subfamilies to *Trichomonas* and be a model in which to study the lesser expansion seen in other eukaryotes. Finally, the size of the expansion, and the concurrent expression of at least one gene, suggests that the combinatorial nature of the adaptin complex may be more important than previously expected.

19A

The range of protists in Mono Lake, a hypersaline soda lake in the eastern sierras. LLOYD A. DAVIDSON and ARDEN E. DAVIDSON, Biology Department, Loyola University, 6525 N. Sheridan Rd., Chicago, IL 60626, USA.

In spite of the controversies surrounding the environmental deterioration of Mono Lake, and the significant amount of research being done on this remarkably productive body of water, nobody has ever attempted to survey the protists that live beneath its surface. Beginning in July of 2003, we have been trying to remedy this situation and we spent 2 months there studying the protistan species that reside in its waters and since then have been receiving regular sample collections for analysis. Based on the metazoan eukaryotes that live in this lake, which include only *Artemia monica*, *Ephydra hians*, rare rotifers and a nematode, we were warned to expect the lake to be depauperate of protists as well. However, we are happy to report that, while not as rich in species as most freshwater lakes, Mono Lake does support a robust population of phagotrophic and heterotrophic protozoa in addition to its many species of unicellular algae, including diatoms. It contains ciliate species representing at least the genera *Cyclidium*, *Frontonia*, *Lacrymaria*, *Litonotus*, *Euplotes* and *Aspidisca*, and the stalked suctorian *Tokophrya*. Additionally, there are species representing the heliozoa, helioflagellates, choanoflagellates, *Pseudobodo*, *Cercomonas* and *Chilomastix*, among other flagellates. The gymnamoebae are represented by *Vanella*, *Nuclearia*, *Mayorella* and *Hartmanella*, and, from the *Heterolobosea*, probable *Vahlkampfia* and *Gruberella* species. Micrographs of many of these species will be presented. Work on these species is still very much in its preliminary stages and a more detailed analysis and identification of the protists found in this extreme environment is advancing.

This work was partially supported by an NSF grant (MCB 99-77901) to Robert Jellison.

20A

The *Tetrahymena thermophila* genome sequence. 2. Relating macronuclear sequence scaffolds to corresponding macronuclear chromosomes. CHRISTINA deTORO, SONDRRA WILLIAMSON, EDUARDO ORIAS and EILEEN P. HAMILTON, MCD Biology Department, University of California, Santa Barbara, CA 93106, USA.

Tetrahymena's highly fragmented macronuclear genome contains ~300 chromosomes that have recently been whole-genome-shotgun sequenced at TIGR. We had genetically mapped Random Amplified Polymorphic DNAs (RAPDs) to *Tetrahymena*'s micro- and macronuclear genomes. We sized the MAC chromosomes that carry RAPDs by cloning and labeling RAPD DNA and probing Southern blots of macronuclear DNA separated by pulsed-field electrophoresis. We now report using cloned RAPD sequences to blastn-search the TIGR *Tetrahymena* scaffold database (<http://www.tigr.org/tdb/e2kl/tgg/>); each RAPD sequence matched a sized scaffold. Comparison of physical sizes of MAC chromosomes (as determined by hybridization) to corresponding scaffold sizes showed that:

- Every complete scaffold (assembled from telomere to telomere) matches in size with its corresponding MAC chromosome within 10% experimental error.
- No incomplete scaffolds (telomere-capped at one or neither end) exceeded the size of its corresponding MAC chromosome.

This work is allowing us to relate the genetic, physical and sequence maps of the micro- and macronuclear genomes to one another. It has also provided strong evidence for the quality and accuracy of the *Tetrahymena* genome sequencing and assembly carried out at TIGR.

(Supported by NIH grant RR02931 and contract to TIGR NIGMS *Tetrahymena* genome sequencing grant. Sequence data were obtained from The Institute for Genomic Research website.)

21A

Molecular markers for assessing community diversity of coastal ciliates. MARY DOHERTY*, BARBARA COSTAS**, OONA L. O. SNOEYENBOS-WEST*, GEORGE MCMANUS** and LAURA KATZ*, *Smith College Biological Sciences Department, Paradise Road, Northampton, MA, USA, **University of Connecticut Department of Marine Sciences, 1080 Shennecossett Road, Groton, CT, USA.

Biogeography of microbial eukaryotes is widely debated. While some claim that all microbial organisms have a cosmopolitan distribution, other data suggest evidence of endemism. To assess these hypotheses for the distribution of ciliates in the orders Choreotrichia and Oligotrichia, coastal community samples were taken from Long Island Sound. We are obtaining sequences of SSU rRNA, ITS rRNA, and mitochondrial genes to use as molecular markers for assessing gene flow across time and space. These data will serve as a basis for continued study of phylogeographic distribution of Choreotrichia and Oligotrichia.

22A

Protozoology's lost legacy. M. F. DOLAN, Department of Geosciences, UMASS Amherst, Amherst, MA 01003, USA.

A review of the contents of the Journal of Protozoology/Journal of Eukaryotic Microbiology from its founding in 1954 to the present was undertaken to determine if the type specimens of new taxa named in the journal have been deposited in museum collections. Type specimens were deposited in museum collections for 36% of these new species. No information was given on the designation of type specimens or the deposition of specimens for 64% of the taxa named, mostly prior to 1980. Type specimen designations have improved greatly in recent years. All species described in the journal in the last 5 years have been deposited as type specimens or type cultures. Linda Cole's Catalog of Type Specimens in the International Protozoan Type Collection (1994) at the National Museum of Natural History was cross-checked, and a final list of the missing organisms, grouped by phylum, was prepared. The Harold Kirby and L. R. Cleveland collections at the American Museum of Natural History are examples of how valuable the resource protozoan type specimen collections can be to researchers. Neither of these researchers had made plans to deposit their specimens in a museum. Their collections were saved only by the assistance of their students and colleagues. The author would like to coordinate an effort to find missing type specimens and facilitate their deposition in suitable museum collections.

23A

A brief guide to identifying bacteria in the field, for protistologists. BETSEY DEXTER DYER, Department of Biology, Wheaton College, Norton, Massachusetts 02766, USA.

Bacteria are notably the most abundant and most diverse of all organisms but also the most easily overlooked and even ignored. Yet, almost all of the major groups of bacteria are identifiable in the field without a microscope. Odors, textures, colorful slimes, mats, and scums are among the bacterial field marks that provide clues as to which microbes are present. The rules-of-thumb and tricks-of-the-trade for identification are often passed down to field microbiologists through graduate schools, but are seldom published for general use in other fields. Protistologists may find this brief introduction to bacterial field marks helpful for placing their own analyses and interpretations of habitats and communities into a greater context. Further information and details may be found in A Field Guide to Bacteria (Cornell 2003).

24A

Morphological and molecular analysis of bromeliad-inhabiting ciliates in Ecuador. MICAH SAUL DUNTHORN, Department of Geosciences, University of Massachusetts-Amherst, MA, USA.

The microbial inhabitants of Neotropical forests are poorly known. According to Finlay (2002, *Science*, 296:1061–1063) the majority of taxa in tropical forests should be previously described elsewhere. Conversely, Foissner (1999, *Protist*, 150:363–368) points to examples of endemic protists, and states that while we can identify the common species there are few trained taxonomists to identify the rare ones in large sampling investigations. To test these hypotheses, I am following up on an initial study of Foissner et al. (2003, *Europ. J. Protistol.*, 39:365–372) of the ciliates inhabiting waters in canopy, sub-canopy and terrestrial bromeliad tanks in two Ecuadorian forests. Using morphological (protargol, silver nitrate, and silver carbonate stains; scanning electron micrographs) and molecular methods (SSU and ITS sequencing) I will concentrate my investigations on the colpodid taxa found free-living in the bromeliad tanks in two Ecuadorian forests. If I find many new taxa, Finlay's hypothesis may not be applicable to at least bromeliad-inhabiting ciliates.

25A

Identification of rumen ciliates using small subunit ribosomal RNA (18S-rRNA)-targeted oligonucleotide probes and fluorescence in situ hybridization (FISH). JOHANNES FRIED*, SILVIA WEBER*, WOLFGANG LUDWIG*, NEIL McEWAN**, JAMIE NEWBOLD***, TADEUSZ MICHALOWSKI****, NADINE THOMAS**, SVETLANA KISIDAYOVA*****, JOHANNES HACKSTEIN***** and KARL HEINZ SCHLEIFER*, *Technische Universität München, Lehrstuhl für Mikrobiologie, Am Hochanger 4, D-85350 Freising, Germany, **Rowett Research Institute, Division of Microbiology and Immunology, Greenburn Road, Bucksburn, AB21 9SB Aberdeen, Scotland, UK, ***University of Wales, The Institute of Rural Science, Llanbadarn Fawr, Aberystwyth, Ceredigion SY23 3AL, UK, ****Polish Academy of Sciences, Kielanowski Institute of Animal Physiology and Nutrition, Instytutcka 3, PL-05-110 Jablonna, Poland, *****Slovak Academy of Sciences, Institute of Animal Physiology, Soltesovej 4-6, 04001 Kosice, Slovak Republic, ****Catholic University of Nijmegen, Faculty of Science, Department of Evolutionary Microbiology, Toernooiveld 1, NL-6525 ED Nijmegen, The Netherlands.

Up to half of the biomass in the rumen can be represented by ciliates, which play an important role in the digestion processes of their hosts. In the literature, very little information can be found on determination of the diversity of complex rumen ciliate communities. The identification of these fast moving protists is mainly based on live observations and comparisons of their highly variable cell morphologies. It makes accurate identification and quantification of rumen ciliates very difficult if not impossible. The development of fluorescence in situ hybridization (FISH), established as a technique to identify prokaryotes and eukaryotes using ribosomal RNA-targeted fluorescently labeled oligonucleotide probes, is a promising approach to identify and investigate rumen ciliates. The present study, part of CIMES (ciliates as monitors for environmental safety), a project sponsored by the European Commission, shows the problems of applying FISH on rumen ciliates and how to solve them. Tests resulted in a new protocol, which recommends para-formaldehyde and formaldehyde at 1–2% final concentration to preserve the ciliates before applying FISH. Furthermore, seven new oligonucleotide probes could be developed and successfully be tested to identify different rumen ciliate taxa of the order Entodiniomorpha (class Litostomo-

matea) by applying FISH. It is also shown, how FISH together with confocal laser scanning microscopy can improve analyses of ciliate cell morphologies. Thanks to Peter Pristas, Peter Javorsky, Ralf Einspanier, Susanne Ulbrich (providing rumen samples), Seung Yeo Moon-van der Staay, Georg van der Staay (providing unpublished 18S-rDNA sequences), the European Commission (financial support, project QLK3-CT-2002-02151).

26A

Phylogeny of lobose amoebae based on actin and small-subunit ribosomal RNA genes. JOSÉ F. FAHRNI*, IGNACIO BOLIVAR*, CÉDRIC BERNEY*, ELENA NASSONOVA**, ALEXEY SMIRNOV*** and JAN PAWLOWSKI*, *Department of Zoology and Animal Biology, University of Geneva, Geneva, Switzerland, **Laboratory of Cytology of Unicellular Organisms, Institute of Cytology RAS, St. Petersburg, Russia, ***Department of Invertebrate Zoology, St. Petersburg State University, St. Petersburg, Russia.

Lobose amoebae are abundant free-living protists and important pathogenic agents, yet their evolutionary history and position in the universal tree of life are poorly known. Molecular data for lobose amoebae are limited to a few species, and all phylogenetic studies published so far lacked representatives of many of their taxonomic groups. Here we analyse actin and small-subunit ribosomal RNA (SSU rRNA) gene sequences of a broad taxon sampling of naked, lobose amoebae. Our results support the existence of a monophyletic Amoebozoa clade, which comprises all lobose amoebae examined so far, as well as the amitochondriate pelobionts and entamoebids, and the slime molds. Both actin and SSU rRNA phylogenies distinguish two well-defined clades of amoebae, the "Gymnamoebia sensu stricto" and the Archamoebae (pelobionts + entamoebids), and one weakly supported and ill-resolved group comprising some naked, lobose amoebae and the Mycetozoa.

27A

Modulation of proteins in *Naegleria fowleri* amoebae by bacteria. ANGELA E. FRITZINGER and FRANCINE MARCIANO-CABRAL, Department of Microbiology and Immunology, Virginia Commonwealth University, 1101 E. Marshall Street, Richmond, VA, USA.

Naegleria fowleri are free-living amoebae found in soil and freshwater habitats that cause a fatal disease in humans called Primary Amoebic Meningoencephalitis. In the natural environment, amoebae feed on bacteria. In the infected host, the amoebae lyse and ingest nerve tissue. Proteins that lyse and degrade bacteria and mammalian cells have been described and are termed as naegleriapores or pore-forming proteins (PFP). In order to prevent autolysis from PFPs, *Naegleria* may have developed mechanisms that enable the amoebae to survive in the presence of cytolytic molecules. Recently, we have established that *N. fowleri* express a "CD59-like" surface protein, but the function of this protein in amoebae has not been elucidated. In mammalian cells, CD59 is a complement-regulatory protein that inhibits complement-mediated lysis of cells expressing this protein. In the present study, expression of the "CD59-like" protein in response to bacteria and bacterial toxins was investigated. Co-culture of *N. fowleri* with log phase *Escherichia coli* or *Pseudomonas aeruginosa* resulted in differential expression of the "CD59-like" protein. Treatment of the amoebae with bacterial toxin also affected expression of the protein. Scanning and transmission electron microscopy demonstrated morphological changes in *N. fowleri* following co-incubation with the bacteria.

Under all conditions examined, the amoebae remained intact and viable. The results of our study implicate a possible protective role of the "CD59-like" protein in response to bacterial predators and bacterial toxins.

28A

Protistan diversity in the Ross Sea, Antarctica: linking genotype, morphology and physiology. REBECCA J. GAST*, DAWN M. MORAN*, MARK R. DENNETT*, JULIE M. ROSE** and DAVID A. CARON**, *Woods Hole Oceanographic Institution, Biology Department, MS#32, Woods Hole, MA 02543, USA **University of Southern California, Department of Biological Sciences, 3616 Trousdale Parkway AHF 301, Los Angeles, CA 90089-0371, USA.

Traditional microscope-based studies of protistan assemblages in the Ross Sea, Antarctica have contributed significantly to our understanding of the microbial biogeography and food web structure in this extreme cold-water environment. However, these investigations have neither been able to characterize the genetic diversity of the communities, nor have they necessarily determined the abundances or trophic contribution of the dominant organisms. Resolution of the latter issue is hindered by the fact that physiological studies are often conducted on opportunistic species that respond to enrichment culture rather than species truly representative of intact communities. Therefore, we sought to determine the genetic diversity of microbial eukaryotes in ice, water and slush samples from the Ross Sea, and establish both morphologic and physiologic links between enrichment cultures and the genetic data. Denaturing gradient gel electrophoresis and ribosomal clone library analyses indicate that the protistan communities are very diverse, but those present in similar habitats are more alike than those in different habitats at the same site. We have been able to identify several protists from our enrichment cultures as being genetically represented in the original samples. General studies of their physiology have been initiated and methods for determining their abundances are being developed.

29A

The hydrogenosomes of the chytrid fungus *Neocallimastix frontalis*—Characterizing the functions of an anaerobic cousin of mitochondria through proteome analysis. GABRIEL GELIUS-DIETRICH and KATRIN HENZE, Institut für Botanik III, Heinrich-Heine Universität Düsseldorf, Universitätsstrasse 1, 40225 Düsseldorf, Germany.

Hydrogenosomes are anaerobic forms of mitochondria in at least four independent groups of anaerobic eukaryotes, the parabasalids, the heterolobosea, the ciliates and the anaerobic chytridiomycetes. Their hydrogenosomes generate ATP from the fermentative decomposition of pyruvate but lack the hallmark pathways of mitochondrial metabolism, the TCA cycle and membrane-associated electron transport. They also, with one exception, do not contain their own genome. We are analysing the proteome of the hydrogenosomes from the chytridiomycete *Neocallimastix frontalis* through 2D electrophoresis and tandem mass spectroscopy to assess the functional capacities of these organelles and to further characterize their relationship with mitochondria.

30A

Observations of asexual and sexual processes in *Protoparidium depressum* (Dinophyceae). KRISTIN E. GRIBBLE and

DONALD M. ANDERSON, Biology Department, MS 32, Woods Hole Oceanographic Institution, Woods Hole, MA 02543, USA.

Life cycle stages of the heterotrophic dinoflagellate *Protoperidinium depressum* were determined for a culture isolated from the west coast of Ireland. Division, sexuality, DNA content, gamete morphology, and identity of morphologically distinct life-history stages were revealed for the first time using a suite of morphological and molecular tools. Asexual division occurred by eleutheroschisis within a temporary, immotile cyst, yielding two daughter cells. Morphological features, including apical and antapical horns, were not fully formed upon emergence from the division cyst, but developed some time after cell division. Daughter cells were half to two-thirds the size of parent cells and swam attached in an epitheca-to-hypotheca orientation before separating. Stages indicating a completed life cycle, including gametes, planozygotes, and nuclear cyclosis were observed. Small cells, believed to be gametes, formed in repeated clonal and non-clonal re-isolations of the culture. In addition to their small size (40–50 μm), blunt apical horn, small antapical horns, and lack of pink pigmentation, these putative gametes had a thecal plate morphology different from that described for *P. depressum*. The LSU rDNA sequence (D1–D3 region) of small cells was, however, the same as that of *P. depressum*. Gametes were never witnessed to feed, but did form fusing, isogamous pairs. When isolated to well plates, gamete pairs failed to complete fusion to form a planozygote. Thus, the morphology of the initial planozygote remains uncertain. Large cells believed to be planozygotes had “typical” *P. depressum* morphology, but possessed two longitudinal flagella and had 2C DNA based on DAPI. Nuclear cyclosis was documented in one such tri-flagellated, motile cell. The cell remained motile after the completion of nuclear cyclosis, but did not progress to the next life stage. Dormant hypnozygotes were not observed.

31A

The shell or the kernel: biodiversity estimates of benthic foraminifera. ANDREA HABURA, STEVEN D. HANES, STEPHEN P. ALEXANDER and SAMUEL S. BOWSER, Wadsworth Center for Laboratories and Research, NYSDOH, Albany, NY 12201, USA.

Estimates of protist biodiversity and distribution have until recently relied on the morphological identification of individual cells. The advent of environmental-DNA-based surveys of these protist communities has broadened our understanding of species richness and diversity. A “total-evidence biodiversity” approach gives two benefits. First, it allows “back-comparison” to previous studies of ecosystems, permitting estimates of the amount of cryptic diversity in different sampling regimes (and by extension, the fossil record). Second, it will provide the best possible estimate of the total species richness of current study sites. Here, we use a well-studied benthic protist community, the foraminiferal assemblage of McMurdo Sound, Antarctica, to determine the effective parameters of both detection methods. We find that molecular-based techniques detect well-characterized calcareous species at depths too great to allow calcification, suggesting that these forms may be able to survive without their distinctive tests. “Fragile” taxa also appear to be better represented in molecular surveys than in morphological ones. The strengths of a combined morphological–molecular approach are demonstrated by an assessment of the distribution of two species in the genus *Notodendrodes*; morphological detection excels in the identification of distinctive, patchily distributed adults, whereas molecular detection with taxon-specific probes enabled identification in less favorable sampling conditions. Such a “broad-plus-deep” ap-

proach appears to maximize detection efficiency in this Antarctic setting.

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32A

Foraminiferal evolution as revealed by β -tubulin data. ANDREA HABURA*, LAURA WEGENER**, JEFFREY L. TRAVIS** and SAMUEL S. BOWSER*, *Wadsworth Center, NYSDOH, Albany, NY 12201-0509, USA, **State University of New York at Albany, Albany, NY 12222, USA.

Foraminiferal taxonomy is traditionally based on morphological characteristics, particularly the structure and composition of the test (shell). Recently, a comprehensive SSU-based molecular phylogeny confirmed some multi-chambered foraminiferal groups while enhancing our understanding of the single-chambered, or the so-called “allogromiid”, taxa. However, some relationships, such as the origin of the foraminiferal order Miliolida, remain unresolved, suggesting an approach involving multiple data sources may improve understanding. Here, we discuss the evolutionary implications of β -tubulin genes from several species of foraminifera. Foraminiferal β -tubulin is highly divergent, possibly indicating a period of accelerated evolution of this gene at the base of the foraminiferal lineage, but it is well-conserved within the group. We show its utility in helping to resolve the phylogenetic position of *Miliammina fusca*, a foraminiferan whose proper classification has been debated for 70 years. *M. fusca* is a morphological “chimera”, and we show that its placement in SSU trees is not particularly robust. However, β -tubulin data assigns *Miliammina* to the Miliolida, some of whose members show a similar chamber coiling pattern. This placement agrees with an earlier observation that *M. fusca*’s actin isoforms more closely resemble those of traditional miliolids. This finding implies that test formation can revert from calcareous to agglutinated, suggesting a reexamination of some relationships inferred from the fossil record.

Supported by NSF OPP0003639.

33A

The *Tetrahymena thermophila* genome sequence: 3. Completing macronuclear chromosome sequence scaffolds and ordering and aligning them along the micronuclear genome. EILEEN P. HAMILTON, SONDRRA WILLIAMSON, CHRISTINA del-TORO and EDUARDO ORIAS, MCD Biology Department, University of California at Santa Barbara, Santa Barbara, CA 93106, USA.

In Ciliates, the diploid, mitotic, genetically silent micronucleus (MIC) is the cell’s germline, while the macronucleus (MAC) is the exclusive site of gene expression. During (postzygotic) MAC differentiation in *T. thermophila*, several hundred chromosomes are generated by site specific fragmentation of the five germline-derived chromosomes at highly conserved chromosome breakage sites. Telomeres are added to every newly generated end, and chromosomes are amplified to $\sim 45\text{N}$. The *Tetrahymena thermophila* macronuclear genome was recently whole-genome-shotgun sequenced to nine-fold redundancy at TIGR. The sequence (publicly available at <http://www.tigr.org/tdb/e2k1/ttg/>) is assembled into contigs (contiguous segments of overlapping sequence reads) linked into scaffolds. Here, we report here the completion of the assembly of many scaffold ends by linking them to their corresponding telomere, using diverse bioinformatic or experimental approaches. 120 chromosomes are now assembled from telomere to telomere and 125 additional scaffolds are telomere capped at one end. The 120 complete scaffolds

encompass ~45% of the ~105-Mb assembly, suggesting a total of ~290 MAC chromosomes. By using paired sequence tags flanking chromosome breakage sites, we are linking unambiguously ordered and oriented MAC chromosome DNA sequence into MIC genome assemblies.

Supported by NIH grant RR02931. Preliminary sequence data were obtained from The Institute for Genomic Research website.

34A

Remarkable salt tolerance of naked amoebae from the Salton Sea and beyond. GWEN HAUER and ANDREW ROGERSON, Oceanographic Center of Nova Southeastern University, Dania Beach, Florida 33004, USA.

This study documents the salinity tolerance of naked amoebae from the Salton Sea, California, from intertidal sand in Dania Beach, Florida, and from freshwater. Amoebae were collected from hypersaline ponds around the perimeter of the Salton Sea, where salinities averaged 160‰, and directly from the shoreline waters of the Sea where salinities were generally between 44 and 48‰. Naked amoebae were also collected from the intertidal zone of a Florida beach, a habitat subject to salinity fluctuations in the range 6–85‰. From these sites, six clones of amoebae were isolated for subsequent salinity tolerance experiments (two marine beach isolates, two Salton Sea isolates, and two hypersaline pond isolates). A seventh clone, *Acanthamoeba polyphaga*, a common freshwater/soil amoeba, was obtained from the Culture Collection of Algae and Protozoa (CCAP), England, UK. The culture experiments compared the effects of gradually changing salinity versus no salinity acclimatization and used growth rate and culture yield as indices of effect. Generally, amoebae were tolerant over a wide range of salinity conditions and were not markedly influenced by pre-conditioning to salinity throughout the experiments. For example, one Salton Sea species, *Platyamoeba pseudovannellida* n.sp. Hauer, Rogerson & Anderson 2001, was able to survive and grow well without pre-adaptation over the range 0–150‰. Overall, the freshwater amoeba *Acanthamoeba* grew between 0‰ and 12‰, the marine clones grew in the range of 0–110‰, and the Salton Sea clones reproduced between 0‰ and 150‰. The hypersaline clones were the most resilient and grew between 0‰ and 270‰. The results suggest that most amoebae are essentially unaffected in terms of growth and yield by moderate and severe salinity changes. This might partly explain the high abundances of amoebae found by the authors in salt-challenged environments. Amoebae in the Salton Sea reached 237,120 cells/liter while those in intertidal sand ranged from 181 to 8,473 amoebae/cm³. The survival and activity of large populations of naked amoebae in sites subject to salinity fluctuations suggest that they should be considered in future studies to better understand their, as yet, undefined ecological role.

35A

Chemosensory responses of *Trichomonas* to polyamines. INGRID HE, NIGEL YARLETT and MICHAEL LEVANDOWSKY, Haskins Laboratories, Pace U., New York, NY 10038, USA.

Trichomonas vaginalis, a sexually transmitted parasitic flagellate of the human urogenital tract, requires the polyamine spermine, which is present in high concentrations in the vagina and the epididymus. We investigated the possibility that these parasites might use spermine or other polyamines as chemical signals in migrating to sites of infection, which in turn would suggest a possible class of chemotherapeutic targets. We tested this possibility with a novel assay method. Log-phase cells were placed in one of two 2-ml tubes connected by a glass capillary; the other

tube contained a test chemical. Numbers of cells migrating through the capillary into the second chamber were monitored over time by sampling the second tube. Significant migration responses were found to be 0.02 mM for spermine, putrescine, and citrulline. Lesser responses were found to spermidine, arginine and ornithine. Other amino acids did not differ from buffered salt solution controls. Cell behavior and motility were studied by videomicroscopy. A larger percentage of motile cells was observed in the presence of the active polyamines than in controls, where more cells attached to surfaces. These results suggest that this parasite does have chemosensory responses to polyamines specifically, and that the behavioral mechanism may involve a simple kinesis, though the latter possibility needs to be tested further.

36A

Characterization of the fenestrin, a cytoskeletal protein involved in *Tetrahymena* cell polarity and of its coding gene. EWA JOACHIMIAK*, JACEK SIKORA**, JANINA KACZANOWSKA*, MAURYLA KIERSNOWSKA*, SABRINA BARCHETTA***, CRISTINA MICELI*** and ANDRZEJ KACZANOWSKI*, *Department of Cytophysiology, Warsaw University, Warsaw, Poland, **Department of MCA Biology, University of Camerino, Camerino, Italy, ***Laboratory of Mass Spectrometry, Institute of Biochemistry and Biophysics, Polish Academy of Science, Warsaw, Poland.

Fenestrin is a cytoskeletal protein, which participates in conjugation and cell cycle regulation of *Tetrahymena* by formation of specific structures during these processes. In cell cycle it appears at the anterior poles of future proter and opisth cells. By 2-D electrophoresis, we found that in starved *T. thermophila* cells fenestrin appears in four isoforms that have the pI in the range of 5.2–5.5. The differences of pI of about 0.1–0.3 between isoforms could suggest that they differ by the phosphorylation state. By mass spectrometry, we identified some of the peptides derived from the enzyme digestion of fenestrin. These peptides were used for searching through *T. thermophila* genome, available at the site www.tigr.org. A nucleotide sequence coding one of the peptides was initially found. Then, by extension, the sequence of all fenestrin gene appeared, containing the other peptides identified by spectrometry. By RT-PCR we characterized one intron in this gene. The analysis of the amino acid sequence of the exons revealed that fenestrin is composed from at least two domains: N-terminal leucine-rich domain and C-terminal glutamic acid-rich domain. The BLAST search on general databases showed the existence of human and mouse proteins homologous to fenestrin in sequence, molecular mass and domain organization. Thus, fenestrin appears to be a protein correlated to cell polarity with an evolutionarily conserved structure. Its function should be cleared by mutational analysis and gene knock out.

37A

Effect of the micronucleus on cellular structures in several hypotrichous ciliates. LIPEI JIN, XINYING LIU and YONGZHEN MAO, The School of Life Sciences, Zhongshan University, Guangzhou 510275, China.

It is commonly believed that there is a sharp functional demarcation between the two types of nuclei of ciliated protozoa: the micronucleus for reorganization of the nuclear apparatus during sexual reproduction and the macronucleus for controlling somatic functions. This traditional view, however, has been critically re-examined recently and correlative researches have shown that the micronucleus plays an essential role in the development of some ciliates. Here we report some morphological evidences about the

effect of the micronucleus on cellular structures from amicro-nucleate cell lines of five hypotrichous ciliates (*Pseudourostyla cristata*, *Pseudokeronopsis monilata*, *Paraurostyla weissei*, *Uroleptus piscis* and *Oxytricha faurei*). All amicro-nucleate cell lines were generated by amputation and investigated by protargol staining. These observations manifest that micronucleus is indispensable to vigor and viability for *P. weissei*, to maintaining the integrity of adoral zone of membranelles (AZM) and/or undulating membranes (UM) for *P. cristata*, *P. monilata* and *U. piscis*, and to keeping regularity of cell shape as well as of the number and the structure of the macronuclei for *O. faurei*. These results suggest that the manifestations of the somatic function of micronucleus for different ciliates are varied and the close species might have a similar form of expression.

This work was supported by a grant from NSFC (No: 30370210) to L. P. JIN.

38A

Highly divergent SSU rRNA genes found in the marine ciliates *Myrionecta rubra* and *Mesodinium pulex*. MATTHEW D. JOHNSON*, TORSTEIN TENGS**, DAVID W. OLDACH**, CHARLES F. DELWICHE*** and DIANE K. STOECKER*, *Horn Point Laboratory, University of Maryland, Center for Environmental Science, Cambridge, MD, USA, **Institute of Human Virology, University of Maryland, School of Medicine, Baltimore, MD, USA, ***Department of Cell Biology and Molecular Genetics/Plant Biology, University of Maryland, College Park, MD, USA.

Myrionecta rubra and *Mesodinium pulex* are among the most commonly encountered planktonic ciliates in coastal marine and estuarine regions throughout the world. Despite their widespread distribution, both ciliates have received little attention by taxonomists. In order to understand the phylogenetic position of these ciliates better, we determined the SSU rRNA gene from cultures of *M. rubra* and *M. pulex*. Partial sequence data were also generated from isolated cells of *M. rubra* from Chesapeake Bay. The *M. rubra* and *M. pulex* sequences were very divergent from all other ciliates, but shared a branch with 100% bootstrap support. Both species had numerous deletions and substitutions in their SSU rRNA gene, resulting in a long branch for the clade. This made the sequences prone to spurious phylogenetic affiliations when using simple phylogenetic methods. Maximum likelihood analysis placed *M. rubra* and *M. pulex* on the basal ciliate branch, following the removal of ambiguously aligned regions. Fluorescent in situ hybridization probes were used with confocal laser scanning microscopy to confirm that these divergent sequences were both expressed in the cytoplasm and nucleolus of *M. rubra* and *M. pulex*. We found that our sequence data matched several recently discovered unidentified eukaryotes in Genbank from diverse marine habitats, all of which had apparently been misattributed to highly divergent amoeboid organisms.

39A

Evolution and implications of genome rearrangements in ciliates. LAURA A. KATZ, Department of Biological Sciences, Smith College, Northampton MA, and Program in Organismic and Evolutionary Biology, University of Massachusetts, Amherst, MA, USA.

We are exploring genome evolution in diverse ciliates to assess whether patterns of protein evolution correspond to the presence of genome duality, and to levels of genome rearrangements among ciliates. Ciliate genomes contain an estimated 25,000–35,000 genes, a number higher than that of other characterized

microbial eukaryotes and on a similar scale to that of humans and *Arabidopsis thaliana*. We have previously hypothesized that the divergence among ciliate proteins, and the accumulation of paralogs within ciliate taxa, is due to differential selection on the functional macronucleus and the transcriptionally inactive micronucleus. To assess this hypothesis, we are expanding our analyses of protein-coding gene sequences to compare the fate of proteins in ciliates to that of other eukaryotes, and to test whether the most divergent ciliate proteins are found in ciliates with extensively processed genomes. These analyses, combined with a phylogenetic perspective that reveals multiple origins of extensive fragmentation in ciliates, provide a possible explanation for the large genome size and diversity of proteins in ciliates.

40A

Molecular evidence for a single common origin of chromalveolate plastids. PATRICK J. KEELING, NICOLA J. PATRON and MATTHEW B. ROGERS, Department of Botany, University of British Columbia, 6270 University Boulevard, Vancouver, BC, Canada V6T 1Z4.

Plastids (photosynthetic organelles of plants and algae) are known to have spread between eukaryotic lineages by secondary endosymbiosis, or the uptake of a eukaryotic alga by another eukaryote, but the number of times this has taken place is controversial. This is particularly so in the case of eukaryotes with plastids derived from red algae, which are numerous and diverse. Despite their diversity, it has been suggested that all these eukaryotes share a recent common ancestor and that their plastids originated in a single endosymbiosis, the so-called ‘‘chromalveolate hypothesis’’. Here, we describe a novel molecular character that supports the chromalveolate hypothesis. Fructose-1,6-bisphosphate aldolase (FBA) is a glycolytic/Calvin cycle enzyme that exists as one of two non-homologous types, class I and class II. Red algal plastid-targeted FBA is a class I enzyme related to homologues from plants and green algae, and it would be predicted that the plastid-targeted FBA from algae with red algal secondary endosymbionts should be related to this class I enzyme. However, we show that heterokonts, cryptomonads, haptophytes and dinoflagellates (all photosynthetic chromalveolates) encode two paralogous class II enzymes, one of which strongly supports a specific relationship between chromalveolates. Exactly where chromalveolates got their plastid-targeted FBA genes is presently uncertain, but their common possession of this unexpected plastid characteristic is compelling evidence for their close relationship and a common origin for their plastids.

41A

Modelling the mechanism of gregarine gliding using bead translocation. CONRAD KING* and JOHN SLEEP**, *Biology Department, University College London WC1E.6BT and **School of Biomedical Sciences, King's College, London SE1.1UL, UK.

The gliding motility of gregarine protozoans has intrigued biologists for several decades but more recently key roles for actin and myosin have been established. Using the trophozoites/gamonts of *Gregarina* cell surface translocation of added 2 µm occurred with similar properties to those displayed by overt gregarine gliding including motility rates of between 0.1 and 10 µm/s. Although this movement has been viewed as a smooth process, gliding (and bead translocation) is an erratic process. Thus a single 2-µm bead translocation observed over 30 s at 0.2-s intervals demonstrated movement for only 26% of this time (39/150). At the low Reynolds Nos operating here bead movement can

only occur when force is developed at the cell surface (no consideration of inertia could apply). A laser trap was used to position a bead at the cell surface. Subsequently, the cell surface motor was able to carry the bead posteriorly out of this force trap of 50 pN. Using established figures for myosin energetics we estimate that the number of myosin heads at the interaction site to be in excess of 200. *Porospora gigantea* demonstrated gliding/ bead translocation rates of up to 60 $\mu\text{m/s}$ = four times the speed of myosin movement along the actin filaments of a muscle sarcomere!

42A

A high percentage of *Euplotes* genes require a +1 translational frameshift for expression. LAWRENCE A. KLOBUTCHER*, MATTHIAS MOLLENBECK**, MICHAEL C. GAVIN* and PHILIP J. FARABAUGH***, *Department of Molecular, Microbial, and Structural Biology, University of Connecticut Health Center, Farmington, CT 06032, USA, **Institute of Cell Biology, University of Witten/Herdecke, 58453 Witten, Germany, ***Department of Biological Sciences, University of Maryland Baltimore County, Baltimore, MD 21250, USA.

A previous survey of *Euplotes* genes available in GenBank indicated that >5% required a +1 translational frameshift for expression, much higher than observed in other organisms. Because this sample of sequences was non-random and biased towards highly expressed genes, we have randomly selected and sequenced 25 *Euplotes crassus* macronuclear DNA molecules to more accurately assess the frequency of frameshifting. Three new +1 frameshift genes were identified, and all have sequence features similar to the previously postulated frameshift sites. The results indicate that the frequency of *Euplotes* +1 frameshift genes is indeed high, and likely >10%. We have also examined the evolution of one of two frameshift sites present in the telomerase reverse transcriptase (*TERT*) genes of *E. crassus*. *TERT* gene segments were isolated from five additional *Euplotes* species. FS2 was found only in *E. crassus* and the closely related *E. vannus*, and phylogenetic analyses indicate that the site arose late in the evolution of the *Euplotes* group. Moreover, a novel frameshift site (FS3) was identified solely in the *TERT* gene of *E. minuta*. These results indicate that frameshift sites have arisen within genes during the recent history of *Euplotes*, and that it may be an ongoing process. Overall, the data suggest to us that the appearance of a frameshift site in a *Euplotes* gene is selectively neutral, or near-neutral. This would require an efficient mechanism of frameshifting in *Euplotes*. A model will be presented linking frequent frameshifting to a second unusual genetic feature of *Euplotes*, stop codon reassignment.

43A

Characterization of *Vorticella convallaria* calcium-binding centrin proteins. KATARZYNA KONIOR, SUZANNE M. McCUTCHEON and HOWARD E. BUHSE, Jr., Department of Biological Sciences, University of Illinois at Chicago, Chicago, IL 60607, USA.

The stalked ciliate, *Vorticella convallaria*, is a good model system to study mechanochemical motility because its contractile organelles (spasmoneme and myonemes) use a mode of contraction that differs from most other eukaryotic motile systems. Since calcium triggers this contraction, we have undertaken the molecular characterization of the calcium-binding proteins associated with these organelles. We have isolated and identified seven unique centrin-like cDNAs from *V. convallaria*. Each encodes an acidic protein of approximately 20-kDa, containing a unique

N-terminus and four potential calcium-binding domains. We predict that each centrin has a distinct function within the cell. To define these functions, we have initiated immunofluorescence localization studies utilizing various anti-centrin antibodies. Western analysis indicates that each antibody recognizes a distinct protein or subset of proteins in *Vorticella*. Using these antibodies, we have localized centrin to various structures within the cell; myonemes, spasmoneme, and the oral apparatus. Because each of these antibodies recognizes a different protein on Western analysis, we conclude that a number of calcium-binding proteins are associated with the contractile organelles. To further characterize this gene family, we have initiated immunolocalization at the ultrastructural level. This will permit subcellular localization of all *Vorticella* centrin and enable us to dissect the function of this multi-gene family.

44A

Protistan communities of the the Great Barrier Reef. G. A. LANGLOIS, Department of Science and Technology, Bryant College, Smithfield, RI 02917, USA.

Studies conducted at the Heron Island Research Station, located on the Capricorn region of the Great Barrier Reef, characterized complex protistan communities. Heron Reef is home to around 72 percent of the coral species found on the Great Barrier Reef. The paper will emphasize the diversity of ciliates found in benthic and epiphytic samples and will focus on the interactive role of the protistan component of the reef, including ciliates implicated in coral diseases that occur following bleaching episodes. Samples were taken by direct capture, observed with phase contrast and epifluorescent microscopy, and recorded by video and photomicrographic methods.

45A

Characteristics of phosphatidylinositol synthase activity from *Tetrahymena*. TANYA A. LANSLEY, BRANDY R. MOYE, EDMOND R. CHAN and PHILLIP E. RYALS, Department of Biology, The University of West Florida, 11000 University Parkway, Pensacola, FL 32514, USA.

Properties of phosphatidylinositol synthase (PtdIns-synthase) from the ciliate *Tetrahymena* were investigated using microsomal fractions enriched for the enzyme. Optimum conditions for PtdIns-synthase activity as assayed in a Triton X-100/CDP-DAG mixed micelle system with [^3H] *myo*-Ins as co-substrate were a pH of 8.0 using 50 mM Tris-HCl buffer and a temperature of 30 °C. Incubation of PtdIns-synthase at pre-selected temperatures prior to assay at 30 °C showed that the enzyme was stable for at least 20 min at 40 °C. A sharp decline in activity was seen after incubation at 50 °C for 20 min. Activity was stimulated by 2 mM Mg^{2+} , and stimulated to a slightly lesser degree by 1 mM Mn^{2+} ; however, these two ions in combination were synergistic. Other metal cations (Ca, Co, Cu, Fe, Mo, Zn, and Li) at a concentration of 2 mM had no appreciable effect on activity. Under optimum assay conditions, *Tetrahymena* PtdIns-synthase activity was linear over at least 20 min and was linear with increasing concentrations of the enzyme. The enzyme was capable of utilizing both [^3H] *scyllo*- and [^3H] (1D) *chiro*-Ins as substrate to form phosphatidyl-[^3H] *scyllo*- and phosphatidyl-[^3H] (1D) *chiro*-inositol, respectively, with (1D) *chiro*-inositol being the least preferred substrate. The in vitro synthesized *myo*- and *scyllo*-inositol-containing phosphoinositides were susceptible to hydrolysis by *Bacillus cereus* phosphatidylinositol-specific phospholipase C, but the (1D) *chiro*-inositol phosphoinositide was resistant. We hypothesize that putative DAG and *scyllo*-IP₃ second mes-

sengers produced by the cleavage of polyphospho-scylo-inositides may function in novel transmembrane signaling cascades.

46A

Isolation and characterization of the “zooxanthellae” from soritid foraminifera and the giant clam *Tridacna maxima*. JOHN J. LEE, MEGAN CEVASCO and GÉRALDO MÉDOR, Biology, City College of CUNY, Convent Ave at 138 St., New York, NY 10031, USA.

Various methodological approaches are being used to characterize *Symbiodinium* spp isolated from *Marginopora vertebralis* and *Amphisorus hemprichii* and *Amphisorus* sp. from the Indo-Pacific. While most of the cells in culture are non-motile vegetative cells, dinospores occur in batch cultures. We find consistent variance among the isolates in the duration and time of appearance of motile forms. Nitrate (0.146 mM) supports higher populations of the isolates than NH_4 (0.146 mM). Higher concentrations of NH_4 inhibit growth. Although Fensome et al. (1993) characterized the Order Suessiales (and Family Symbiodiniaceae) as gymnodiniophycideans in which the amphiesmal vesicals are arranged in 7–10 longitudinal series we were unable to detect any plates on the surfaces of our isolates in the SEM. We were unsuccessful in revealing plates on non-motile forms by freeze fracture but they were present after treatment with various chemical agents (e.g. H_2O_2 , HOCl^-) and enzymes (cellulase, chitinase). They are not polygonal or arranged in the patterns anticipated for the Symbiodiniaceae. Polyclonal antibody studies of the *Symbiodinium*-like isolates are in agreement with molecular systematic studies in the sense that there is a common relation. The foram endosymbiotic dinoflagellates share some common surface antigens with each other and with the few invertebrate isolates we have tested. Studies of the foraminiferal symbionts' plastids revealed considerable morphological diversity. We isolated a diversity of endosymbionts from the mantle of *Tridacna maxima* from the Red Sea and the Marshall Islands. These included *Symbiodinium*, *Amphidinium*, *Tetraselmis*, and an unidentified chlorophyte.

47A

Assessment of damaged DNA and cell structure in *Giardia lamblia* trophozoites. SCOTT LENAGHAN and CHRISTINE SUNDERMANN, Department of Biological Sciences, Auburn University, AL 36849, USA.

Control of protozoal pathogens in water and food destined for human use employs chemical and physical agents for inactivation of trophozoites and cysts. γ radiation, when given in certain doses, damages DNA as well as other cell components. Previously, we characterized damage in irradiated *Giardia lamblia* trophozoites by using Nomarski interference and scanning electron microscopy. To further examine cellular damage, irradiated trophozoites were viewed with transmission electron microscopy. Degranulation of cytoplasm at higher irradiation doses was observed, however, axonemes, adhesive discs, and peripheral vesicles appeared morphologically similar in treated as well as control cells. In order to determine damage to the DNA, the comet assay was employed. The comet assay is a single cell gel electrophoresis assay that measures the number of single and double strand breaks in DNA. This was achieved by embedment of live cells in low melting point agarose, lysis of cells, and then addition of an alkali solution to allow the DNA to unwind. After ample time for unwinding, the cells were electrophoresed, thus pulling the smaller broken strands away from the supercoiled DNA. We used Sybr GreenR, a DNA specific fluorescent dye, and a signal-enhancing

camera to visualize the DNA. Lengths of comet tails (corresponding to number of strand breaks) were measured for quantification of damage.

48A

A preliminary report of protozoan diversity in the Joshua Tree National Park desert biome. DAVID C. LENNARTZ, Department of Biological Sciences, Mt. St. Mary's College, 12001 Chalon Road, Los Angeles, CA 90049, USA.

Joshua Tree National Park represents an extreme environment for soil protists. The region is a mid-latitude desert subject to temperature ranges of 0–41 °C or more and annual rainfall of approximately 1.73 cm, with relatively rare flooding episodes that may deluge the area with as much as 14 cm within 1–2 h. The soil of this region has not been the subject of a systematic survey of protist diversity for at least 40 years. The goal of this study is to begin a systematic inventory of the soil protists of this region by collecting and analyzing soil samples by microscopy and other methods. Of particular interest is the soil within the rhizospheres around individual Joshua Trees (*Yucca brevifolia*) and creosote bushes (*Larrea tridentata*)—two of the most common plant genera in the region. Core samples allowed to dry for 1–2 months were subjected to the “non-flooded Petri dish” culture method (Foissner 1987) to obtain specimens. Standard brightfield and phase contrast microscopy, selected staining methods and observation of motile behavior were used to establish identification of organisms. This preliminary study focuses on ciliate and amoeboid protists though heterotrophic flagellates were observed to be very abundant in most samples. An initial species diversity list along with representative images is presented.

49A

Gene expression in the dinoflagellate *Karenia brevis*: Analysis of an expressed sequence tag (EST) library and development of an oligonucleotide microarray. KRISTY B. LIDIE^{*,**}, JAMES C. RYAN^{*} and FRANCES M. VAN DOLAH^{*,**}, ^{*}Marine Biotoxins Program, NOAA Center for Coastal Environmental Health and Biomolecular Research, Charleston, SC, USA, ^{**}Marine Biomedicine and Environmental Sciences Program, Medical University of South Carolina, Charleston, SC, USA.

Karenia brevis (Davis) is a unicellular dinoflagellate responsible for nearly annual “red tides” in the Gulf of Mexico. Insight into the molecular mechanisms that control growth, toxicity, and adaptive mechanisms in *K. brevis* is critical to understanding the formation and persistence of these toxic blooms. However, little information is available on the molecular biology *K. brevis*. Therefore, we constructed a cDNA library from which to gain insight into its expressed genome and to develop tools for studying gene expression. Large-scale sequencing of 9728 clones yielded 7001 high-quality expressed sequence tags (ESTs) for further analysis. The highest expressed gene accounts for only 1% of the total ESTs. Approximately 29% of ESTs were found to have homology to known sequences in other organisms after BLASTx similarity comparisons to the Genbank nonredundant database ($p < 10^{-4}$). Using a minimum identity of 95% within a 50-bp region of overlap, the 7001 ESTs were assembled into 5054 contigs. Of the 5054 contigs, 4399 contained only a single sequence. Of those containing >2 ESTs, approximately 40% displayed single nucleotide polymorphisms, suggesting the presence of multiple gene copies in this haploid organism. Gene-specific 60-mer oligonucleotides were then designed for each of the 5054 contigs using a Parcel clustering package, version 2.2.8 (Agilent). The resulting oligonucleotides and controls were

printed on glass slides to yield an 8000 feature array using inkjet printing technology (Agilent). The 8000 features included 5054 unique features and 2946 secondary probes. Validation experiments to assess system noise, reproducibility, log ratio accuracy, and differential gene expression are underway using Agilent Feature Extraction tools and Rosetta Luminator Gene Expression Analysis software.

50A

Cloning of mitochondrial cytochrome b gene from *Pseudourostyla cristata* by heat shock. XINYING LIU*, YONGZHEN MAO*, KAM-LEN DANIEL LEE** and LIPEI JIN*, *The School of Life Sciences, Zhongshan University, Guangzhou 510275, China, **Department of Applied Biology and Chemical Technology, Hong Kong Polytechnic University, Hong Kong.

The total RNA was extracted from *Pseudourostyla cristata* after 2 h of heat shock, and the cDNA was synthesized directly using Smart LD-PCR. The amplified cDNA products showed a strong band around 1.5 kb. The products were purified, cloned and sequenced. The resulting full cDNA sequence had a total length of 1538 bp including 1068 bp of cytb gene and UTR at both sides and showed all of the expected features for a fully functional CYB. The predicted protein is 49% (30% identical) similar to *Paramecium tetraurelia*. The sequence predicts a UAA stop codon 207 nucleotides from the end of the transcript suggesting a 3' UTR, unlike that of *Paramecium tetraurelia*. Prediction of transmembrane helices in the analysis of proteins indicates the expected eight transmembrane domains. The cytb gene expression profiles after heat shock and before heat shock were generated using quantitative real-time PCR. After 2 h of heat shock, the cytb gene expression was significantly enhanced. This is the first report that shows complete cytb gene among hypotrichous ciliates. Currently, we are investigating whether the method of cloning cytb rapidly are fit for other hypotrichous ciliates so as to analyze their phylogenetic relationship.

This work was supported by a grant from NSFC (No: 30370210) to L. P. JIN.

51A

Identification of mRNAs expressed differentially between micronucleates and amiconucleates of *Pseudourostyla cristata* during starvation. XINYING LIU*, YONGZHEN MAO*, KAM-LEN DANIEL LEE** and LIPEI JIN*, *The School of Life Sciences, Zhongshan University, Guangzhou 510275, China, **Department of Applied Biology and Chemical Technology, Hong Kong Polytechnic University, Hong Kong.

The evident character of amiconucleates of *Pseudourostyla cristata*, as compared with the normal micronucleates, was that a substantial proportion of cells exhibited an abnormal AZM. Moreover, the percentage of abnormal AZMs in amiconucleates during starvation was much higher. To understand how the molecular events of the micronucleus govern this mode, we employed suppression subtractive hybridization (SSH) to identify candidate genes which express in micronucleate cells but not in amiconucleate cells during starvation. A total of 284 transcripts were screened by cDNA array and 17 clones were confirmed to be differentially expressed among micronucleate cell. All 17 ESTs were analyzed by blaster matching in GeneBank at deduced amino acid sequence: 10 clones, no homologous proteins; 4 clones, myoglobin protein; 1 clone, cytochrome P450 monooxygenase; 1 clone, cytosolic malate dehydrogenase; the last clone, homologs of hsp90, which is interesting. Traditionally, heat shock proteins are known for their function during cellular stress as

molecular chaperones responsible for the folding, refolding and transport of newly synthesized protein. Why was it significantly enhanced in micronucleates but not in amiconucleates during starvation? Further research including RACE, immunoblotting and gel mobility shift will be carried out. This work was supported by a grant from NSFC (No: 30370210) to L. P. JIN.

52A

Symbiotic cooperation between dinos and ciliates in "Sym City". CHRISTOPHER S. LOBBAN, Biology Program, College of Natural & Applied Sciences, University of Guam, Mangilao 96923, USA.

Two new ciliates—a heterotrich and a hypotrich—have recently been discovered in the shallow benthos of coral reefs on Guam. They are unusual for ciliates in having zooxanthellae, but they share this characteristic with corals and many other reef animals. The heterotrich was a new genus, named *Maristentor dinoferus*, the hypotrich a *Euplotes* sp., and the dinoflagellate partner a *Symbiodinium* sp. Apart from their interest as dual-alveolate symbioses, the heterotrich has abundant, fluorescent pigment with intriguing possibilities for comparison with stentorin and blepharismisins from freshwater ciliates that serve in both phototropisms and chemical defense. Comparison with deep-water corals also raises the possibility that the cortical pigment enhances photosynthesis of the zooxanthellae.

53A

Alveolates and stramenopiles in the coral reef microbenthos. CHRISTOPHER S. LOBBAN, Biology Program, College of Natural & Applied Sciences, University of Guam, Mangilao 96923, USA.

Protists are notably absent from reef biodiversity checklists, with the exception of forams. A recent comprehensive compilation for Guam listed 303 forams, which indicate the level of biodiversity, yet there were only two other protozoans and no lists for benthic microalgae. My studies of the microbenthos began with some tube-dwelling diatoms and colonial *Pelagophyceae* that form macroscopic, seaweed-like colonies. More recently, I have become interested in two ciliates with symbiotic zooxanthellae, and in looking for these have begun to notice other protists. This presentation will display a "rogue's gallery" of microbenthos, beginning with the alveolates (ciliates—*Condylostoma* sp. and *Euplotes* spp. in particular; dinos and the symbioses), then some stramenopiles: diatoms and pelagophytes. The audience will be invited to identify the mystery ciliates and I hope the stills and videos will whet your appetite to study tropical reefs.

54A

Phylogenetic position of the apostome ciliates tested using the small subunit rRNA gene sequence. DENIS H. LYNN*, MICHAELA STRUEDER-KYPKE* and PHYLLIS C. BRADBURY**, *Department of Zoology, University of Guelph, Guelph, ON, Canada N1G 2W1, **Department of Zoology, NC State University, Raleigh, NC 27695-7617, USA.

Apostome ciliates are exclusively symbionts, typically of crustaceans. Their life cycles are complex, but show remarkable similarities to ophyrogenines and to other symbiotic oligohymenophoreans, particularly those that undergo microstome-macro-stome transformations. The highly modified structures around the cytostome—the rosette and x, y, and z kinetics—have made the assessment of homologies to other groups of ciliates difficult. A description of the apostome somatic kinetids with well-developed

overlapping kinetodesmal fibrils suggested affinities with hymenostomes (Bradbury 1966. *J. Protozool.*, 13:591). Further analysis of the ultrastructural features of the oral region of the tomite of *Hyalophysa* demonstrated homologies to the paroral and adoral ciliature of hymenostomes, confirming these affinities (Bradbury 1989. *J. Protozool.*, 36:95). To test this hypothesis, we proposed to sequence the small subunit rRNA (SSrRNA) genes of apostomes. Specimens of *Gymnodinioides* sp. were collected from molts of the amphipod *Marinogammarus obtusatus* collected off the coast of Eastport, ME. Seawater was decanted from the substrate upon which tomonts had settled, and the dish was flooded with 70% ethanol. The ethanol-fixed ciliates were collected by a micropipette, rinsed with distilled water, and DNA was extracted using the modified Chelex[®] protocol. The SSrRNA genes were amplified by PCR, and sequences obtained in both directions. Initial BLAST searches of the GenBank databases clearly demonstrated affinities with oligohymenophorean ciliates, particularly scuticociliates. In preliminary phylogenetic analyses which use likelihood, distance and parsimony methods, this relationship was confirmed. However, the apostome sequence diverged basal to the Subclasses Astomatia and Scuticociliatia consistent with the assignment of a Subclass Apostomatia within the Class Oligohymenophorea. We have made collections of *Hyalophysa chattoni* and anticipate that the SSrRNA gene sequences of this genus will confirm the relationship of apostomes to the oligohymenophoreans.

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55A

Ciliate diversity related to shrimps and blue crab. HONGWEI MA and ROBIN M. OVERSTREET, Department of Coastal Sciences, The University of Southern Mississippi, Ocean Springs, MS 39566, USA.

Observations of ciliates using the Chatton–Lwoff wet silver method, Wilbert's protargol impregnation method, and silver carbonated method revealed the infraciliature and nuclear apparatus of species collected from the blue crab (*Callinectes sapidus*) and brown shrimp (*Farfantepenaeus aztecus*) from Mississippi, white shrimp (*Litopenaeus setiferus*) from South Carolina, northern shrimp (*Pandalus borealis*) from Maine, and water associated with each. Specimens were measured and photographed under an Evolution MP 5.0 RTV digital camera system. More than 26 ciliate species, belonging to 21 families in 11 orders, were observed. Among them, five species occurred on or in the blue crab, four on the brown shrimp, two on the white shrimp, and three on the northern shrimp. Of those, only three infested more than one host species. We have seen additional symbiotic species on other specimens of the same hosts but not on those reported here. A total of 16 free-living ciliate species occurred in the systems associated with the decapods. Among them, scuticociliates were the most important ones that could infect the decapods. One of these, *Uronema marinum*, also occurred in the hemolymph of the blue crab and impaired its health.

The research was supported by the USDA, CSREES Grant No. 2002-38808-01381 and the Blue Crab Advanced Research Consortium (BCARC), NOAA, NA17FU2841.

56A

Mechanisms of pathogenesis of the CNS pathogen, *Naegleria fowleri*. REBECCA MACLEAN and FRANCINE MARCIANO-CABRAL, Virginia Commonwealth University School of Medicine, 1101 East Marshall Street, Richmond, VA, USA.

Primary Amebic Meningoencephalitis is a fatal disease of the central nervous system (CNS) caused by *Naegleria fowleri*, an ameboflagellate found in freshwater habitats throughout the world. Pathogenic and nonpathogenic *Naegleria* have been isolated from the environment, but mechanisms of pathogenicity are not fully understood. Activation of complement in the CNS, induction of proinflammatory cytokines, as well as direct damage to brain tissue by the amebae may contribute to the pathogenesis of infection. We examined pathogenic *N. fowleri* in brain tissue from mice by scanning electron microscopy. Trophozoites with prominent food cups were abundant on lysed brain tissue. Ribonuclease protection assays of brain microglial cells cocultured with amebae revealed that pathogenic *N. fowleri* induced the production of proinflammatory cytokine mRNAs. Additionally, we have demonstrated that pathogenic *N. fowleri* activate the alternative pathway of complement but are resistant to complement-mediated lysis. Previous studies in our laboratory have determined that an 18–20 kDa protein on the surface of *N. fowleri* is cross-reactive with antibodies to mammalian CD59, a complement regulatory protein that prevents lysis of target cells. Also, cDNA encoding CD59 from human erythroleukemic cells was shown to hybridize with RNA and gDNA isolated from complement-resistant *N. fowleri* but not with nonpathogenic complement-sensitive *N. gruberi* in Northern and Southern blots. The amebae “CD59-like” protein may be a virulence factor and play a role in regulating complement and also serve to protect the ameba from cytolytic molecules.

57A

Food uptake in the mixotrophic *Dinophysis acuminata*. LUCIE MARANDA, EMILY W. GRASON and JAN E. RINES, Graduate School of Oceanography, University of Rhode Island, Narragansett, RI 02882, USA.

Evidence of food uptake in the photosynthetic genus *Dinophysis* comes solely from the presence of food vacuoles, as no photosynthetic cells have ever been observed in the act of feeding. We examined the feeding ecology of *D. acuminata* in natural populations and under laboratory conditions. Using depth-integrated sampling of the water column, we determined the frequency of food vacuolated cells at 2-h intervals over a 24-h period in a shallow marine embayment. Food vacuoles in preserved cells were enumerated using Nomarski differential interference contrast microscopy; ultrastructural characters were recorded by transmission electron microscopy. A peak in the feeding activity was observed toward dusk for an abundant June population, with 26% of cells with at least one food vacuole. Mechanisms of concurrent carbon acquisition were evident from the presence of chloroplasts with starch grains and food vacuoles within the same cell. Vacuole content could not be identified. In a preliminary 2-wk long simulated grazing experiment, a mixture of two hypothesized preys, *Rhodomonas salina* and *Dunaliella tertiolecta*, was offered to *D. acuminata*; the *Dinophysis* populations decreased steadily and at the same rate, whether food was present or not. The evaluation of the food vacuole frequency will be repeated in the coming season to verify the observed pattern, while grazing experiments will include a variety of food items and incubation conditions. Our current inability to successfully culture any photosynthetic *Dinophysis* limits ecophysiological approaches, either at the population or cellular level, to manipulation of field samples.

Supported by National Institutes of Health Grant GM62126-01A1.

58A

Analysis of centrin-, actin-, and tetra- containing filaments in the *Tetrahymena vorax* macrostomal oral apparatus and their

relation to the microtubule cytoskeleton. NEIL B. MCLAUGHLIN, SUZANNE M. MCCUTCHEON and HOWARD E. BUHSE, JR., Department of Biological Sciences, University of Illinois at Chicago, 840 West Taylor Street, Chicago, IL, 60607.

Our laboratory studies how the contractile cytoskeleton contributes to the process of phagocytosis. Because of its larger size and ease of manipulation, we chose the macrostomal cell of *Tetrahymena vorax* as our model for analysis of the distribution of proteins of prominent filamentous structures within the large oral apparatus (OA). Previous work in our laboratory identified centrin as a component of the fine filamentous reticulum (FFR) and actin and tetra as colocalizing components of the coarse filamentous reticulum (CFR) and cross-connectives (CC) (*J. Eukaryot. Microbiol.*, 51:253–257). Our new data also show that actin coimmunoprecipitates with tetra proteins, confirming our actin–tetra colocalization results. Because of its positioning around the cytostome, the actin-containing CFR/CC is a logical candidate for involvement in phagosome “pinch-off” following prey ingestion. We have analyzed this process by employing an assay that uses the addition of calcium to induce phagosome formation. We show that inhibitors of actin are able to block this event, indicating that actin is necessary for phagosome “pinch-off”. The OA also contains precisely arranged arrays of microtubules. We have examined the spatial relationship of the microtubular arrays to the distribution patterns of centrin, actin and tetra, and found that tubulin and centrin fluorescences overlap in the region of the undulating membrane (UM). Tubulin fluorescence overlaps with actin and tetra labeling at the inner edge of the CFR, where the CFR and UM converge. In addition to ciliary and oral rib labeling, tubulin antibodies also recognize the outer microtubule bundle (OMB), which delineates the right and posterior boundary of the OA.

59A

In the land of the blind, one-eye is king: ecology of the mixotrophic ciliates *Strombidium oculatum* and *Strombidium stylifer*. GEORGE MCMANUS*, HUAN ZHANG*, SENJIE LIN*, LAURA KATZ** and KATARZYNA A. PIROG**, *Department of Marine Sciences, University of Connecticut, Groton, CT, USA, **Biology Department, Smith College, Northampton, MA, USA.

Strombidium oculatum and *Strombidium stylifer* live in tide pools at mid-latitudes on both sides of the North Atlantic Ocean. These “green” ciliates obtain chloroplasts from macroalgae, in particular from members of the Ulvophyceae, which they use for photosynthesis. Both ciliates also contain orange-pigmented granules that are organized into an eyespot. These granules are probably also obtained from their prey, the motile swarmer cells of the algae. *Strombidium stylifer* has been cultivated in our laboratory on pieces of Ulva, as well as a variety of microalgae. *Strombidium oculatum* has resisted cultivation to date. *S. stylifer* is an obligate mixotroph, incapable of sustained growth either starved in the light or fed in the dark. The presence in the ciliates of algal-derived organelles of photosynthesis and photodetection raises interesting questions about the behavior of both predator and prey, with light possibly being a critical integrating cue for their interaction.

60A

Species diversity patterns of choanoflagellates in the Pettaquamscutt River Estuary, Rhode Island, USA. SUNSHINE MENEZES and PAUL E. HARGRAVES, Graduate School of

Oceanography, University of Rhode Island, South Ferry Road, Narragansett, RI 02882, USA.

Choanoflagellates are a large group of heterotrophic protists that play an important role in carbon transfer through the microbial loop, as they consume bacterioplankton and have also been found to be a major food source for macrozooplankton. Although many choanoflagellate species are cosmopolitan, species diversity of the acanthoecid choanoflagellates has been shown to vary substantially between study areas around the globe. A seasonal investigation of acanthoecid species diversity was conducted along the salinity gradient in the Pettaquamscutt River Estuary (PRE), Rhode Island, USA. The taxonomy of the Acanthoecidae is based upon the morphology of a complex basket-like lorica that surrounds the protoplast. Only one previously published study has investigated the diversity of this choanoflagellate family in the western North Atlantic, specifically in the coastal waters of Newfoundland. The Rhode Island survey found that maximum species richness occurred between July and October, and some species exhibited a salinity preference in their distributions, while others had a seasonal shift in occurrence. *Acanthocorbis apoda* (Leadbeater) Hara and Takahashi, 1984 occurred most consistently throughout the study (55% of samples), while *Diaphanoeca grandis* Ellis, 1930, *Diplothea costata* Valkanov, 1970, *Savillea micropora* (Norris) Leadbeater, 1975, *Stephanoea cupula* (Leadbeater) Thomsen 1988, and *Stephanoea supracostata* Hara 1996 were each present in 36–45% of samples. The PRE survey found a four-fold increase in species richness of acanthoecid choanoflagellates over the Canadian study, providing further evidence of the large regional variability in choanoflagellate distributions.

61A

Is it the tortoise or the hare? Ciliates grazing on dinoflagellates in boom–bust dynamics. DAVID J.S. MONTAGNES, School of Biological Sciences, University of Liverpool, BioSciences Building, Crown Street, Liverpool, UK.

This mini-review examines the maximum growth rate and threshold levels (the food level where growth rate is zero) of marine planktonic ciliates and heterotrophic dinoflagellates. There are strong indications that ciliates will out-compete dinoflagellates, as they have higher maximum growth rates, lower threshold levels and are able to rapidly exploit both harmful and non-harmful dinoflagellates as prey. I suggest that ciliates are neither the tortoise (slow and steady) nor the hare (fast and foolish), but the best combination of both. To some extent, this argument is raised as a “straw-man” to be knocked down by discussion; the session is intended to serve as a platform for evaluating the relative importance of ciliates and heterotrophic dinoflagellates in marine ecosystems.

62A

Antarctic amoebae adventures. DAWN M. MORAN*, O. R. ANDERSON**, MARK R. DENNETT*, DAVID A. CARON*** and REBECCA J. GAST*, *Biology Department, Woods Hole Oceanographic Institution, Woods Hole, MA, USA, **Lamont-Doherty Earth Observatory of Columbia University, Palisades, NY, USA, ***Department of Biological Sciences, Southern California University, Los Angeles, CA, USA.

Ice, slush, sediment and water samples were collected from the Ross Sea, Antarctica and enriched with a variety of inorganic and organic nutrients. All cultures were maintained at an ambient temperature of 1 °C, and amoebae were observed to occur in the slush, sediment and water enrichments. Amoebae were isolated

into clonal or monocultures, and their characterization was accomplished using a combination of molecular and morphological methods. Full-length 18S ribosomal DNA sequence data indicated that seven of the isolates represented four different amoebae of the Vexilliferidae and Vannellidae families. Acquisition of 18S ribosomal sequences from the parasomes of two amoebae further confirmed their identification as *Neoparamoeba* species. Light microscopy, fluorescence microscopy and TEM observations were accomplished and further support the molecular data. To determine whether these amoebae represented psychrophilic strains, their temperature tolerances were tested. Cultures were inoculated at 1 °C and the temperature was raised 1 °C/day (to a total of 5°), and then held at that temperature for 6 days. This pattern was repeated in 5°-increments, up to 20 °C. The cultures were observed frequently for death and possible cyst formation. Three (2 types of Vannellids) out of seven amoeba cultures survived up to 20 °C. Whereas, the two different Vexilliferids died at a range of temperatures between 10 °C and 20 °C. These data indicate that amoebae found in the Antarctic are varied in their physiological adaptation for growth at cold temperatures.

63A

The twilight of sun-animalcules. SERGEY NIKOLAEV*, CÉDRIC BERNEY**, JOSÉ FAHRNI**, ALEXANDER P. MYLNIKOV***, NIKOLAI B. PETROV* and JAN PAWLOWSKI**, *A. N. Belozersky Institute of Physico-Chemical Biology, Department of Evolutionary Biochemistry, Moscow State University, Moscow, Russian Federation, **Department of Zoology and Animal Biology, University of Geneva, Geneva, Switzerland, ***Institute for Biology of Inland Waters, RAS, Yaroslavl'skaya obl., Borok, Russian Federation.

Molecular sampling of the taxonomic diversity of the living world is nowadays a task of paramount importance. Heliozoa represents one of the major eukaryotic taxa, which remain significantly underrepresented in molecular databases. The term Heliozoa was coined to embrace organisms with a rounded body and stiff pseudopodia. Despite evidences from ultrastructural studies, which conclusively show the polyphyly of selected heliozoan groups, contemporary morphological systems retain Heliozoa as a monophyletic taxon. From the perspective of reconstructing the true phylogeny of Eukaryota, molecular approaches to analyse relationships within this large protist group are evidently necessary. Phylogenetic analysis of our data shows that the four heliozoan taxa branch either independently or within different eukaryotic phyla. The actinophryids (*Actinosphaerium*, *Actinophrys*) appear as a lineage of stramenopiles, while the desmothoracids (*Clathrulina*, *Hedriocystis*) branch within "core Cercozoa". The position of both groups is strongly supported in all analyses and is congruent with ultrastructure-based taxonomic revisions. The centrohelids (*Chlamyaster*, *Heterophrys*, *Pterocystis*, and *Raphidiophrys*) do not seem to be related to any particular eukaryotic phylum, in agreement with a recent molecular study. The taxopodid *Sticholonche* was found to branch between Polycystinea and Acantharea, two classes of radiolarians. Results obtained in this study suggest that the heliozoan body form cannot be used as a diagnostic argument to unite Heliozoa. Instead, we discriminate between the three heliomorphic taxa of independent origin, Actinophryida, Desmothoracida and *Sticholonche*, and propose the novel higher rank taxon Centrohelida. The term Heliozoa should thus be used solely in historical context.

64A

A hypothesis of genome structure in marine phytoplankton. MATTHEW J. OLIVER***, ZOE V. FINKEL***, OSCAR M.

SCHOFIELD*** and PAUL G. FALKOWSKI***, *Environmental Biophysics and Molecular Ecology, Rutgers University, 71 Dudley Rd. New Brunswick, NJ 08901, USA **Coastal Ocean Observation Laboratory, Institute for Marine and Coastal Sciences, Rutgers University, 71 Dudley Rd. New Brunswick, NJ 08901, USA, ***Geological Sciences Wright Geological Laboratory, Rutgers University 610 Taylor Road, Piscataway, NJ 08854, USA.

The genome size of marine phytoplankton is in the order of 106–1011 base pairs. Within the phytoplankton it has often been assumed that dinoflagellates have extraordinarily large genomes that significantly contribute to their nutrient requirements. We test this hypothesis by compiling cell size and genome size data from across many phytoplankton lineages, and apply a simple regression model. Our results suggest that dinoflagellates do not have anomalously large genomes, but instead scale with cell size with the same slope and intercept as many other diverse phytoplankton taxa. Based on the known correlations of genome size and genome content in diverse taxa, we model the approximate expected genome structure in unsequenced lineages of marine phytoplankton. Based on this model, we hypothesize that retrotransposons play a significant role in genome size in marine phytoplankton.

65A

The *Tetrahymena thermophila* genome sequence: 1. Genome-wide uniformity of macronuclear chromosome copy number in spite of amitotic (random) copy distribution. EDUARDO ORIAS*, STEVEN L. SALZBERG*** and ARTHUR L. DELCHER**, *University of California, Santa Barbara, CA, USA, **The Institute for Genomic Research, Rockville, MD, USA, ***Johns Hopkins University, Baltimore, MD, USA.

In Ciliates, the diploid, mitotic, genetically silent micronucleus is the cell's germline, while the macronucleus is the exclusive site of gene expression. Pre-genomic studies in *Tetrahymena thermophila* have shown that

- The macronucleus contains 250–300 chromosomes, derived from the five germline chromosomes;
- G2 copies of each chromosome are distributed amitotically (randomly) during macronuclear division.

In heterozygotes, this results in the segregation of vegetative descendants expressing only one of the two alleles (assortment). Phenotypic assortment rate measurements for a handful of loci suggested a uniform macronuclear chromosome G1 copy number (~45 per macronucleus; rDNA minichromosome excepted). The NIGMS/NSF-supported, whole-genome-shotgun sequencing of the *T. thermophila* macronuclear genome was recently completed at TIGR. (Sequence and scaffold databases, and related resources, are publicly available for blasting or downloading without restriction at <http://www.tigr.org/tdb/e2k1/ttg/>). Analysis of the depth of sequence coverage shows that (rDNA excepted) at least 95%—and possibly all—of the macronuclear genome is in chromosomes with uniform copy number. This implies the existence of regulatory mechanisms that independently correct the number of copies of nearly 300 different chromosomes after random distribution at every cell cycle.

Supported by NIH grant RR02931. Preliminary sequence data were obtained from The Institute for Genomic Research website.

66A

Small subunit ribosomal RNA sequences of Phaeodarea challenge the monophyly of Haeckel's Radiolaria. STEPHANE POLET, CÉDRIC BERNEY, JOSÉ FAHRNI and JAN PAW-

LOWSKI, Department of Zoology and Animal Biology, University of Geneva, Geneva, Switzerland.

In his grand monography of Radiolaria, Ernst Haeckel originally placed Phaeodarea within the class Radiolaria, together with Acantharea and Polycystinea. Cytological and ultrastructural studies, however, questioned the monophyly of Radiolaria, suggesting the independent evolutionary origin of the three taxa. Some recent molecular analyses based on small subunit ribosomal RNA (SSU rRNA) sequences challenged this classification, supporting the sisterhood of Acantharea and Polycystinea. In order to further test the monophyly of all Haeckel's Radiolaria, and as no data on Phaeodarea were available, we sequenced the complete SSU rRNA gene of three Phaeodarea and three Polycystinea. Our analyses show that the monophyletic Phaeodarea clearly branch among Cercozoa, and confirm that Acantharea and Polycystinea share a common history. This result enhances the morphological variability within Cercozoa, a phylum already containing very heterogeneous groups of protists. Our study suggests that the ability to secrete SrSO_4 and the organization of microtubules inside axopodia are better phylogenetic markers than the simple presence of a central capsule and axopodia.

67A

Microarrays: a tool for ecologists and protozoologists!. P. A. RUBLEE, V. C. HENRICH, M. M. MARSHALL, R. AMOS and L. BALSER, University of North Carolina at Greensboro, NC, USA.

A major difficulty in microbial ecology is identifying and enumerating the members of the microbial community, both prokaryotes and eukaryotes. The advent of PCR and molecular tools offers new windows into microbial communities that do not require cultivation, thus bypassing some selective artifacts and the effort required to maintain cultures. Our goal is to develop community microarrays that have the capability to assess thousands of taxa within a single sample, including important members of the microbial community. We describe the basic approach to microarray development and preliminary results from three lakes that are providing insight into microbial communities.

68A

The evolutionary relationships amongst excavates: a concatenated protein analysis. ALASTAIR G.B. SIMPSON*, YUJI INAGAKI** and ANDREW J. ROGER***, *C.I.A.R. Program in Evolutionary Biology, Department of Biology, Dalhousie University, 1355 Oxford Street, Halifax, NS, Canada, **Department of Bio-science Nagahama Institute of Bio-science and Technology 1266 Tamura, Nagahama, Shiga 526-0829, Japan, ***C.I.A.R. Program in Evolutionary Biology, Department of Biochemistry and Molecular Biology, Dalhousie University, 5850 College Street, Halifax, NS, Canada.

Morphological studies and phylogenies of ribosomal RNA, taken together, suggest that excavate protists may be related to each other, but many of the deepest level relationships amongst these organisms remain poorly understood. We have assembled a data set of six slowly evolving nuclear-encoded protein genes that include nine of the 10 recognized excavate groups. Maximum likelihood analyses demonstrate that diplomonads and *Carpodomonas* than parabasalids are related to each other. They also confirm that *Trimastix* is specifically related to oxymonads, forming the taxon Preaxostyla. There is strong support for a clade of Euglenozoa, Heterolobosea and jakobids, but, unexpectedly, jakobids and Heterolobosea are robustly recovered as sister taxa. *Malawimonas* is placed either as sister to Preaxostyla or as sister

to the (Euglenozoa, Heterolobosea, jakobid) clade. The original data set strongly supports an association between the (diplomonad, *Carpodomonas*, parabasalid) clade and Opisthokonts. However, this grouping is not recovered when α -tubulin is excluded from the analysis, suggesting that the signal for this relationship lies within this one protein and might be suspect. All other important nodes in the tree are, by contrast, robust to the removal of any one gene. With α -tubulin excluded, excavates tend to form just two clades, with no strong nodes separating them. Jakobids, with their apparently ancestral bacterial-type mitochondrial RNA polymerase are nonetheless nested within a clade with normal phage-type RNA polymerases, complicating any understanding of deep-level mitochondrial evolution. The position of jakobids also seriously challenges the now well-accepted concept of a taxon Discicristata.

69A

Colchicine induces nuclear death in *Tetrahymena thermophila*. SITA SINGHAL and JASON WOLFE, Wesleyan University, Middletown, CT 06459, USA.

We have induced the resorption of the micronucleus in living conjugants of *Tetrahymena*. After pairing, but prior to meiosis, the micronucleus undergoes an extraordinary elongation, expanding from a sphere of about 1- μm diameter to a ribbon-shaped structure whose length is greater than that of the cell. Previous EM studies showed an organized basket of microtubules subjacent to the nuclear envelope in the elongating micronucleus. We exposed early pairs to colchicines (1–5 mM), a microtubule inhibitor, and found that micronuclear elongation was prevented. Further, colchicine caused the collapse of already elongated micronuclei. Unexpectedly, with increased time of exposure to colchicine, micronuclei gradually disappeared from view; by 5 h ~ 60% of cells in pairs lost their micronuclei. Direct observation ruled out ejection from the cell. We present evidence that micronuclei are degraded by the lysosomally mediated process of autophagy. Degrading micronuclei in living pairs exposed to both Hoechst 33342 and Acridine Orange stain yellow—blue (for DNA) and orange (for acidification)—as expected for an autophagic vacuole. Further, 3-methyladenine, an inhibitor of autophagy, blocked acidification and prevented degradation. Also, based on the requirement of actin filaments to form autophagosomes, we tested Cytochalasin B, an actin inhibitor. Cytochalasin prevented acidification and blocked micronuclear degradation. Finally, degrading micronuclei, uniquely, fluoresced red in cells whose lysosomes were pre-loaded with Texas Red Dextran, indicate that material was transferred from lysosomes to nuclei-containing vacuoles. These data suggest that nuclear degradation is achieved by autophagy. The mechanism by which colchicine triggers micronuclear autophagy remains to be explored.

70A

Genome evolution in Microsporidia. CLAUDIO H. SLAMOVITS and PATRICK J. KEELING, Canadian Institute for Advanced Research, Department of Botany, University of British Columbia, 3529-6270 University Blvd., Vancouver, BC, Canada, V6T 1Z4.

Microsporidia are eukaryotic intracellular parasites that evolved from fungi. However, they are highly derived and specialized, and lack several typical eukaryotic features, such as canonical mitochondria, flagella and peroxisomes. They also exhibit seemingly "primitive" traits, probably due to their extreme sequence divergence and reductive evolution, which long obscured their true phylogenetic affinity. Gene content and geno-

mic architecture in Microsporidia are known from *E. cuniculi*, the only species whose complete genome has been sequenced. A genomic sequence survey of *A. locustae*, a second and distantly related species, carried out in our laboratory provided valuable data for many studies that improved our knowledge on the biology and evolution of Microsporidia. We present here the latest results of our survey, emphasizing comparative analyses between *A. locustae* and *E. cuniculi* in order to explore the dynamics of genome evolution and compaction in this intriguing group of parasites. We found that, in spite of their extreme rates of sequence evolution, microsporidian genome structure evolves slowly compared to other eukaryotic groups and we propose that it is a consequence of compaction. We also report the discovery and characterization of a photolyase in *A. locustae* that protects the spore from DNA damage. In addition, comparative analysis allows the identification of several important genes that would otherwise not be recognised due to the extreme level of sequence evolution. Finally, we compared intergenic DNA between both species aiming to understand the structure of regulatory sequences in these highly compacted genomes.

71A

Several mixotrophic dinoflagellates preferentially ingest ciliates as an alternate source of nutrition. GABRIELA W. SMALLEY, Georgia Institute of Technology, 10 Ocean Science Circle, Savannah, Georgia 31411, USA.

Mixotrophy is a common nutritional strategy among dinoflagellates. Especially in enriched coastal regions where dinoflagellates are often abundant, mixotrophy may constitute an important microbial pathway. In Chesapeake Bay, several bloom-forming dinoflagellates are mixotrophic, including *Ceratium furca*, *Akashiwo sanguinea*, and *Gymnodinium instriatum*. While these three species are mainly photosynthetic, all three are capable of ingesting prey. Analysis of food vacuole content shows that they predominantly feed on oligotrich ciliates. Their grazing impact on prey populations can be substantial at times, with both *C. furca* and *A. sanguinea* capable of removing over 100% of the <20- μ m-oligotrich standing stock. The ingested ciliate carbon, nitrogen, and phosphorus may contribute significantly to the dinoflagellates' nutritional requirements. For example, *C. furca* could meet up to 65% of its C, 92% of its N, and 57% of its P requirements for reproduction through phagotrophy, theoretically leading to a 40%-reduction in doubling time. Feeding in these dinoflagellate species is influenced by a number of factors, including prey abundance, irradiance, and nutrients. In *C. furca*, internal carbon, nitrogen and phosphorus pools and their relative concentrations seem to play a role in feeding regulation. When cellular nutrient ratios deviate significantly from those required for optimum growth, either due to limited inorganic nutrient availability or reduced carbon fixation, feeding is induced in an attempt to balance the nutrient budget. The ability of these dinoflagellates to exploit an organic nutrient source (i.e. ciliates) may thus give them a competitive advantage over purely photosynthetic species, especially when inorganic nutrients are limited.

72A

Hidden diversity in the tide pool ciliate *Strombidium occulatum* as revealed by multiple molecular markers. OONA L.O. SNOEYENBOS-WEST*, KATARZYNA A. PIROG, AUTUMN GRIFFIN*, GEORGE MCMANUS**, BARBARA COSTAS** and LAURA A. KATZ, *Department of Biological Sciences, Smith College, Northampton, MA 01063, USA, **Department

of Marine Sciences, University of Connecticut, Groton, CT 06340, USA.

We find tremendous genetic diversity underlying the morphological entity *Strombidium occulatum*. We used multi-locus sequence data to examine molecular diversity within *S. occulatum* across space and time so as to explore the biogeography and biodiversity of this marine ciliate species. We are sequencing the internally transcribed spacer regions (ITS1 and ITS2) of the SSU rDNA gene, Mitochondrial SSU and Histone H4 from *S. occulatum* populations collected on either side of the North Atlantic Ocean: from tide pools along the western coast of the United Kingdom and the southern coast of New England. Samples were collected over short temporal (tidal) and both small and large spatial scales in order to explore the level of intraspecific genetic variation. Analyses of our sequence data show that a high level of DNA polymorphisms and genetic heterogeneity (e.g. up to 13% divergence at the ITS locus) underlies *S. occulatum*, indicating the existence of extensive cryptic speciation in this ciliate. We propose that the high levels of genetic diversity in *S. occulatum* arose during times of population subdivision (allopatry), perhaps driven by glacial retreat and sea-level change during the Middle Miocene (~15 mya) in Europe.

73A

Molecular phylogeny of phyllopharyngen ciliates and their novel Group I introns. OONA L.O. SNOEYENBOS-WEST*, JEFFREY COLE**, ANNA CAMPBELL*, D. WAYNE COATS*** and LAURA A. KATZ, *Department of Biological Sciences, Smith College, Northampton, MA 01063, USA, **Protistology Collection, American Type Culture Collection, Manassas, VA 20110-2209, USA, ***Smithsonian Environmental Research Center, PO Box 28, Edgewater, MD 21037, USA.

We analyze small subunit ribosomal DNA (ssu-rDNA) sequences to evaluate both the monophyly of the ciliate class Phyllopharyngea de Puytorac et al. (1974), and relationships among subclasses. Classifications based on morphology and ultrastructure divide the Phyllopharyngea into four subclasses: the Phyllopharyngia, Chonotrichia, Rhynchodia and Suctorina. Our analyses of ssu-rDNA genealogies derived from sequence data collected from diverse members representing three of the four subclasses of Phyllopharyngea provide strong support for the monophyly of the Phyllopharyngea, and show that the Chonotrichia emerge from within the Phyllopharyngia. Based on this initial sampling, suctorian budding types are monophyletic, and exogenous budding appears to be basal to evaginative and endogenous budding. Further, we report the discovery of a group I intron at position 891 in the *Suctorina Acineta* sp. and *Tokophrya lemnae*, and a second group I intron at position 1506 in *T. lemnae*. These introns represent only the second examples of group I introns in a ciliate ribosomal gene, since the discovery of ribozymes in the LSU rRNA gene of *Tetrahymena thermophila* 20 years ago. Phylogenetic analyses of Group I introns suggest a complex evolutionary history involving either multiple losses or gains of introns within endogenously budding Suctorina.

74A

Understanding genome complexity and protein evolution in ciliates: insights from mitochondria and nuclei. OONA L.O. SNOEYENBOS-WEST*, ELYSE LASSER*, JESSICA C. CARDILLO* and LAURA A. KATZ, *Department of Biological Sciences, Smith College, Northampton, MA 01063, USA.

We are exploring the evolutionary implications of genome duality in ciliates. Ciliates possess both a somatic macronucleus

(MAC) and a germline micronucleus (MIC). Only the MAC is transcriptionally active, and divides by amitosis rather than mitosis and meiosis. We have previously hypothesized that genome duality is the driving force behind the heterogeneous and unusually elevated rates of amino acid substitutions we find in some protein coding genes. Does the dimorphic nature of the ciliate genome allow them to “cross the valleys in the adaptive landscape” of protein evolution? To test this hypothesis, we are analyzing additional protein genes as well as mitochondrial genes. Mitochondrial genes serve as an excellent comparison to those of the nucleus because phylogenetic information lost by the rapidly evolving nuclear genome may be retained in the mitochondria.

75A

Molecular evolution of eukaryotes using structural genomic data. ALEXANDRA STECHMANN^{*,**} and THOMAS CAVALLIER-SMITH^{*}, ^{*}Department of Zoology, University of Oxford, South Parks Road, Oxford OX1 3PS, UK, ^{**}Department of Biochemistry and Molecular Biology, Dalhousie University, 5850 College Street, Halifax, NS, Canada B3H 1X5.

One of the most challenging evolutionary problems is locating the root of the eukaryote tree. We know that the root lies somewhere within the kingdom Protozoa, but the widespread view that early eukaryotes were amitochondrial has recently been dramatically overturned. Systematic biases in sequence evolution prevent the reliable inference of the eukaryote root from single-gene trees. Concatenated sequence trees should be more reliable, but leave many possibilities open. In principle deletions/insertions or gene fusions should be superior for this purpose whenever ancestral and derived states are clearly distinguishable. Using a derived gene fusion between dihydrofolate reductase (DHFR) and thymidylate synthase (TS), genes we were able to greatly narrow down the position of the root. This gene fusion clearly divides eukaryotes into two clades: Amoebozoa plus opisthokonts (unikonts, which are ancestrally uniciliate) and bikonts, which are ancestrally biciliate. Another gene fusion between the first three enzymes of the six enzyme pyrimidine synthesis pathway supports this division. This derived three gene fusion is seen in Amoebozoa and opisthokonts (unikonts) but not in bikonts. A third gene fusion in the phosphofructokinase gene which is so far only found in opisthokonts and Amoebozoa further lends support to the eukaryote unikont/bikont divide. Independent support comes from sequence trees based on concatenated protein alignments. Our results show that the primary divergence of eukaryotes was probably between unikonts (animals, Fungi, Choanozoa and Amoebozoa) and bikonts (plants, chromists, all other Protozoa: alveolates, excavates, Rhizaria, Apusozoa) and that the root of the eukaryote tree lies between these two groups.

76A

Microzooplankton grazing on harmful dinoflagellate blooms: Are ciliates or heterotrophic dinoflagellates important? DIANE K. STOECKER, MATTHEW D. JOHNSON, MATTHEW REAUGH and DANIEL E. GUSTAFSON, JR., University of Maryland Center for Environmental Science, Horn Point Laboratory, PO Box 775, Cambridge, MD 21664, USA.

Microzooplankton grazing can be important in regulating growth of dinoflagellate populations, including species responsible for harmful algal blooms. In the Chesapeake Bay region, microzooplankton community grazing coefficients on small cell-size dinoflagellates are often greater than potential gross growth coefficients of dinoflagellates, and thus grazing may prevent bloom formation. Who are the major microzooplankton grazers

on small dinoflagellates? Ciliates or other dinoflagellates? Data from Chesapeake Bay and its tributaries indicate both; sometimes ciliates and sometimes dinoflagellates are the major grazers. The importance of ciliates and heterotrophic dinoflagellates varies with season and location, but often one group dominates the microzooplankton assemblage. Specific clearance and division rates of ciliates are higher than that of heterotrophic dinoflagellates, thus it could be expected that ciliates would be the dominant microzooplankton grazers. However, during summer, small heterotrophic dinoflagellates are often the dominant grazers on small dinoflagellates in the mesohaline Bay. Differential predation by copepods on ciliates may be responsible for this pattern. When microzooplankton community grazing is less than dinoflagellate cell division, red tides may result. Thus, it is important to understand the factors controlling both ciliate and heterotrophic dinoflagellate populations and their grazing impacts.

77A

Functional characterization of the *T. gondii* ortholog of parafusin by an in vivo *Paramecium* ortholog assay. S. C. TUCKER, L. LIU and B. H. SATIR, Department of Anatomy and Structural Biology, Albert Einstein College of Medicine, Bronx, NY, USA.

Parafusin (PFUS) is a phosphoglycoprotein of *Paramecium* that functions in Ca^{2+} -regulated exocytosis. The ortholog of this protein in *Toxoplasma gondii*, called Parafusin Related Protein 1 or PRP1, is associated with micronemes and functions in Ca^{2+} -regulated secretion (Matthiesen et al., 2003). We have demonstrated that PRP1 colocalizes with PFUS to the *Paramecium* dense core vesicles (trichocysts) by means of a new assay. In this assay, recombinant, fluorescently tagged PRP1 was electroporated directly into either wild type or mutant live *Paramecium* cells. After a brief recovery period, the protein was observed on trichocysts docked below the cell membrane. In control experiments using labeled actin and pyruvate kinase, no localization to trichocysts was seen. Additionally, PRP1 could only be detected on undocked trichocysts that had accumulated in the cytoplasm of *tam8*, a *Paramecium* exo-mutant that fails to transport trichocysts to the surface. Furthermore, PRP1 fluorescence disappeared from the trichocysts of wild-type *Paramecium* that were stimulated with lysozyme for synchronous secretion, in a manner similar to that observed for PFUS. Cumulatively these data suggest that PRP1 of *T. gondii* can be properly modified, transported and localized after electroporation and that fluorochrome labeling of recombinant PRP1 did not adversely affect localization. The assay is ideal for assessing strategic residues and domains within PRP1 and other orthologs for their structural and functional relevance in exocytic processes in living cells.

78A

The alternative oxidase from *Euglena gracilis*. URSULA THEISSEN, PEER HEBING and WILLIAM MARTIN, Institute of Botany III, University of Düsseldorf, Düsseldorf 40225, Germany.

The alternative oxidase (AOX) is an ubiquinol oxidase of the mitochondrial respiratory chain. AOX is an additional terminal oxidase; the electron flux to AOX branches from the classical electron-transport chain at the level of the ubiquinone pool leading to the direct reduction of oxygen to water in a single four-electron transfer step. O_2 reduction catalyzed by AOX is not coupled to proton pumping and hence not ATP-producing. The energy is released as heat. AOX is specifically inhibited by salicylhydroxamic acid (SHAM), but is cyanide resistant. AOX was discovered in plant thermogenic tissues and all plant species

tested to date can express the alternative pathway. It was long thought that AOX is specific to higher plants, but it is now turning up in many non-photosynthetic eukaryotes. For example the bloodstream form of the protist *Trypanosoma brucei brucei* that lacks functional cytochromes, limits most of its energy metabolism to glycolysis while passing the reducing equivalents produced to water via a plant-like alternative oxidase. It has also been reported that oxidative phosphorylation in the protist *Euglena gracilis* is supported by an alternative respiratory pathway in which an AOX is involved. We have cloned the mitochondrial AOX from *Euglena* and have investigated the evolutionary history of this enzyme in prokaryotes and eukaryotes.

79A

Not the most basal eukaryotes: challenges and pitfalls affecting the placement of *Giardia* and *Trichomonas*. GISELLE WALKER* and T. MARTIN EMBLEY**, *The Natural History Museum, London, UK and Københavns Universitet, København, Denmark, **University of Newcastle, Newcastle, UK.

Diplomonads (including *Giardia*) and parabasalids (including *Trichomonas*) have traditionally been seen as the deepest-branching eukaryotes, on the evidence of apparently simple ultrastructural morphology, and small subunit ribosomal RNA trees. However, this position does not appear to be well supported, especially in the light of the discoveries of apparent mitochondrial remnants in *Giardia* and *Trichomonas*. We systematically re-analysed protein data, and show here that their traditional placement may be due to a combination of factors acting on data sets (narrow taxon sampling) and on genes (rate heterogeneity and compositional heterogeneity, mutational saturation). Our data found no consistent support for either *Giardia* or *Trichomonas* being basal eukaryotes.

80A

Filling in the gaps: EST surveys of under-represented chromalveolates. ROSS F. WALLER, NICOLA J. PATRON and PATRICK J. KEELING, CIAR, Evolutionary Biology Program, Department of Botany, University of British Columbia, 3529–6270 University Boulevard, Vancouver, BC, Canada V6T 1Z4.

The “chromalveolate hypothesis” posits the monophyly of alveolates (apicomplexans, dinoflagellates and ciliates), haptophytes and cryptomonads. One implication of the monophyly of chromalveolates is that the plastids in this group could share a common ancestor—derived from a single endosymbiosis. Members of all chromalveolate groups, with the exception of ciliates, possess a secondary plastid originating from a rhodophytic symbiont (although the rhodophytic origin of the apicomplexan plastid remains contentious). A single origin of this chromalveolate plastid represents the most parsimonious hypothesis for plastid acquisition. Growing support for the “chromalveolate hypothesis” has relied on a relatively small number of molecular studies to date, and a more thorough examination of this alliance has been undermined by a paucity of broad molecular data from several of its members, despite complete genome sequences from apicomplexans and ciliates. To address this deficit we have undertaken EST (Expressed Sequence Tag) surveys of several taxa, representing dinoflagellates, haptophytes and cryptomonads. Using EST data we are undertaking broad molecular comparisons of these groups through molecular phylogenies, analyses of common biochemical pathways, and studies of organelle evolution. This genome-based approach provides a more complete view of the major chromalveolate groups, and a chance to more thoroughly test the chromalveolate hypothesis.

81A

Growth of the vacuoleless mutant of *Tetrahymena thermophila* NP1 in phytate. SAMANTHA WEBB*,** and HARRIETT E. SMITH-SOMERVILLE*, *Department of Biological Sciences, University of Alabama, Tuscaloosa, AL, USA, **Stillman College, Tuscaloosa, AL, USA.

Phytate, the salt form of phytic acid, is the major store of phosphate in seeds and grain. Since non-ruminant farm animals poorly digest phytate, it is also a source of environmental phosphate contamination in agricultural areas. We are using *Tetrahymena*, a ciliated protist with multiple routes for nutrient assimilation, as a model to investigate the contribution of heterotrophic protists to the environmental cycling of phosphate from phytate. This ciliate has the ability to grow on phytate as the sole phosphate source (Ziemkiewicz, H. T., Johnson, M. D. & Smith-Somerville, H. E. 2002. *J. Eukaryot. Microbiol.*, 49:428). *Tetrahymena thermophila* NP1, a temperature-sensitive vacuoleless mutant (ATCC #50202), provides a way to separate membrane transport from uptake through phagosomes, and to assess the importance of each mechanism. This cell grows equally well at the permissive and non-permissive temperatures with either phytate or inorganic phosphate as the phosphate source. Our results demonstrate that phagosomes are not required to use the phosphate from phytate.

82A

Divergent β -tubulins from foraminifera: Implications for microtubule assembly. LAURA WEGENER*, JEFFREY L. TRAVIS*, ANDREA HABURA** and SAMUEL S. BOWSER**, *State University of New York at Albany, Albany, NY 12222, USA, **Wadsworth Center, New York State Department of Health, POB 509, Albany, NY 12201-0509, USA.

Foraminifera utilize an unusual and very rapid microtubule assembly/disassembly system: a state transition between microtubules and helical filaments. The helical filament is an assembly polymorph that forms when the microtubule lattice uncoils into a ribbon of tubulin dimers connected only by lateral subunit/subunit bonds. An unusual β -tubulin sequence, which may be implicated in this process, has previously been reported from the foraminiferan *Reticulomyxa filosa*. In order to determine the taxonomic distribution and possible significance of this tubulin isoform, we sequenced and analyzed β -tubulin genes from a broad taxonomic range of foraminifera. These genes contain a large number of substitutions, conserved within the group, which may alter the biochemical properties of β -tubulin, especially in regions involved in subunit/subunit binding between α - and β -tubulin in the microtubule lattice. In particular, the M-loop, which is intimately involved in lateral subunit binding, shows only 30% homology to the consensus sequence for eukaryotic β -tubulins. Multiple regions identified to be involved in longitudinal contacts with α -tubulin are also highly substituted. In contrast, the foraminiferan α -tubulin is well conserved with its homologs in other organisms. We suggest that these alterations to the consensus eukaryotic β -tubulin sequence may help to explain the unusual assembly properties of foraminiferan microtubules.

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83A

Genes for infectivity and endocellular lifestyle of symbiotic X-bacteria (Candidatus “*Legionella jeonii*”) in *Amoeba proteus*. GYONG S. YANG, KEONG M. LEE and TAE I. AHN, School of biological sciences, Seoul National University, Seoul 151–742, Korea.

Candidatus “*Legionella jeonii*” is an obligate endocellular symbiont in *Amoeba proteus* and cannot be cultivated in vitro. In order to detect genes of the bacteria related with intracellular survival and integration to the host, we performed comparative genomic analysis. A total of 486 non-redundant clones from the genomic DNA library of 3×10^6 pfu were tagged by nucleotide sequencing. To date, we have tagged 1023 genes in 1.5 Mbp and identified 383 genes of assigned functions and 180 conserved hypothetical genes. The remaining 460 tags had low homology. The 69% of high- and 28% of low-homology tags had matches in *Legionella pneumophila* database. We predicted 26 genes to be related with infection and endocellular lifestyle of *L. jeonii* including stress proteins (*dpsA*, *dnaK*, *dnaJ*, *mutT*), virulence system (*icmK*, *icmTS*, *dotA*, *lvrB*, *lvrC*), infection and pathogenicity factors (*rcp*, *shpA*, *rhuM*, *rpfB*), drug resistance proteins (*norM*, *mdfA*), transporter proteins (*ceaA*, *prop*, *necA*), type IV secretion system (*virB3*), a global response regulation sensor kinase (*grrS*), a GTP-binding protein (*rabB*) and ppGpp-regulated growth inhibitors (*mazEF*). The large number of putative insertion elements (26 genes) in the bacterial genome was unusual. To date, about a half genome of *L. jeonii* has been tagged in comparing with the genome of *L. pneumophila* (3.4 Mbp). These results suggest that *L. jeonii* still contains most of those genes for infectivity and endocellular lifestyle of pathogenic bacteria. The symbiotic *L. jeonii* in *A. proteus* could be a model system for molecular pathogenicity and symbiosis.

84A

Developmental genome processing and protein evolution in the ciliate *Chilodonella uncinata*. REBECCA A. ZUFALL* and LAURA A. KATZ*, *Department of Biological Sciences, Smith College, Northampton, MA 01063, USA.

Ciliates are microbial eukaryotes with dual genomes, one present in the transcriptionally inactive germline micronucleus (MIC) and the other in the somatic macronucleus (MAC). In the development of the MAC from the MIC, ciliates process their genomes by chromosomal fragmentation, excision of internal excised sequences (IESs), and amplification of chromosomes. *Chilodonella uncinata* is in a class of ciliates, Phyllopharyngea, that undergo extensive processing to generate MACs containing thousands of gene-sized chromosomes. Previous analyses suggest that sequences involved in this processing are highly variable among ciliate lineages. In this study, we examine cis-acting signals involved in the elimination of IESs in *C. uncinata* in order to understand the phylogenetic level at which processing signals are conserved. In addition, we are testing the hypothesis that the differential selection on dual genomes in ciliates allows unusually rapid divergence among paralogs of protein coding genes.

British Section of the Society of Protozoologists
42nd Annual Meeting
March 25–27, 2004
Lancaster, UK

85A

The response of protist and metazoan communities in permeable pavement structures to high oil loadings. S. J. COUPE and H. G. SMITH, School of Science and the Environment, Coventry University, Coventry CV1 5FB, UK.

Permeable pavement structures (PPS) have been demonstrated to provide an efficient and sustainable method of controlling urban

derived hydrocarbon contamination. Until recently, laboratory PPS mesocosm models have used crushed granite as the load bearing sub-base material. However, the use of virgin stone may not be the optimal choice of substrate, as this is not necessarily sustainable or cost effective in the long term when compared to the use of recycled materials. However, recycled materials such as waste concrete may change the environmental conditions in PPS mesocosms, and the characteristics of the eukaryotic community may become different from those which have been previously described. In the current experiment, granite and recycled concrete sub-base materials were compared for their ability to retain 900 g/m² of clean mineral oil applied to the mesocosm surface. It was observed that, even at this very high oil loading, 99.95% of the applied oil was retained within granite and concrete-based structures, but the effluent was two pH units more alkaline in concrete mesocosms than granite. The eukaryotic microfauna in the effluent from both mesocosm types showed a ten-fold increase in protist abundance, and a doubling in the number of protist genera, compared with earlier work using only 18 g/m² of applied oil. Five genera of testate amoebae not previously recorded in PPS were identified, these included *Arcella*, *Assulina*, *Cryptodiffugia*, *Cyclopyxis* and *Diffugia* in addition to the three genera observed previously using the lower oil application. Metazoan abundances increased from 1.5×10^1 organisms per ml using the lower oil loadings to 2.0×10^3 /ml in the current experiment. Rotifers and nematodes were the most numerous, but tardigrades were also observed in both concrete and granite-based mesocosms. Despite the differences in effluent pH, it was apparent that there were only marginal differences in the eukaryotic microbiology of the two mesocosm types. This was thought to be due to the layered structural arrangement of the pavement and the location of the highly oil-retentive polypropylene geotextile and extensive biofilm layer positioned above the concrete sub-base. Work is now underway to find oil loadings that will adversely affect the abundance and diversity of eukaryotic organisms in PPS mesocosms.

86A

Interactions of *Acanthamoeba polyphaga* with *Listeria monocytogenes* and MRSA. S. A. HUWS, M. R. W. BROWN and A. W. SMITH, Department of Pharmacy and Pharmacology, University of Bath, Claverton Down, Bath, UK.

Bacterivorous protozoa and bacteria have been in co-existence since the origin of life. This co-existence has led unequivocally to the evolution of many different co-interactions. Most bacteria are ingested and digested, but many escape ingestion for various reasons. Others are ingested but evade digestion, and a few, notoriously *Legionella pneumophila*, even have the capacity of multiplying within the protozoan host. The aims of this study were to elucidate the interactions of various multi-drug-resistant *Staphylococcus aureus* (MRSA) strains, *Listeria monocytogenes* sv4b, and *Escherichia coli* K12 with the amoeba, *Acanthamoeba polyphaga*. To evaluate the interactions, we set up co-cultures in Neffs' amoebic saline, at a multiplicity of invasion (MOI) of 1:100 of amoeba to bacteria, and a temperature of 37 °C, although the effects of MOI and temperature were also assessed. Survival of bacteria and amoeba was checked at regular intervals, coupled with microscopy. It was discovered under our test conditions, that *E. coli* was ingested and digested by *A. polyphaga*, but in contrast, *L. monocytogenes*, had the capacity to flourish in the presence of *A. polyphaga*. We also report, for the first time, that all six MRSA isolates tested, survived and replicated in association with *A. polyphaga*, in comparison to conditions where amoebae were absent. Indeed, we also have evidence suggesting that increases

in MRSA, in the presence of *A. polyphaga*, may be attributable to intracellular survival and replication. These findings have profound implications for the hospital environment, where *Acanthamoeba* sp., are also commonly isolated. In conclusion, this study illustrates the significance of protozoa as vehicles augmenting the survival of MRSA and *L. monocytogenes* in the environment.

87A

Identifying new vahlkampfiid amoebae by ITS and 5.8S ribosomal DNA sequencing. J. F. DE JONCKHEERE* and S. BROWN**, *Christian de Duve Inst. Cellular Pathology, Brussels, B-1200, Belgium, **Centre for Ecology and Hydrology, Dorset, Winfrith Technology Centre, Dorchester, Dorset DT2 8ZD, UK.

Small subunit ribosomal DNA (SSU rDNA) comparisons have revealed that not only species but even genus identification based on morphology is unreliable in the family Vahlkampfiidae, and SSU rDNA sequences have been used to redefine vahlkampfiid genera. However, it is time consuming to use SSU rDNA sequences for species identification. The ITS region (including the 5.8S rDNA) has the advantage that it is much shorter than the SSU rDNA and thus easier and quicker to obtain. Initially, 29 strains of 13 species belonging to seven vahlkampfiid genera were investigated. It was found that each species has a specific ITS sequence, which appears to be conserved within a species. The same had previously been found in the vahlkampfiid genus *Naegleria*. Consequently, we have demonstrated that the ITS region (including the 5.8S rDNA) can be used to identify vahlkampfiid amoebae. In the current investigation, the ITS sequences of 16 undescribed vahlkampfiid isolates were determined. Based on these sequences, seven isolates were identified as belonging to described species, while nine represent six new species. Four of the new species belong to the genus *Tetramitus*, and one each to the genera *Vahlkampfia* and *Paravahlkampfia*.

88A

Protozoa and pathogenic bacteria: lessons learned from *Legionella pneumophila*. A. W. SMITH, Department of Pharmacy and Pharmacology, University of Bath, Claverton Down, Bath, UK.

The microhabitats favoured by protozoa are dynamic ecosystems comprising a heterogeneous microflora of bacteria and fungi as well as protozoa in a complex biofilm. Many protozoa feed by grazing on microbial biofilms, but they do not do so indiscriminately. A dynamic equilibrium will exist and the nature of the relationship changes depending on environmental conditions. The view of protozoa as the hunter and bacteria as the hunted is now recognised as overly simplistic. Some bacteria not only resist predation by protozoa, but also have become endosymbionts of free-living protozoa, demonstrating adaptation to the otherwise hostile intracellular environment. The concept that protozoa can be reservoirs for some pathogens is more widely appreciated and the catalyst for much of this work came from the observation that *Legionella pneumophila* infects and multiplies within some species of free-living amoebae. In addition to amplifying cell numbers, the phenotype of intra-protozoal-grown cells is quite distinct from their free-living counterparts cultured on complex laboratory media: altered surface properties, decreased susceptibility to chemical inactivation and antibiotics, and enhanced invasiveness for mammalian cells. There is significant conservation in the genes used by *L. pneumophila* to parasitize protozoa and macrophages and it has been proposed that co-evolution of bacteria and lower order eukaryotic cells has equipped bacteria both for environmental survival and virulence towards higher order eukaryotes.

Other authors have even described protozoa as an "evolutionary gym" enabling *L. pneumophila* to "train" for its role in invading higher eukaryotic organisms.

A growing list of bacteria and fungi that interact with protozoa confirm that the co-evolution between *L. pneumophila* and protozoa is not unique in nature. An emerging theme amongst many of the micro-organisms associating with protozoa is that they typically are recognised as opportunistic pathogens and many lack a recognised animal host. These features are suggestive of a relationship much closer to symbiosis which becomes parasitic only in special circumstances. The question remains why this is so? The answer could lie in co-evolution with lower order eukaryotic organisms present in biofilms.

Groupement des Protistologues de Langue Française (GPLF) 42nd Annual Meeting May 12–14, 2004 Montpellier, France

89A

Prolyl oligopeptidase from *Trypanosoma brucei* hydrolyzes collagen and is involved in parasitic growth. I. M. D. BASTOS*, J. M. SANTANA** and P. GRELLIER*, *Laboratoire de Biologie Fonctionnelle des Protozoaires, USM504, Département Régulations, Développement, Diversité Moléculaire, Muséum National d'Histoire Naturelle, 61 rue Buffon, 75231 Paris Cedex 05, France, **Laboratório Multidisciplinar de Pesquisa em Doença de Chagas, CP04536, Universidade de Brasília, Brasília, Brazil.

Proteases play important roles in several processes of the biology of parasites including interactions with their hosts. We have reported that prolyl oligopeptidase from *Trypanosoma cruzi* (POPTc80) is associated with the entry of trypomastigotes into mammalian host cells. In this study, the gene coding to prolyl oligopeptidase of *Trypanosoma brucei* (POPTb) was identified and characterized. It is represented by a single copy per haploid genome of the parasite and its deduced amino acid sequence shares 77% identity with POPTc80. Secondary structure predictions demonstrated that POPTb shows the highly preserved secondary structure composition and arrangement of prolyl oligopeptidases. Active recombinant POP Tb, produced in *E. coli*, displayed enzymatic activity on peptides containing Pro at P1 position and collagen at a slightly alkaline pH. Its enzymatic activity was highly sensitive to POPTc80 inhibitors. Furthermore, these inhibitors arrested growth of procyclic and bloodstream *T. brucei* forms in a dose-dependent manner. These data suggest that POPTb activity is required for normal parasite development.

90A

A new real-time RT-PCR method allowing reliable absolute quantification of human cytokine mRNA in paediatric malaria clinical samples. P. BOEUF*, I. VIGAN*, D. JUBLOT*, J.-C. BARALE**, B. GOKA***, J. KURTZHALS****, B. D. AKANMORI****, L. HVIID****, O. PUIJALON* and C. BEHR*, *Parasites Molecular Immunology Unit, CNRS URA 2581, Pasteur Institute, Paris, France, **Biology of Host-Parasite Interactions Unit, CNRS URA 2581, Pasteur Institute, Paris, France, ***Korle-Bu Teaching Hospital, Accra, Ghana, ****Centre for Medical Parasitology, Rigshospitalet, Copenhagen.

gen, Denmark, ****Noguchi Memorial Institute for Medical Research, Legon, Accra, Ghana.

In endemic areas, clinical manifestations of *Plasmodium falciparum* infection range from asymptomatic parasitaemia to life-threatening severe syndromes. Immune differences that could account for this disparity are poorly understood. Using tight criteria to classify patients into non-overlapping clinical categories, we showed that cerebral malaria and severe anaemia were distinct immunological syndromes and that a proper quantitative description of cytokine profiles in the various clinical groups is essential to the understanding of the activation of immunocompetent cells.

Due to the limited size of paediatric blood samples, we chose to measure cytokine mRNA using real-time RT-PCR. We showed that RT efficiency displayed intra- and intergenic variations that have to be taken into consideration for reliable absolute RNA quantification when comparing clinical cases. We thus developed a SYBR Green I-based real-time RT-PCR method using synthetic external RNA standards specific for each gene. Absolute RNA quantification is achieved by reverse transcribing known copy numbers of this RNA standard in parallel with cellular RNA. Strictly specific primers were designed to allow the quantification of any RNA in the same thermocycling parameters for future automation. Our method gave similar results for a lower cost when compared with TaqMan, and led to reproducible and reliable absolute RNA quantification. We validated it in vitro on naïve PBMC stimulated by LPS and ex vivo on PBMC from malaria patients. This new method raises the unprecedented possibility to compare cytokine mRNA levels between different clinical groups and is a powerful tool to further study the inflammation processes associated to malaria.

91A

Thymidine kinase and reverse genetics in *Plasmodium falciparum*. S. BONNEFOY*, M. DIEZ*, P. R. PENDYALA*, C. BLANC**, M. GUILLOTTE* and O. MERCEREAU-PUIJALON*, *Immunologie Moléculaire des Parasites, Institut Pasteur, Paris, France, **Parasitologie Biomedicale, Institut Pasteur, Paris, France.

In the past few years, genetic manipulation of *Plasmodium falciparum* has allowed functional studies of many parasite proteins. However, a careful interpretation of gene manipulation results is required, as chromosomal events may occur in vitro independently of the experimental modifications (gene invalidation, overexpression...), and because of the intrinsic mutagenic potential of genetic manipulation, or pleiotropic consequences related to the insertion of exogenous sequences. To ensure that mutant phenotypes arise from the planned mutations, it is necessary to obtain revertants re-expressing the wild-type gene. On this line, we have developed a positive/negative selection strategy allowing in a first step, gene targeting, then excision of the resistance marker. We converted the *Toxoplasma gondii* DHFR/TS into a tri-functional enzyme following fusion with the Herpes virus type I thymidine kinase (HSV-TK1). Positive selection corresponds to pyrimethamine resistant development, while negative selection acts in presence of the pro-drug ganciclovir. We used this tool to study RESA, a dense granule protein that associates with spectrin following invasion. *Resa* gene invalidation was obtained using a targeting plasmid allowing expression of the fusion DHFR/TS/HSV-TK1, and molecular and phenotypic analyses of mutant parasites were performed. Next, wild-type *resa* gene expression was restored after excision of all the exogenous sequences following phosphorylation of ganciclovir by the HSV thymidine kinase into a toxic metabolite. Phenotypic studies of revertants confirm some functions attributed to RESA protein, and

validate the positive/negative strategy in this reverse genetic approach, a useful tool for development of 'hit and run' mutagenesis in *P. falciparum*.

92A

Intestinal microsporidiosis among HIV-positive patients in Libreville (Gabon): *Enterocytozoon bienewisi* isolates show a very high diversity of genotypes. J. BRETON****, M. KOMBILA*, M. OKOME-NKOUMOU**, H. RINDER*** and M. THELLIER****, *Department of Parasitology-Mycology, Faculté de Médecine, B.P. 861, Libreville, Gabon, **Infectious Diseases Unit, Fondation Jeanne Ebori Hospital, Libreville, Gabon, ***Department of Infectious Diseases and Tropical Medicine, Munich University, Leopoldstrasse 5, 80802, Munich, Germany, ****Service of Parasitology-Mycology, Pavillon Laveran, CHU de la Pitié-Salpêtrière, 47 Boulevard de l'Hôpital, 75013 Paris, France.

Intestinal microsporidiosis caused by *Enterocytozoon bienewisi* and *Encephalitozoon intestinalis* are opportunistic infections occurring in severely immuno-depressed patients, and may cause severe chronic diarrhea. Due to the extent of the HIV pandemic, microsporidia could pose a serious threat to public health in Africa. This study was conducted in Libreville, Gabon, central Africa, among HIV-positive patients between September 2002 and December 2003. Criteria for inclusion were being HIV positive (HIV1 and/or HIV2), regardless of the CD4 cell count, stage of the disease or gastrointestinal symptoms. Nine hundred and ninety-three stool samples from 836 patients (mean age 38 years, range 18 months-74 years) were screened for microsporidia by immunofluorescence using species-specific monoclonal antibodies and confirmed by PCR. Twenty seven cases of *E. bienewisi* infection were found in the cohort, a prevalence of 2.7%. These patients were in the most severely immuno-depressed group, 84% had a CD4 cell count below 200/mm³ compared to 57% of the cohort (mean = 55 CD4 + cells/mm³ compared to a mean of 88/mm³ for the microsporidia negative patients with less than 200 CD4/mm³).

The ribosomal DNA ITS sequence has been determined for several *E. bienewisi* isolates. A very high diversity of genotypes was found: eight different sequences were found from only 12 samples. This is surprisingly high since in previous studies conducted in Europe and America only 15 of the 49 *E. bienewisi* genotypes described have been found to infect humans. Only three genotypes from our samples have already been described: type K (two specimens), previously reported in human and cat, type D (one specimen) in human, pig and rhesus monkey, type E (1 specimen) in man, pig, racoon and muskrat. The five other genotypes are new and have been named Gabon-1 to 5. The phylogenetic analysis of the sequences shows that genotypes Gabon-1,2,3 and 4 (one specimen each) are close to types E, WL12, D and K, respectively, which belong to a group comprising most genotypes described so far. However, the Gabon-5 genotype (four specimens) is only distantly related to this group and is most apparent to the group of genotypes (WL1-3) recently described from racoons in North America. It was thus suggested that this group could represent another species or subspecies of *E. bienewisi* which are host-adapted and do not infect humans. However, the finding of genotype Gabon-5 shows that even distantly related genotypes infect severely immuno-depressed humans.

This work was supported by the French National Agency for AIDS Research (ANRS-1264).

93A

Characterization and Functional Analysis of HMG Proteins in *Plasmodium falciparum*. S. BRIQUET, M. GISSOT, C.

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Little is known about regulation of gene expression in *Plasmodium*. Indeed, only few regulatory elements and no clearly defined transcription factors have been reported, so far. In order to evaluate the interplay between *cis*- and *trans*-acting elements controlling the regulation of transcription, we annotated several High Mobility Group proteins of the HMGB superfamily containing one or two HMG-box(es) of 80 aa consisting of three α -helices arranged in L-shaped configuration. These computational analyses suggest that the *Plasmodium* proteins (PfHMG1-4) might function as architectural factors through their ability to bind DNA in a structure-specific manner and to induce bending of DNA. PfHMG1 (97 aa) was cloned from the genomic DNA of the 3D7 clone. The protein was expressed in *E. coli* and antibodies were raised in mice. A messenger of 1.3 kb was characterised by Northern blotting of 3D7 total RNA. Temporal expression of the transcript and protein was examined in two *Plasmodium* clones, 3D7 and F12 its gametocyte-less derivative. The transcript appeared to be developmentally regulated throughout the erythrocytic cycle of *Plasmodium* as well as the protein present in the nuclear extracts. The functional analysis of the recombinant PfHMG1 via EMSA with synthetic fourway DNA junctions and ligase-mediated circularisation assays showed that the factor is a genuine architectural protein. In addition, the functional analyses of two other factors PfHMG3 and PfHMG4 are underway. The sequences of PfHMG1 and PfHMG3 are quite similar; however, *pfhmg1* transcript is mainly expressed during erythrocytic asexual cycle in contrast to *pfhmg3* preferentially expressed in the gametocytes.

94A

Proteomics of the Microsporidia *Encephalitozoon cuniculi*: from Proteins of unknown Function to New Cell Wall Ones. D. BROSSON, C. TEXIER, G. PRENSIER and C. P. VIVARES, Parasitologie Moléculaire et Cellulaire, Laboratoire de Biologie des Protistes, UMR CNRS 6023, Université Blaise Pascal, 63117 Aubière Cedex, France.

The microsporidia *Encephalitozoon cuniculi*, a human opportunist pathogen, is an obligate intracellular parasite, the nuclear genome of which has been fully sequenced. Our descriptive proteomic analysis of the late developmental stages showed that one-third of the most abundant proteins are of unknown function. The spore, stage of resistance and dissemination, is protected by a wall, which plays the major role for survival in the environment. In order to find new spore wall candidates among the unknown proteins, a bioinformatic screening on the basis of the prediction of secretion, membrane anchoring and eventually of some domain homology was performed. In a first validation test, we used proteomics data to identify the target of a polyclonal antibody directed against the cell wall. The identified protein matched with criteria of selection and presented homologies with the polysaccharide deacetylase family, suggesting its implication in cell wall biogenesis. Its localization and the timing of its expression were studied by TEM immunocytochemistry. Screening was then applied to 44 expressed proteins of unknown function. Four of the nine potential cell wall components were demonstrated as being either endospore or exospore proteins, using polyclonal antibodies directed against GST recombinant proteins. The analysis of these new wall proteins should facilitate further bioinformatic screenings and should help in the better understanding of the structure and function of the microsporidian spore wall.

95A

Plasmodium falciparum Genes Differentially Expressed During Merozoite Morphogenesis. S. CHARNEAU, I. FLORENT and P.

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While the *Plasmodium falciparum* fully annotated genome has now been released, functional analysis of its expected about 5300 genes remains to be carried out to understand their roles during the parasite development, in particular for about 60% of genes devoid of significant similarities in other species. Here, a suppression subtractive hybridisation (SSH) screen was used to identify genes coding for proteins expressed during merozoite morphogenesis. The resulting library yielded clones with cDNA inserts ranging from a few hundreds to about a thousand base pairs. We report here the analysis of 50 independent cDNA clones and of the corresponding 40 *P. falciparum* genes. While 27.5% of them turned out to be previously described genes, all known to code for proteins expressed in schizont/merozoite stages (EBA-175, MSP-1, MSP-3, MSP-6, ABRA/MSP-9, coronin, myosin A and the CLAG/RhopH1 protein), the remaining 72.5% were "putative" and "hypothetical" genes, coding for a variety of proteins among which we found several protease orthologs and orthologs of cytoskeleton-associated and/or actin-interacting proteins such as dynamin and formin-2 as well as other rho-try proteins. In addition to providing experimental evidence for their expression in schizont/merozoite stages, we discuss how these genes represent original candidates to investigate the molecular events required to elaborate infective merozoites.

96A

Effect of the Disruption of RNA Pol II on the Transcription by RNA Pol I by *Trypanosoma brucei*. S. DEVAUX, L. LECOR-DIER, L. VANHAMME and E. PAYS, Laboratory of Molecular Parasitology, Institute of Molecular Biology and Medicine, University of Brussels, Gosselies, Belgium.

Trypanosomes exist in two major distinct forms: the procyclic form in insects in which the surface molecule is procyclin and the bloodstream form in mammals in which the surface molecule is VSG. The promoter of these locus recruits RNA Pol I. Despite the use of this polymerase, the transcripts are spliced and polyadenylated exactly as the mRNAs synthesised by RNA Pol II. The fact that mRNA production from these sites must involve a concerted action between RNA Pol I and a "RNA factory" normally linked to RNA Pol II, led to the proposal that the control of these Pol I transcription units would involve a limiting component bridging RNA Pol I and an elongation/processing complex normally associated with RNA Pol II. To study the effects of the disruption of RNA Pol II on VSG expression level, we cloned the *T. brucei* homologue of the rpb9 subunit of RNA Pol II, which plays a role in RNA elongation in other eukaryotic cells. In PF, disruption of Tbrpb9 by conditional RNAi led to a strong transcriptional stimulation in the beginning of the ESs. This effect was linked to the inhibition of procyclin transcription. Thus, the disruption of RNA Pol II abolished the stage-specific regulation of the transcription units for the two major surface antigens, both of which recruit RNA Pol I.

97A

Revisiting Putative Functional Properties of the *Plasmodium falciparum* Pf155/RESA Protein Using Genetically Engineered Parasites. M. DIEZ*, M. GUILLOTTE*, C. LE SCANF**, H. CONTAMIN**, P. DAVID*, B. COOKE***, O. MERCERAU-PUIJALON* and S. BONNEFOY*, *Unité d'Immunologie Moléculaire des Parasites, Département de Parasitologie, Institut Pasteur, Paris, France, **Laboratoire d'immunologie Parasitaire,

Institut Pasteur de la Guyane, Cayenne, French Guiana, ***Molecular & Cellular Rheology Laboratory, Department of Microbiology, Monash University, Victoria, Australia.

Pf155/RESA (Ring Infected Erythrocyte Surface Antigen), is a *Plasmodium falciparum* protein discharged by the dense granules of merozoites during invasion and exported to the inner face of the erythrocyte membrane, where it interacts with spectrin in the young stages. Chromosome 1 subtelomeric deletion, eliminating amongst others the *resa* gene, may occur during adaptation of parasite isolates to in vitro culture. This is accompanied by erythrocyte membrane modifications of the red blood cell, such as increased adhesion and effect on membrane rigidity. It has been proposed that RESA was likely to contribute to these functional and rheological modifications. To assess this, we have constructed *resa* knock-out parasites in the FUP/CB strain. This results in negating the erythrocyte membrane immunofluorescence of glutaraldehyde-fixed red blood cells (EMIF), indicating that RESA is critical for EMIF reactivity of hyperimmune human sera. Phenotypic and functional analysis of *resa* k.o. parasites indicated that loss of RESA expression neither affects membrane rigidity nor CD36 binding under flow conditions. Furthermore, infection of *Saimiri sciureus* monkeys showed that *resa* gene deletion does not account on its own for the greater adaptation and parasite virulence in this model.

98A

Study of PfMyb1 Transcription Factor Regulation Network during *Plasmodium falciparum* Erythrocytic Cycle. M. GISSOT*, S. BRIQUET*, C. BOSCHET*, P. REFOUR** and C. VAQUERO*, *Inserm U511, CHU Pitie-Salpetriere, 91 boulevard de l'Hopital, 75013 Paris, France, **Department of Immunology and Infectious Diseases, Harvard School of Public Health, 665 Huntington Ave, Boston, MA, USA.

During the complex life cycle of *Plasmodium falciparum*, the regulation of events that occur during the erythrocytic cycle, such as proliferation and differentiation, implies a fine control of transcriptional activities governing the expression profiles of each gene. However, transcriptional regulation and notably its actors, transcription factors and regulation motifs, are poorly described in *Plasmodium*. In order to decipher the mechanisms implicated in transcriptional regulation, we studied a transcription factor belonging to the tryptophan family and showed that the PfMyb1 protein contained in nuclear extracts has a specific DNA binding activity. We took advantage of long *pfmyb1* double stranded RNA (dsRNA) to inactivate the cognate messenger and understand the role of PfMyb1 during the erythrocytic cycle. Culture treated with *pfmyb1* dsRNA exhibited a 40% growth inhibition and mortality during trophozoite to schizont transition when compared to either untreated control or culture treated with unrelated long dsRNA. We have further demonstrated that *pfmyb1* transcript and protein decreased up to 80% in treated trophozoite culture at the time of *pfmyb1* expression peak. Thus, we investigated the effect of this partial loss of transcript and protein using a thematic DNA microarray containing PCR products, representative of *P. falciparum* genes involved in cell cycle and transcriptional regulation. SAM software enabled us to identify several genes over and under-expressed, potentially directly or indirectly regulated by PfMyb1. These alterations of expression were verified by qPCR and Western blotting. We are currently working on the promoters of those genes to decode determinants of gene regulation by Pfmyb1.

99A

A TFIH Homologue Controls the Mutually Exclusive Expression of the Stage-Specific Antigens of *Trypanosoma brucei*. L.

LECORDIER, S. DEVAUX, D. WALGRAFFE, P. POELVOORDE, L. VANHAMME and E. PAYS, Laboratory of Molecular Parasitology, Institute of Molecular Biology and Medicine, University of Brussels, Gosselies, Belgium.

Antigenic variation in *Trypanosoma brucei* relies on a mono-allelic control of the multiple variant surface glycoprotein (VSG) expression sites (ESs). A single ES is active in bloodstream forms (BF) and none are in insect-specific procyclic forms (PF), where procyclin replaces the VSG. In PF, disrupting TbTFIIH by conditional RNAi of any of the Tbp44, TbXPD or TbXPB subunits, or by over-expression of a TbXPD mutant, led to strong transcriptional stimulation in the beginning of the ESs. Both RNA elongation and mRNA production were up-regulated. This effect was linked to inhibition of *procyclin*, and occurred irrespective of the various cellular phenotypes, some of which were lethal due to cell cycle arrest. In BF, stimulation of *procyclin* was linked to inhibition of the active ES. Thus, TbTFIIH is involved in the mutually exclusive expression of the VSG and *procyclin* transcription units, both of which are transcribed by RNA Pol I.

100A

Functional Genomics of *Plasmodium falciparum*: a new approach for identifying therapeutic targets against malaria. K. G. LE ROCH*, Y. ZHOU*, J. R. JOHNSON**, J. YOUNG**, C. KIDGELL** and E. WINZELER*., *Genomics Institute of the Novartis Research Foundation, San Diego, CA 92121, USA, **Department of Cell Biology ICND202, the Scripps Research Institute, 10550 No. Torrey Pines Rd., La Jolla, CA 92037, USA.

In 1996, an international consortium was established to determine the sequence of the 26 Mb *Plasmodium falciparum* genome. Six years later, the genome sequence was published with an estimation of 5,268 genes. However, 60% of these predicted genes encode hypothetical proteins that share little or no sequence similarity with proteins from other organisms. Elucidation of function using traditional forward and reverse genetic techniques with these hypothetical *P. falciparum* genes is proving challenging and time consuming. As a result, high-throughput genomic and proteomic techniques have emerged as important tools to elucidate gene function more rapidly and cost-effectively. Despite the major challenges posed by the wealth of data generated by high-throughput genomics and proteomics, we have developed methods to optimize the analysis of these large data sets. Here, we describe how we interpret the data to generate biologically relevant conclusions, which will improve the general understanding of the biology of the malaria parasite as well as lead to the identification of new promising targets for drug and vaccine developments.

101A

Protective Role of $\gamma \delta$ T Lymphocytes during *Plasmodium falciparum* Infection. S. LOIZON*, H. CONTAMIN*., E. BOURREAU**, J.-C. MICHEL*, O. PUIJALON* and C. BEHR*, *Unite d'Immunologie Moleculaire des Parasites, URA CNRS 2581, Institut Pasteur, Paris, France, **Unite d'Immunologie Parasitaire et Centre de Primatologie, Institut Pasteur de la Guyane, Cayenne, France.

Human T $\gamma \delta$ lymphocytes, bearing V γ 9 V δ 2 chains, constitute a small proportion (1–5%) of the lymphocytes in the blood. These cells recognize nonpeptidic molecules such as phosphoantigens, intermediary metabolites products by microorganisms, including parasites. These antigens are recognized in the absence of molecules CMH. Our previous results have shown that *Plasmodium falciparum* synthesize phosphoantigen, intermediary products of

the Rohmer pathway. These phosphoantigens are able to induce the proliferation of T γ δ lymphocytes, producing pro-inflammatory cytokines and providing a parasitotoxic activity against parasitized red blood cells. Also, during the acute phase of infection, these lymphocytes increase both in percentage and numbers. Today, the biological role of T γ δ lymphocytes remains poorly understood. Indeed, these lymphocytes are found in primates but there is no equivalent in rodents. We decided to study this population in the infection model *Plasmodium falciparum*/spleen intact *Saimiri sciureus*. In this monkey, we found a population of TV γ 9 (representing 0.2–1% of total lymphocytes in the peripheral blood) which react with phosphoantigens as do human TV γ 9 cells. Activation kinetics of these cells in infected monkeys show an early activation occurring on the 6th day of infection, while the CD4 T activation starts at day 10. Also, we observed an important increase of numbers and percentages of these TV γ 9. Their maximal peak coincides with the beginning of parasite clearance. Together, these data associated with parasitotoxic activity in vitro argue for a role of TV γ 9 cells in parasitemia control, particularly during primary infection.

102A

Membrane skeleton in *Paramecium*: a functional analysis of epiplasmins by RNAi. S. POMEL, M. DIOGON, P. BOUCHARD, D. BAYLE, V. RAVET, C. HENOU and B. VIGUES, Laboratoire de Biologie des Protistes, UMR CNRS 6023, Univ. Blaise Pascal, 63177 Aubière, France.

The membrane skeleton, also called epiplasm, is a prominent cortical structure in the ciliate *Paramecium*. In the present study, we have developed RNAi of Ep31 and Ep34, two homologous epiplasmic proteins (epiplasmins) for a functional analysis of epiplasmic membrane skeleton in *P. tetraurelia*. A severe phenotype is evident 20 h after the induction of RNAi. In this phenotype, anlagen of daughter cells are formed but are unable to separate from each other at time of cytokinesis. This abortive division is repeated every 4–5 h leading to “monstrous” organisms that will die 70–75 h after induction. Immunofluorescence microscopy was performed in order to compare organization and dynamics of cytoskeletal systems in wild types and in cells processed for RNAi of epiplasmins. Microtubule-based systems are not affected as judged by examination of rows of basal bodies (bb) and bb-associated fibers. Likewise, new oral apparatuses and pulsatile vacuoles are correctly positioned in anlagen of daughter cells. The main abnormalities concern the epiplasmic membrane skeleton. Immunolabeling of epiplasmins revealed significant disorganization of the epiplasm in regions close to aborted division furrows. Western blot analysis confirmed co-silencing of Ep31 and Ep34 epiplasmins. These results suggest that epiplasm-mediated elongation of the cell body is essential for growth of the division furrow and accomplishment of binary fission in *Paramecium tetraurelia*.

103A

Identification of a Novel Basal Body Protein of *Trypanosoma brucei*: a Role in the Cell Cycle? L. PRADEL, M. BONHIVERS and D. R. ROBINSON, Laboratoire de Génomique fonctionnelle des Trypanosomatides, ATIP. UMR-CNRS 5162, Université Bordeaux 2, Bordeaux, France.

All eukaryote cells have cell cycle stages through which they must pass in order to undergo growth, development and division (e.g. G1, S, G2, M, and C). Transition through these stages is controlled by numerous proteins, including cyclin-dependent kinases. *Trypanosoma brucei* is a parasitic protozoan of global,

medical and agricultural importance that does not have many of the classical cell cycle checkpoints found in other eukaryotes such as cell arrest when mitosis is blocked. However, previous work has identified flagellum basal body separation as a novel cell cycle checkpoint in this organism. Essentially, if basal bodies do not separate cell division is blocked. From in silico database searches we have identified a *T. brucei* protein (we have named TbNEK1), with high homology to NIMA-related kinase family of proteins of *Aspergillus nidulans*. We have cloned, expressed and made antibodies to this protein, which is located to the basal bodies. In vivo over-expression of TbNEK1 in *T. brucei* induces cell cycle modifications resulting in the formation of multiple basal bodies, loss of kinetoplast segregation, multinucleated cells and cell division block. These results suggest the role of TbNEK1 is, in part, to modulate basal body replication or function, which subsequently leads to secondary effects that block progression through the cell cycle.

104A

Trichomonas and GFP, the absolute misfit...? G. BRICHEUX and G. COFFE, Equipe cytosquelette, Biologie des Protistes, UMR 6023, Université Blaise Pascal, 63177 Aubière, France.

Trichomonas vaginalis belongs to a group of parasitic protists with a significant impact on human health, such as *Giardia duodenalis* and *Entamoeba histolytica*. This microorganism is able to assume a dynamical behavior towards its host; in contact with the vaginal epithelium cells, the swimming form of *Trichomonas* modifies its sub-membraneous cytoskeleton, and changes to an amoeboid and adherent form to the host cells. Using this adherence mechanism, *Trichomonas* is able to invade the vaginal mucous membrane while remaining an extracellular parasite. Thus, there is a direct link between cytoskeletal dynamics and pathogenicity. We have characterized some of the cytoskeletal proteins from *Trichomonas*: actin, α -actinin, coronin, and more recently gelsolin. Also, the genome sequencing of this parasite is under way, and will bring numerous data, particularly putative proteins that will need further functional identification. We are presently working to tune up some in vivo analysis methods in order to study these new proteins. The use of the GFP (green fluorescent protein) as a tag to study gene expression and protein localization becomes more and more widely reported. However, this method requires precise minimal conditions in order to satisfy the physico-chemical constraints of the chromophore and in the same time to respect the physiology of the eukaryotic cell.

105A

Multigenic Family Analysis among the *Encephalitozoon* Genus. N. DIA^{*,**}, B. S. TOGUEBAYE^{**}, C. P. VIVARES^{*} and E. CORNILLLOT^{*}, ^{*}Equipe Parasitologie Moléculaire et Cellulaire, UMR 6023 CNRS—Université Blaise Pascal, 63177 Aubière cedex, France, ^{**}Laboratoire de Parasitologie Générale, Département de Biologie Animale, FST, Université Cheikh Anta Diop, Dakar, Senegal.

Encephalitozoon cuniculi is a parasite presenting a wide range of hosts among mammals (including human). Bioinformatic analysis of the genome sequence (11 chromosomes for a haploid genome size of 2.9Mbp) revealed the presence of multigenic families in a subterminal position on the chromosomes (in 5' of the rDNA copies on 22 extremities). One of these families was called InterB and clustered together about 30 genes within 10 homology groups. According to the predictions, the typical InterB protein presents in its N-terminal part a signal peptide, followed by two transmembrane domains. The core of this protein is

characterized by an α hypervariable region rich in glutamine and lysine. The C-terminal region presents several conserved motifs (GGAR, YHL and FQI).

A set of specific and generic oligonucleotides was designed to study the InterB genomic repertoire. The migration profile of the generic primers PCR products presents five to six bands. This profile is well conserved for the *E. cuniculi* isolates as well as for the different *Encephalitozoon* species. Some polymorphism is evident with amplicons of 300 bp in average. Sequence analysis revealed a higher level of polymorphism, especially in the α region. The InterB repertoire of transcripts was studied by reverse transcription. All InterB genes should be transcribed. There is no known homologue for InterB but, considering the position of the genes, we speculate on the possible role of InterB proteins. Similar multigenic family organizations have been described for proteins involved in pathogenicity of parasites such as *Trypanosoma*, *Plasmodium*, *Giardia* or *Pneumocysti*.

106A

Identification of a dipeptidyl peptidase from *Trypanosoma brucei*. D. GIRARD, I. M. D. BASTOS and P. GRELLIER, Laboratoire de Biologie Fonctionnelle des Protozoaires, USM504, Département Régulations, Développement, Diversité Moléculaire, Muséum National d'Histoire Naturelle, 61 rue Buffon, 75231 Paris Cedex 05, France.

Dipeptidyl peptidases (DPP) are ubiquitous enzymes. Several DPPs have been described and the most known among them is dipeptidyl peptidase IV, a type II multifunctional cell surface protein that belong to the prolyl oligopeptidase family of serine proteases. DPP IV specifically cleaves dipeptides from the N-terminus of peptides with a Pro, Hyp or Ala in the penultimate position. DPP IV substrates include certain chemokines, growth factors (e.g. growth hormone, growth hormone-releasing factor, glucagon-like peptides 1 and 2) and neuro- and vasoactive peptides including neuropeptide Y, peptide YY and substance P. In this work, we present the identification of a dipeptidyl peptidase from *Trypanosoma brucei* (DPPTb). Its gene was isolated by PCR using primers designed from *T. brucei* ESTs sequences that matched with human DPP IV. However, the cloned and sequenced insert obtained displays similarity mainly with human DPP8 and DPP9. Active recombinant DPP Tb was expressed in insect cells using the baculovirus system. We observed by immunofluorescence, using anti-DPPTb antibodies, that native DPPTb is localized in vesicles in the cytoplasm from procyclic forms of *T. brucei*.

107A

Observation on a *Enteromyxum leei* (Myxozoa Myxosporaea) Parasitosis on Farming Sea Bream *Sparus aurata*. E. JUBLANC*, N. ELKIRIC*, M. TOUBIANA*, J. SRI WIDADA*, A. LE BRETON**, G. LEFEBVRE***, C. SAUVEGRAIN*** and A. MARQUES*, *EPI, Ecolag UMR 5119, Case 093, University of Montpellier II, F 34095 Montpellier 05, France, **Fish Health Consultant, F 31330 Grenade sur Garonne, France, ***Aquanord Terre des Marins, F 59820 Gravelines, France.

Enteromyxum leei parasitic infestation was first identified in 2001 on the site. Two years of investigations on this disease, when the infestation become detectable by light microscopic observation of the spores, it was observed that rearing parameters have very little influence on the evolution of the pathology, which then leads to heavy losses. This parasitic infestation is characterized by the location of the myxosporidia in the host (the digestive track), the absence of cyst formation and the occurrence of an amplifica-

tion of the parasite development. PCR techniques allow to diagnose the first stages of the infestation on fish juvenile stages but have so far not allowed the detection of a potential intermediate host. The epidemiological observations on site allowed recommending some actions to interfere with the life cycle of the parasite (cleaning of water channels should limit the prevalence of potential intermediate hosts) and adapting health management to reduce the pressure of the pathogen (date of introduction of fingerlings on site, water temperature, oxygen level, flow rates). Two different hypotheses can explain the development of the disease in the present case on this site: (1) either a population of parasites in extension with a sexual reproduction and a complete life cycle on the site (2) or an asexual reproduction, such as a clone developing within the rearing structure and whose expansion may be related to transports of live fish and farming activities. The set up of health management procedures able to efficiently control the disease is strongly related to the understanding of this specific point.

108A

Microsporidia of Fishes: Characterization of a New Species and Development of the Infectious Focus. L. MANSOUR*,**, G. PRENSIER*, O. K. BEN HASSINE**, C. P. VIVARES* and E. CORNILLLOT*, *Equipe Parasitologie Moléculaire et Cellulaire, UMR 6023 CNRS—Université Blaise Pascal, 63177 Aubière cedex, France, **Laboratoire de Biologie, Ecologie et Parasitologie des Organismes Aquatiques, Faculté des Sciences, 1092 Tunis, Tunisia.

Fish parasites microsporidia are distributed among fifteen genera. In most cases, the infectious focus present a cyst-like structure. According to the stage of evolution, the infection focus is either a xenoma or a granuloma. Specific host-parasite interaction is at the origin of a large diversity of foci form and tissue specificity.

In this context, we describe *Microgemma tincae* n. sp. in the peacock wrasse, *Symphodus tinca*, fished near the Tunisian coasts. *M. tincae* is found in the liver of its host. Parasite characterization is based on the ultrastructural description of the life cycle (Transmission Electronic Microscopy) as well as on a part of the rDNA sequence (840 bp of the *ssu* gene). The inflammatory reaction responsible for the transition from xenoma to granuloma is carefully described. Prevalence is high and small fishes present high level of infection (several foci). One of the parameters responsible for the distribution of the number of foci per fish is the ratio between secondary and new infection.

We studied the *Spraguea lophii* infection to address this question at the molecular level in the angler fish, *Lophius* spp. We wanted to know if the reaction set up by the host was sufficient to stop the parasite propagation. The analysis was made in two-dimensional (2D) PFGE at an infection focus scale (1-mm³ samples). We compared 2D genome images from samples isolated from different parts of one or several infection foci of the same or different fish origin. We conclude that the fish immune system cannot avoid secondary infection.

109A

Preservation of *ptp1* and *ptp2* Gene Synteny in Two Microsporidia Parasite of Insects: *Nosema locustae* and *Nosema grylli*. V. POLONAIIS, F. DELBAC and C. P. VIVARES, Parasitologie moléculaire et cellulaire, Laboratoire Biologie des Protistes, UMR CNRS 6023, Université Blaise Pascal, 63177 Aubière cedex, France.

Microsporidia are obligate intracellular parasites infecting a wide range of invertebrate and vertebrate hosts including human. The spore contains a specialized organelle called the polar tube involved in host cell invasion. Three polar tube proteins (PTP1-3) have been identified in *Encephalitozoon cuniculi*. The genes coding for *ptp1* and *ptp2* are localized on the same chromosome and are at a distance of 860 bp. A similar synteny organization is found in two other *Encephalitozoon* species. Comparison of the *E. cuniculi* genome with that of *Nosema locustae*, an microsporidia parasite of grasshoppers, indicates that several genes are positioned in the same order and orientation. This allowed the identification of two ORFs distant of 1117 bp, that despite divergent sequence features, encode proteins having common characteristics with *Encephalitozoon* PTP1 and PTP2 (signal peptide, proline-rich internal repeats for PTP1, lysine-rich protein for PTP2 ...). To test whether they correspond to PTP, polyclonal sera were raised against two recombinant proteins expressed in *E. coli*. In indirect immunofluorescence, the extruded polar tubes of *N. locustae* spores are specifically labelled. NIPTP1 and NIPTP2 have 355 and 287 amino acids, respectively, and are only soluble in the presence of high concentrations of reducing agent.

Aptp1 and *ptp2* containing cluster have been also identified in *Nosema grylli*. PTP1 from *N. locustae* and *N. grylli* present a strong variability in the proline-rich internal repeats, whilst PTP2s are highly conserved. It would be interesting to show whether this gene synteny conservation is related to a specific interaction between PTP1 and PTP2.

Scandinavian Section Society of Protozoologists
25th Annual Meeting
May 6, 2004
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110A

Effects of *Ginkgo biloba* extract and its constituents on chemotaxis and cGMP-dependent protein kinase activity in *Tetrahymena thermophila*. F. CHEN^{*,**} and V. LEICK^{*,**}, ^{*}Food Science and Technology College, Huazhong Agricultural University, Wuhan 430070, Hubei Province, China, ^{**}Institute of Medical Biochemistry and Genetics, Department B, University of Copenhagen, Copenhagen, Denmark, ^{**}The Panum Institute, Blegdamsvej 3C, 2200, Copenhagen N, Denmark.

Ginkgo biloba extract (GBE), which is one of the most sold herbal extracts in the world, is considered as a multifunctional material, which can promote radical scavenging activity and improve brain functioning. Although much research has carried out on its mechanisms against diseases, there is a need for further investigation for understanding molecular mechanisms, potential health benefits, and possible health risks. Recently, we have been developing an alternative screening method for studying phytochemical materials bioactivities with *Tetrahymena thermophila* as experimental organism. In this paper, the effects of GBE and its constituents were systematically investigated on chemoattraction and cGMP-dependent protein kinase (PKG) in *T. thermophila*. GBE and its constituents exerted significant inhibitions of chemoattraction and PKG. The minimal concentrations to completely inhibit chemotaxis of *T. thermophila* were 2 mg/ml, 50, 25, 12, 100, 100, 400, 500, 300 and 400 μ M for GBE, quercetin, kaempferol, isorhamnetin, myricetin, genistein, rutin, quercitrin, isoquercitrin and quercetin-3-O-galactoside, respectively. The IC₅₀ values for PKG were 0.15 mg/ml, 26, 22, 19, 160, 132, 81 and

186 μ M for GBE, myricetin, kaempferol, quercetin, isorhamnetin, genistein, rutin and isoquercitrin, respectively. The results indicate that the inhibitions of GBE and its constituents of chemotaxis of *T. thermophila* and their effects on PKG are rather similar. This suggests that the ciliate of *T. thermophila* may be a potential experimental organism in screening such bioactive phytochemicals.

111A

Paramyxea in Scandinavia. J. I. R. LARSSON^{*} and M. KØIE^{*,**}, ^{*}Department of Cell and Organism Biology, University of Lund, S-22362 Lund, Sweden, ^{**}Marine Biological Laboratory, DK-3000 Elsinore, Denmark.

Paramyxea is a small phylum of marine parasites, using molluscs, crustaceans or polychaetes as hosts. Probably the group is cosmopolitan, but the distribution known so far is restricted to southern Europe, Australia and East Asia. Polychaetes of the genus *Nephtys*, collected from Danish waters, are infested with a paramyxean parasite in the gut epithelium. The parasite develops intracellularly and is in close contact with the cytoplasm of the host cell. When harbouring mature spores gut cells are lysed and spores are released into the digestive tract.

Reproduction is by endogeneous budding, typical for *Paramyxea*, and the only reproduction observed is sporogony. A stem cell (cell 1) yields two or four sporonts (cell 2), each of which produces four tetracellular spores (cells 3–6). The mature spore is fusiform, about 10 microns long, and surrounded by an individual sac filled with fibrous material. The sacs are united to a spherical body with prominent striated terminal projections. Haplosporosomes, of three different types, are present in cells 1, 4 and 5. The species, which is new to science, is the first find of *Paramyxea* from Scandinavia.

112A

Tetrahymena cells and the quantitative two-phase assay: visual assessment of chemokinesis and gravikinesis. J. D. OLESEN, P. HELLUNG-LARSEN and V. LEICK, Copenhagen University, Panum Institute, Institute of Medical Biochemistry and Genetics, Dpt. B, Blegdamsvej 3, Copenhagen, Denmark.

A two-phase assay was used to assess chemokinesis and gravikinesis of *Tetrahymena* cells. METHOD: *Tetrahymena thermophila* cells were cultured without stirring at 36 °C in a medium containing proteose peptone, yeast extract, glucose, and MOPS. Growing cells or cells in the stationary phase (1 million cells/ml) were used. Before assay, the cells were diluted 1+1. A semi-microcuvette contained a 1-ml bottom-phase with NycoprepTM Universal (5% w/v) and MOPS. Carefully, a 1-mL top-phase in the same buffer was layered on top of the bottom-phase. Cells were either in the top- or the bottom-phase. A commercially fed supplement for animals was used as chemoattractant in the phase opposite to the cells (Bioprotein, BP; Norferm, Norway). Changes in cell concentrations were monitored as optical density at 600 nm. RESULTS: (1) cells in top-phase: – BP, 0.0001/30 min; +BP, 0.12/4 min. (2) cells in bottom-phase: top-phase exposed to a stream of nitrogen gas: – BP, 0.07/2 min; +BP 0.05/2 min; top-phase not exposed to a stream of nitrogen gas: – BP, 0.08/2 min; +BP, 0.04/2 min. CONCLUSIONS: The BP proved to be an effective chemoattractant. Gravikinesis was observed with and without the presence of BP. Aerotaxis was not observed. The two-phase assay is a sensitive, reliable, and quick procedure for the assessment of chemokinesis and gravikinesis.

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113A

Body height, body mass index, waist-hip ratio, fluctuating asymmetry and second to fourth digit ratio (2D:4D) in subjects with latent toxoplasmosis. J. FLEGR*, M. HRUSKOVÁ**, Z. HODNY*** and J. HANUSOVÁ*, *Department of Parasitology, Faculty of Science, Charles University, Vinicna 7, 128 43 Prague 2, Czech Republic, **Department of Anthropology and Human Genetics, Faculty of Science, Charles University, Vinicna 7, 128 43 Prague 2, Czech Republic, ***Department of Cellular Ultrastructure and Molecular Biology, Academy Sciences of the Czech Republic.

Between 20% and 60% of the population of most countries are infected with the protozoan *Toxoplasma gondii*. Subjects with clinically asymptomatic life-long latent toxoplasmosis differ from those who are *Toxoplasma* free in several behavioral parameters. Case-control studies cannot decide whether these differences already existed before infection or whether they were induced by the presence of *Toxoplasma* in the brain of infected hosts. Here, we searched for such morphological differences between *Toxoplasma*-infected and *Toxoplasma*-free subjects that could be induced by the parasite (body weight, body height, body mass index, waist-hip ratio), or could rather correlate with their natural resistance to parasitic infection (fluctuating asymmetry, 2D:4D ratio). We found *Toxoplasma*-infected men to be taller and *Toxoplasma*-infected men and women to have lower 2D:4D ratios previously reported to be associated with higher prenatal testosterone levels. The 2D:4D ratio negatively correlated with the level of specific anti-*Toxoplasma* antibodies in *Toxoplasma*-free subjects. These results suggest that some of the observed differences between infected and noninfected subjects may have existed before infection and could be caused by the lower natural resistance to *Toxoplasma* infection in subjects with higher prenatal testosterone levels.

114A

Diagnostics of *Neospora caninum* antibodies of bovine milk by ELISA. L. HURKOVÁ*, D. HARAGÁLOVÁ* and D. MODRY***, *Department of Parasitology, Faculty of Veterinary Medicine, University of Veterinary and Pharmaceutical Sciences Brno, Palackého 1–3, 612 42 Brno, Czech Republic, **Institute of Parasitology, Academy of Sciences of the Czech Republic, Branisovská 31, 370 05 České Budejovice, Czech Republic.

Neospora caninum (Apicomplexa: Sarcocystidae) is a coccidian parasite with a two-host life-cycle. Dogs and coyotes are definitive hosts. A wide spectrum of mammals serves as intermediate hosts (i.a. ruminants, horses, canids). The reproductive failures associated with abortion in cattle have an economic impact. A milk ELISA is a simple serological method for diagnostics of anti-*Neospora caninum* antibodies in dairy herds. We used a commercial *Neospora caninum* iscom ELISA kit (Svanova, Sweden) designed to detect bovine *Neospora*-specific antibodies in blood serum. We examined 332 dairy herds in the Czech Republic. Milk serum was examined at dilution 1:100 according to manufacturer's instructions. The number of milking cows in the examined herds varied, but never exceeded 200 per one bulk sample. Four out of 332 examined herds were significantly positive, the positive bulk milk samples were obtained from herds consisting of 7, 19, 35 and 98 cows. Hitherto, we have studied in

detail the smallest herd. The milk and blood samples or only blood samples (in some animals, e.g. calves) were examined individually. We observed the correlation of positivity between milk and blood. Moreover, the mother-descendant positivity/negativity was found. A dog living in the farm was examined, being coprologically negative but serologically positive (IFAT).

This is the next proof of the occurrence of *N. caninum* in cattle in the Czech Republic. In addition, our results refer to an importance of transplacental transmission in cattle.

115A

The effect of heavy metals on the rumen ciliate *Entodinium caudatum*. K. MIHALIKOVÁ, P. JAVORSKY, Z. VÁRADYOVÁ and S. KISIDAYOVÁ, The Institute of Animal Physiology, Slovak Academy of Sciences, Soltésvej 4–6, 04001 Kosice, Slovak Republic.

The effect of three heavy metals on the growth of the rumen ciliate *Entodinium caudatum* was studied in vitro. The *Entodinium caudatum* (*E.c.*) cells were treated with mercury (HgCl_2), copper (CuCl_2) and chromium ($\text{K}_2\text{Cr}_2\text{O}_7$) for a period of 4–5 days. The tested concentrations of mercury and copper were 1, 5, 10, 20, 50 mg/l¹ and concentrations of chromium were 2, 10, 20, 40, 100 mg/l. All experiments were performed under two different conditions: either in presence of intact bacterial population in culture or reduced bacterial population, when culture was treated by antibiotics. Concentration of Hg of 5–20 mg/l¹ significantly decreased the concentration of *E.c.* cells at the end of experiment. The dose of 50 mg/l was lethal. Copper caused significant decrease in the *E.c.* cell concentration after exposure to 10–50 mg/l. No tested concentrations of copper resulted in culture death. Chromium was the most toxic metal. Concentration of 2–20 mg/l caused significant decrease in cell growth and concentrations of 40 and 100 mg/l were lethal. Cultures treated by antibiotics were more sensitive to tested heavy metals. It could be concluded that all heavy metals decreased the cell growth of *E.c.* at concentration from 5 mg/l.

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Protozoan population and fermentation parameters of rumen contents of sheep from heavy metal contaminated area. Z. VÁRADYOVÁ, K. MIHALIKOVÁ, P. JAVORSKY and S. KISIDAYOVÁ, Institute of Animal Physiology, Slovak Academy of Sciences, Soltésvej 4–6, 040 01 Kosice, Slovak Republic.

The rumen contents from Slovak Merino sheep (12 months of age) were used for in vitro study on the influence of area contaminated by heavy metals on rumen fermentation and protozoan activity. Sheep were browsing in the contaminated area of Kal'ava village (Slovakia) and were exposed to 1-year intake of heavy metals. The area of Kal'ava is contaminated by atmospheric pollution from the non-ferrous metal works at Krompachy (Slovakia). Based on the levels of mercury (4.8 mg/kg), copper (232.9 mg/kg), cadmium (1.2 mg/kg), lead (92.5 mg/kg) and arsenic (74.6 mg/l¹) the soil was categorized as profusely contaminated. Grass contamination was below the toxic limits. In the tested materials, copper was present at the highest levels followed by lead and arsenic. Meadow hay was used as a tested substrate of fermentation activity; it was incubated with buffered rumen fluid for 24 h. The significantly decreased values of fermentation parameters (total gas, methane, total VFA, acetate) were associated with a reduced total concentration of protozoans. Significant

decrease was detected in the total rumen ciliate population and population of *c*. No significant differences were observed in the concentration of *Dasytricha ruminantium* and *Ophryoscolex c. tricornatus*.

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Functional frataxin homologue in hydrogenosomes of *Trichomonas vaginalis*. P. DOLEZAL*, A. DANCIS**, M. EMBLEY*** and J. TACHEZY*, *Department of Parasitology, Faculty of Science, Charles University, Vinicna 7, 128 43 Prague 2, Czech Republic, **School of Medicine, University of Pennsylvania, Philadelphia, PA, USA, ***Department of Zoology, Natural History Museum, London, UK.

The cellular structure of *Trichomonas vaginalis* substantially differs from other eukaryotes. Instead of mitochondria it possesses hydrogenosomes where key metabolic reactions are mediated by FeS proteins. The gene expression and the activity of FeS proteins is fully dependent on the iron availability. Although hydrogenosomes contain a high amount of iron, there is no information about hydrogenosomal iron homeostasis and FeS cluster formation. In mitochondria, frataxin plays an unclear role in iron metabolism. The deficiency of frataxin in human causes Friedreich's ataxia, which is associated with an increased level of mitochondrial iron. It has been suggested that frataxin is involved in various functions such as FeS cluster formation or iron storage. We cloned *T. vaginalis* frataxin homologue (TvFTX) coding for a protein of 121 amino acids. Although lacking transcription initiator element in 5'UTR region, the expression of tvftx was verified by a nuclear run-on assay. Moreover, the gene upregulation in iron deficiency was shown. The N-terminal part of TvFTX contains an extension similar to presequences targeting proteins to hydrogenosomes and mitochondria. To identify its cellular localization we transfected *T. vaginalis* with TvFTX fused to HA tag. The immunodetection of HA tag specifically labeled hydrogenosomes and the corresponding cellular fraction. The hydrogenosomal targeting sequence however does not support the translocation into yeast mitochondria. Thus for the complementation of yeast frataxin homologue (YFH1) a mitochondrial targeting sequence is required. When expressed in mitochondria of δ YFH1 cells TvFTX complements the mutant growth defect and restores the activity of aconitase (iron-sulfur protein). Together with recent characterization of *T. vaginalis* cystein desulfurase these data indicate the conserved role of hydrogenosomes and mitochondria in FeS cluster assembly.

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RNA binding proteins MRP1 (gBP21) and MRP2 (gBP25) are essential for editing and stability of a subset of mitochondrial mRNAs in procyclic *Trypanosoma brucei*. E. VONDRUSKOVÁ*, J. BURG**, A. ZÍKOVÁ*, N.L. ERNST***, K. STUART***, R. BENNE** and J. LUKES*, *Institute of Parasitology, Czech Academy of Sciences, and Faculty of Biology, University of South Bohemia, České Budejovice, Czech Republic, **Department of Biochemistry, Academic Medical Center, University of Amsterdam, Amsterdam, The Netherlands, ***Seattle Biomedical Research Institute, and Department of Pathobiology, University of Washington, Seattle, WA, USA.

In trypanosomes, MRP1 and MRP2 (previously called gBPx and gBP_y, in which x and y indicate the MW of these proteins in a

particular species) are guide (g)RNA binding proteins that are part of a large heteromeric complex that may play a role in U-insertion/deletion editing of mitochondrial mRNAs. In order to shed more light on the function of these proteins, we generated procyclic *Trypanosoma brucei* cell lines in which the levels of MRP 1 and/or MRP2 mRNA were downregulated by RNA interference (RNAi). Here we report that the RNAi-mediated knockdown of MRP1 and/or MRP2 resulted in severe growth inhibition and loss of both proteins. This loss occurred even in cells in which only one of the MRPs was targeted by RNAi, indicating a mutual dependence for stability of these proteins. The elimination of the MRPs substantially reduced the levels of edited cytB and RPS12 mRNAs, but resulted in little or no reduction of edited cox2, cox3 and A6 mRNAs, as measured in poisoned primer extension analyses. Surprisingly, we found a five-fold increase in ND7 mRNA editing in MRP1+2 double knockdown cells. In addition, the knockdowns also resulted in reduction in the amounts of mRNAs that do not undergo RNA editing (cox1, ND4 and ND5 mRNAs), but little change was observed for mitochondrial 12S rRNA. Together, the results indicate that in procyclic *T. brucei*, MRP1 and MRP2 play a role in transcript-specific editing and other RNA processing activities.

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Role of IscS in Fe-S cluster assembly in *Trypanosoma brucei*. O. SMÍD*, E. VONDRUSKOVÁ**, V. VILÍMOVÁ*, R. SUT'ÁK*, J. LUKE** and J. TACHEZY*, *Department of Parasitology, Faculty of Science, Charles University, Vinicna 7, 128 44 Prague 2, Czech Republic, **Institute of Parasitology, Czech Academy of Sciences, and Faculty of Biology, University of South Bohemia, 37005 České Budejovice, Czech Republic.

Despite the significance of proteins containing iron-sulfur cluster (Fe-S proteins), the processes of Fe-S cluster assembly and maturation of Fe-S proteins are poorly understood. However, several key proteins involved in the assembly have been identified, notably IscS, a cystein desulfurase, which provides sulfur for Fe-S cluster and IscU, a metallochaperone acting as a scaffold for cluster assembly. In this work, we studied the process of Fe-S cluster biosynthesis in *Trypanosoma brucei* by identifying the homologue of IscS in the *T. brucei* (TbIscS). To address the function of TbIscS, we inhibited its expression by means of RNA interference (RNAi). After RNAi induction, generation time of the TbIscS knock-down cell line was significantly prolonged. All types of mitochondrial ATP production in the cells were severely affected. Analysis of glucose metabolism end products determined pyruvate as major excreted metabolite of the induced cells, while the uninduced cells produced only small amount of this glycolytic end product. These data demonstrate that mitochondrial metabolism is impaired in cells with TbIscS knocked down. To test whether the observed phenomena were results of Fe-S cluster assembly disruption, we examined the Fe-S cluster-dependent activity of aconitase. This enzyme is localized in its active form in mitochondrion as well as in cytosol of *T. brucei*. After RNAi induction we observed the reduction of aconitase activity in both compartments (approx. 70% reduction in cytosol, approx. 30% in mitochondria). Western blots together with the EPR analysis showed that the reduction in cytosolic activity was due to impaired Fe-S cluster formation, while decrease in aconitase activity in mitochondria corresponded to the reduced level of the protein.

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Parasitic Stramenopila. M. KOSTKA, V. HAMPL, I. CEPICKA and J. FLEGR, Department of Parasitology, Faculty of

Science, Charles University, Vinicna 7, 128 43 Prague 2, Czech Republic.

In the vast and diversified eukaryotic group Stramenopila we can find many different life strategies, one of them being parasitism. There are two groups of stramenopiles containing important parasites of vertebrates: the oomycetes (e.g. *Saprolegnia*) and the wholly parasitic group Slopalinida+its sister taxon—*Blastocystis*. The order Slopalinida comprises families Opalinidae and Proteromonadidae. The two families are considered related based on the structure of basal bodies and their appendages and the presence of subpellicular microtubules. However, a robust recognition of phylogenetic affinities of Opalinidae—the peculiar multinucleated intestine commensals of frogs—has been hindered by the absence of reliable molecular data. Up to now, all attempts to sequence opalinid genes failed, as the obtained sequences labeled as *Protoopalina intestinalis*, *Cepedea virguloidea* and *Opalina ranarum* in GenBank apparently originate from a zygomycete contamination. We present the first molecular data for the family Opalinidae—SSU rRNA gene of *Protoopalina intestinalis*. Our phylogenetic analyses undoubtedly show opalinids as a sister group to *Proteromonas* within the Stramenopila clade, confirming the monophyly of Patterson's order Slopalinida. The enigmatic genus *Blastocystis* is resolved with high statistical support as a sister group to Slopalinida. Our analyses clearly demonstrate that Cavalier-Smith's phylum Bigyra, which comprises Oomycetes and their relatives together with Slopalinida and *Blastocystis*, is not monophyletic. We also show, that *Blastocystis* sp. isolate obtained from the red-footed tortoise (*Geochelone carbonaria*), forms a sister group to all *Blastocystis* isolates from birds and mammals sequenced so far.

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Critical analysis of the topology and rooting of the parabasalian 16S rRNA tree. V. HAMPL, I. CEPICKA, J. FLEGR, J. TACHEZY and J. KULDA, Department of Parasitology, Faculty of Science, Charles University, Vinicna 7, 128 43 Prague 2, Czech Republic.

The morphological classification of the protozoan phylum Parabasala is not in absolute agreement with the 16S rRNA phylogeny. However, there are strong indications that tree-construction artifacts play a considerable role in the shaping of the 16S rRNA tree. We have performed rigorous analyses designed to minimize such artifacts using the slow-fast and taxa-exclusion methods. The analyses, which included new sequences from the genera *Monocercomonas* and *Hexamastix*, in most respects confirmed the previously suggested tree topology and polyphyly of order Hypermastigida and family Monocercomonadidae but detected one artificial cluster of long branches (Trichonymphidae, Pseudotrichonymphidae, *Hexamastix* and *Tricercomitus*). The analyses also indicated that the rooting of the phylum Parabasala on the trichonymphid branch, as widely accepted, is doubtful and that reliable rooting on the basis of current data is likely impossible.

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Secondary endosymbioses and evolution of unicellular eukaryotes. M. OBORNÍK* and B. R. GREEN**, *Institute of Parasitology, Academy of Sciences of the Czech Republic, Branisovská 31, 37005 České Budejovice, Czech Republic, **University of British Columbia, Department of Botany, Vancouver BC, Canada V6T1Z4.

Many lines of evidence support the idea that the first chloroplast was the result of an endosymbiotic relationship between a cyano-

bacterium and a non-photosynthetic eukaryote. Most of the cyanobacterial genes were lost, but a few remained in the chloroplast genome and as many as a thousand were transferred to the host nucleus. Genes encoding functions required by the chloroplast had to acquire presequences to target the products to the chloroplast. The situation gets more complicated when we consider the algae with chlorophyll c. They are the product of secondary endosymbiosis, where a putative red algal ancestor was engulfed by another non-photosynthetic eukaryote, which retained the red algal chloroplast but eventually got rid of the rest of the cell. This left the chloroplast surrounded by two additional membranes: one derived from the red algal plasma membrane and the other from the host's phagocytic vacuole. In order for the endosymbiotic relationship to work, there must have been a substantial amount of gene transfer from the red algal nucleus to the host nucleus to support chloroplast functions. In the cryptophytes we even see an intermediate stage in this process, a relict nucleus (nucleomorph) in the periplastidal space between the outer two membranes and the original chloroplast envelope. Now that the draft genome sequence of the diatom *Thalassiosira pseudonana* (Diatom Genome Consortium) as well as genomes of rhodophyte *Cyanidioschyzon merolae* and green plants are available, it is possible to investigate the evolutionary history of plastid localized metabolic pathways. Phylogenetic analyses of nuclear-encoded putatively plastid-targeted enzymes showed that plastids obviously utilize enzymes not only of expected plastid (cyanobacterial) origin. Within the diatom, apicomplexan, plant and rhodophyte genomes, we have identified several enzymes that originate in α -proteobacteria (mitochondria) or even in eukaryotic nucleus, but possess N-terminal plastid-targeting presequences. Although diatoms are, according to multiprotein phylogeny, related to Alveolates, some plastid-related metabolic pathways show substantially different evolutionary pattern as well as, in silico, predicted localizations of involved enzymes.

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Modulation of murine cellular immune responses and cytokine production by sand fly saliva. I. ROHOUSOVA*, **, M. LIPOLDOVA** and P. VOLF, *Department of Parasitology, Charles university, Vinicna 7, 128 43 Prague 2, Czech Republic, **Institute of Molecular Genetics, Academy of Sciences of the Czech Republic, Flemingovo nam. 2, 166 37 Prague 6, Czech Republic.

Sand flies (*Diptera: Phlebotominae*) are hematophagous arthropods and important vectors of *Leishmania* parasites. During blood feeding, both vector saliva and parasites are injected by an infected sand fly. The aim of this study was to characterize how sand fly saliva interferes with host immunity in order to clarify the mechanism underlying enhancing effect of sand fly saliva on *Leishmania* infection. Spleen cells from naive BALB/c mice were incubated in vitro with *Phlebotomus papatasi*, *P. sergenti* or *Lutzomyia longipalpis* salivary gland lysate (SGL). We studied the effect of saliva on cell proliferation and cytokine production in non-stimulated and concanavalin A-stimulated splenocytes. Both spontaneous and mitogen-stimulated lymphocyte proliferation were significantly suppressed with SGLs of all three sand fly species and all SGLs doses tested (equivalent to 1/16, 1/4 and 1 gland per well). The proliferation index (a lysate-treated/lysate-untreated cells ratio) was significantly lower when cells were incubated with SGL and was ranging between 0.52 and 0.30 for non-stimulated splenocytes and 0.69–0.32 for mitogen-stimulated lymphocytes. This result indicates that saliva from different sand fly species is able to suppress host proliferative response even to the potent *Leishmania*-unrelated mitogen. In parallel experiments,

we analysed the effect of SGL (1 gland per well) on cytokine production. The levels of Th1 (IL-2, IFN- γ) and Th2 (IL-4) cytokines were determined in cell culture supernatants by capture ELISA. Exposure to SGL alone did not affect levels of IL-2 or IL-4. The production of IFN- γ was inhibited by *P. papatasi* SGL, whereas the SGLs of other two species were ineffective. In mitogen-stimulated cells both Th1 cytokines were modulated; the level of IFN- γ production was significantly suppressed and IL-2 enhanced by SGL of all three sand fly species. The production of Th2 cytokine IL-4 was inhibited only by *L. longipalpis* SGL; no changes in IL-4 levels were noted after incubation with *Phlebotomus saliva*. This species-specific immunomodulation is in agreement with our previous finding that the composition of sand fly saliva varies among species. Taken together, sand fly saliva modulate cell proliferation as well as cytokine production. These findings further increase our understanding of sand fly–host immune relationship.

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The hyrax (*Procavia capensis*) is reservoir host of *Leishmania tropica* in northern Israel. M. SVOBODOVA*, J. VOTYPKA*, V. KRAVCHENKO**, R. W. ASHFORD*** and A. WARBURG**, *Department of Parasitology, Charles University, Vinicna 7, 128 43 Prague 2, Czech Republic, **Department of Parasitology, Hebrew University, Jerusalem, Israel, ***Liverpool School of Tropical Medicine, Liverpool, UK.

In order to identify a reservoir host of *Leishmania tropica*, we have trapped mammals in a cutaneous leishmaniasis focus in the Galilee region of Northern Israel. During May 2003, racoon traps (in total 275 in 11 nights) and Sherman traps (in total 170 in 8 nights) were put in boulders surrounding four endemic villages. In total, 34 hyraxes (*Procavia capensis*), 32 spiny mice (*Acomys cahirinus*) and one house mouse (*Mus domesticus*) were caught. Biopsies were taken from anaesthetized animals, which were released after recovery. Parasites were isolated from the nose of one hyrax. The strain was typed as *L. tropica* and was identical with previous isolates from patients and sand flies originating from the focus. Biopsies from other hyraxes and mice (ear biopsy) were culture and PCR negative, while PCR from the infected hyrax was positive. Previously, *L. tropica* DNA was detected in two hyraxes trapped in the focus but parasites were not isolated. Thus, we have confirmed the potential of hyraxes to serve as reservoir hosts. After inoculation of the hyrax strain into the ear of spiny mice, 2/4 animals were culture-positive, and 5/5 were PCR positive. Moreover, parasite DNA was detected not only in the inoculation site, but also in the nose and hind foot of one *Acomys*. *L. tropica* is thus able to disseminate in its host. Differences in biology between the two most abundant mammals in the focus are probably the reason why the hyrax, and not the spiny mouse, is a suitable reservoir of *L. tropica*.

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Effects of Time-Variant Extremely-Low-Frequency (ELF) Electromagnetic Fields (EMF) on Cholinesterase Activity in *Dictyostelium discoideum*. ANDREA AMAROLI, FRANCESCA TRIELLI, BRUNO BIANCO*, STEFANO GIORDANO*, ELSA

MOGGIA* and MARIA U. DELMONTE CORRADO, Dipartimento per lo Studio del Territorio e delle sue Risorse, Genova, Italy, *Dipartimento di Ingegneria Biofisica ed Elettronica, Università di Genova, I-16100 Genova, Italy.

Recently, we have shown the presence of molecules belonging to the cholinergic signalling system in the ciliate *Paramecium primaurelia* and in the sarcodine *Dictyostelium discoideum*. Propionylcholinesterase (PrChE) activity has been detected in single-cell amoebae of *D. discoideum*, using cytochemical, electrophoretic, and spectrophotometric methods (Falugi et al., *Chemosphere*, 48:407–414, 2002; Amaroli et al., *Europ. J. Protistol.*, 39:213–222, 2003). It has been suggested that this enzyme activity is involved in cell–cell and cell–environment interactions, as its inhibition by xenobiotic compounds affects cell migration and aggregation. In this work, we have spectrophotometrically evaluated the effect of an ELF-EMF of 300 T, 50 Hz, on PrChE activity after exposure of single-cell amoebae from 1 h up to 48 h at 22 °C. The enzyme activity was inhibited significantly by 1-h- and 3-h exposures, whereas it was similar to the control value after a 4-h exposure. A significant increase in PrChE activity was found after 5- and 24-h exposures, while a decrease in PrChE activity appeared in 48-h-exposed samples. The increased PrChE activity detected in 24-h-exposed cells returned to the control value 24 h after transferring the amoebae to standard conditions. A delay in both migration and aggregation processes was observed in 3-h-exposed cells, corresponding to a decreased PrChE activity. After a 24-h exposure, a decrease in the fission rate and an increase in the cell size were found.

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An alternative approach for the molecular characterization of uncultivable protists retaining morphological information. ILARIA ANDREOLI, FILIPPO FERRANTINI, LARA MANGINI, GIOVANNI SANTANGELO, GIULIO PETRONI and FRANCO VERNI, Dipartimento di Etologia, Ecologia, Evoluzione, Università di Pisa, I-56126 Pisa, Italy.

Recent culture-independent studies based on small-subunit ribosomal RNA (SSU rRNA) gene analysis revealed the existence of completely new clades of protists. The main problems with this approach are to correlate sequences from environmental rRNA genes with the organisms they belong to and then to detect the ecological role of these organisms in the environment. In order to overcome such problems we chose an alternative approach allowing us both a molecular characterization of uncultivable organisms with a low relative abundance in environmental samples, and a morphological analysis, even if restricted. The experimental protocol consists of two steps: an initial observation and photo-taking of the single cell under the DIC (Differential Interferential Contrast) microscope and then PCR amplification and direct sequencing of the 18S rRNA gene of the same cell. The advantages of this method are the possibility to: (1) establish a precise link between morphology and gene sequence; (2) detect the possible occurrence of highly similar species within the studied population; (3) avoid the insertion of *Taq*-polymerase errors in the gene sequence; and (4) detect possible polymorphisms in the gene under examination. Such an approach was used to sequence the 18S rRNA gene of organisms belonging to the class Karyorelictea that comprises several uncultivable ciliates with limited distinctive features. Gene sequences analysis revealed an unexpected genetic variability in trachelocercids and, in particular, the existence of polymorphisms in the SSU rRNA gene of a group of them. Such specimens show a similar morphology and, as a result of phylogenetic analysis, they form a constant clade.

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Primary cultures of rat retina as a cell model for studying *Toxoplasma gondii* interconversion. MARIA C. ANGELICI, SERENA CELANI, FIORELLA MALCHIODI-ALBEDI* and ANDREA MATTEUCCI*, Dipartimento di Malattie Infettive, Parassitarie ed Immunomediate, Roma, Italy, *Dipartimento di Biologia Cellulare e Neuroscienze, Istituto Superiore di Sanità, I-00161 Roma, Italy.

Toxoplasma gondii infection is an important cause of chorioretinitis, that may lead to blindness and glaucoma. The particular tropism of the parasite for retinal tissue suggests that retinal cell cultures, reproducing retinal microenvironment, may represent a suitable cell model for obtaining intracellular replication of the parasite and its stage conversion from tachyzoites to bradyzoites. To verify this hypothesis, we set up primary cultures of embryonic retina, obtained from Wistar rats at the 18th gestational day. Eyes were removed under sterile conditions and the retina was isolated under a stereomicroscope. After dissociation in trypsin, cells were seeded at high density ($1-2 \times 10^6$ cells) onto poly-L-lysine-coated 24-well cell culture plates and grown in MEM with 10% foetal calf serum. Cultures were maintained in a humidified, 37 °C incubator with 5% CO₂. This culturing procedure yields a mixed cell population, where all the principal cell types of the retina are present, including neuronal cells (amacrine cells, photoreceptors, ganglionar cells), macroglial cells (Muller cells, astrocytes) and microglia. One week after *T. gondii* ME49 strain inoculation, cyst-like structures were observed, firstly, enclosed in grape-like clusters of round cells, then free in the medium. In these cysts, bradyzoites were still infective up to 2 months, and they developed to the tachyzoite stage when re-inoculated in a fibroblast monolayer. These results suggest that the microenvironment of retinal cell cultures may favour parasite interconversion from tachyzoite to bradyzoite. Studies are in course to monitor cell-signallings (INF- γ , TNF- α and nitric oxide) that accompanies this phenomenon. This in vitro model may be a valuable tool to study the role of inflammatory process in the pathogenesis of ocular toxoplasma infection and to test new drugs to prevent ocular toxoplasmosis.

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Tetrahymena thermophila as whole cell biosensor for environmental biomonitoring. SABRINA BARCHETTA, ANTONIETTA LA TERZA, FEDERICO BUONANNO, PATRIZIA BALLARINI and CRISTINA MICELI, Dipartimento di Biologia Molecolare Cellulare Animale, Università degli Studi di Camerino, I-62032, Camerino, Italy.

Whole-cell-based biosensors represent an innovative and efficient class of tools useful for environmental monitoring. The majority of these biosensors that have been developed up to now are mainly based on bacteria. However, bacteria sense toxicity in a different way from eukaryotes, thus limiting the extrapolation of results to higher eukaryotes and humans. So far, the most used eukaryotic sensing element is represented by yeast, which is generally viewed as the benchmark eukaryotic microorganism. We propose, as a valid alternative to the yeast-based sensors, the use of genetically engineered cell lines of the ubiquitous freshwater ciliated protozoa *Tetrahymena thermophila*. Ciliates, and in particular *Tetrahymena* species, offer a number of suitable characteristics to be used as environmental sensors: (a) they occupy the first trophic levels and consequently are early warning indicators of cellular suffering; (b) they are available for most of the newly developed molecular genetic techniques; (c) they can be easily cultured and maintained in small volumes; and (d) their cultures can be frozen and maintained in liquid nitrogen.

Moreover, the analysis of the recently sequenced macronuclear genome of *Tetrahymena thermophila* has revealed that this ciliate shares a degree of sequence conservation with human genes higher than that showed by other single-celled eukaryotic model organisms, yeast included. These considerations make *Tetrahymena* an appealing bio-system for toxicity assessment since it can provide information of direct relevance to human health. In this study, *Tetrahymena thermophila* cells were transfected with the circular vector pD5H8 containing the coding sequence of the reporter gene "green fluorescent protein" (GFP) under the control of the homologous hsp70 stress promoter, to generate a fluorescent-inducible bioreporter strain able to detect general toxicity. Several classes of chemicals (inorganic and organic), physical stressors and also real environmental samples from contaminated sites were assayed using the bioreporter strain. The sensitivity of this assay was significantly higher than that measured by lethality tests using the same organism.

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Molecular phylogeny of free-living and symbiotic *Verrucomicrobiales* by using protein-coding genes. ANDREAS BAUER, GIOVANNA ROSATI*, WOLFGANG LUDWIG, KARL-HEINZ SCHLEIFER and GIULIO PETRONI*, Department for Microbiology, Technical University Munich, D-85380 Freising, Germany, *Department of Ethology, Ecology, Evolution; University of Pisa, I-56126 Pisa, Italy.

Verrucomicrobiales are a newly proposed, deep branched division of bacteria from which only a handful of cultured isolates are available. Culture-independent analysis of environmental samples shows that members of *Verrucomicrobiales* are widespread and can be divided into several subdivisions. Symbiotic species of *Euplotidium* (Ciliophora) and Nematodes, epixenosomes and *Xiphinmobacter* spp. respectively, are according to the 16S sequences also members of this group. Up to now, only minimal sequence data mainly focused on 16S rRNA gene sequences are available for this group of organisms. The phylogenetic relatives of *Verrucomicrobiales* are still unclear with *Chlamydiales* and *Planctomycetes* being the most probable candidates. The present study describes preliminary results on phylogenetic relationships of *Verrucomicrobiales* by using protein-coding genes like Hsp70 and RpoC/B. The species which are used for this study are *Prostheobacter vanneervenii*, *P. debontii*, *Opiritutus* sp. VeGlc2, *O. terrae* and epixenosomes. The genes were amplified with designed consensus-degenerated primers and results are available on the Hsp70. The sequences show several differences within the group. *Verrucomicrobium* and *Prostheobacter* sequences cluster together, but in *P. debontii*, two sequences were found, and the second one appears quite divergent. Also the two *Opiritutus* genes are quite atypical and cluster together in a deep position. Preliminary attempt was performed on *Euplotidium*-epixenosome complex. However, amplified and cloned sequences revealed to be closely related to ciliate sequences. Further attempt will be performed applying microtiter-plate-subtractive-hybridization to characterize epixenosome genes.

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Effects of climacostol on normal and tumoral mammalian cell lines. FEDERICO BUONANNO, MASSIMO BRAMUCCI, HIDEO IIO* and CLAUDIO ORTENZI**, Department of Molecular Cellular and Animal Biology, University of Camerino, I-62032 Camerino, Italy, *Department of Material Science, Osaka City University, Osaka, Japan, **Department of Educational Science, University of Macerata, I-62100 Macerata, Italy.

Climacostol, 1,3-dihydroxy-5-[(Z)-2'-nonenyl]benzene, is a natural toxin contained in the extrusomal cortical granules of the heterotrich ciliate *Climacostomum virens*. It is used for chemical defence against predators such as the raptorial ciliate *Dileptus margaritifer* and its cytotoxic activity has been assessed on several species of ciliates such as *Didinium nasutum*, *Paramecium caudatum*, and *Blepharisma japonicum* (Miyake et al. 2003, *Europ. J. Protistol.*, 39:25–36). On the basis of its chemical structure, climacostol may be classified into the large group of natural compounds known as resorcinolic lipids, that show anti-microbial, antiparasitic, and antitumoral activities (Kozubek et al. 2003, *Cell Moll. Biol. Lett.*, 6:351–355). To explore the possibility to use climacostol in medical applications, we examined the effects of chemically synthesized climacostol (Masaki et al. 2004, *Tetrahedron*, 60:7041–7048) on the growth and proliferation of tumoral and normal mammalian cell lines: (1) human promyelocytic leukaemia cells, HL60; (2) human squamous carcinoma cells, A431; and (3) non-tumoral cells derived from mice Leydig cells, TM3. It was observed that (1) a concentration of 10 µg/ml of climacostol exerts a strong cytotoxic activity on all cell lines used; (2) at lower concentrations of 10 ng/ml and 1 ng/ml, the effect of climacostol is limited to the inhibition of the cell growth; and (3) the normal TM3 cells are more resistant to climacostol than the two tumoral HL60 and A431 cell lines. The dose-dependent cytotoxic effects of climacostol encourage further investigation on the potential use of this ciliate toxin as an anti-cancer chemical.

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Euplotes daidaleos and its endocytobionts. SERGEI I. FOKIN, Dipartimento di Etologia, Ecologia ed Evoluzione, Università di Pisa, I-56126 Pisa, Italy.

Among almost 200 endocytobiotic associations between freshwater ciliates and *Chlorella*-like algae (A) one *Euplotes* species—*E. daidaleos* Diller, Kounaris, 1966 is existing. It was not so far under particular investigation. Comparative study of cell morphology and behavior of “green” and derived from it “white” stocks of the ciliate (C), collected in Italy and Russia were carried out, using light and electron microscopy. The green C usually maintain 30–80 units of A and some bacteria (B), belonging to the different types—*Polynucleobacter*-like bacterium, β-subgroup of Proteobacteria (size about $2 - 10 \times 0.4 - 0.5 \mu\text{m}$) and another eubacterium (size $1 - 2 \times 0.6 - 0.8 \mu\text{m}$). The number of the first B was much higher in the white C, but could vary between different host cells. According to data obtained with fluorescent microscopy it looks like these B can produce long chains, which consisted of quite short individuals with only one nucleoid. Association between *Euplotes* and *Chlorella* within the system is rather a close one: loss of the A after long time cultivation of C in darkness usually did not happen. The majority of the ciliates (96–100%) kept A, but this number often dropped down, apparently as a result of digestion by C of some of the A. Positive phototaxis is almost absent in green *E. daidaleos* in comparison with that of *Paramecium bursaria*–*Chlorella* association. The rate of division was not significantly deviated according to A presence. Moreover, cells of the green *Euplotes* did not like high level of illumination and in any cases needed some additional food. This situation is also quite opposite to the *P. bursaria*–*Chlorella* system. In between 2 and 3 months of cultivation, the main part of the white stocks have lost its viability partly because of disturbance in the cirral pattern. In about 90% of cells some frontal, ventral, transversal and caudal cirri disappeared in different combinations. It is the first indication on some connection between A-symbionts and host morphogenesis. Large food va-

cuoles almost all time presented in such cells show some kind of problems with digestion as well. *E. daidaleos* could be considered as three-lateral symbiotic system, promising for further investigations.

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The application of molecular techniques to the surveillance and control of leishmaniasis in Italy: the facts and doubts. MARINA GRAMICCIA, TRENTINA DI MUCCIO, ALDO SCALONE, ERIKA ROSSI, GIOIA BONGIORNO, MICHELE MAROLI, GIOVANNI POGLAYEN* and LUIGI GRADONI, Reparto di Malattie Trasmesse da Vettori e Sanità Internazionale, Dipartimento MIPI, Istituto Superiore di Sanità, I-00161 Roma, Roma, Italy, *Dipartimento di Sanità Pubblica Veterinaria, Università di Messina, I-98123 Messina; Italy.

Leishmaniasis are parasitic diseases caused by various species of the protozoan *Leishmania*. In Italy human visceral and cutaneous leishmaniasis are caused by *Leishmania infantum*, which has the dog as domestic reservoir, and different Phlebotominae species as the vector. Several molecular techniques are currently used with success for a rapid, very sensitive and non-invasive detection of *Leishmania* organisms in specimens of infected organs. The high sensitivity, however, may raise doubts on the significance of target DNA detection, which may represent either an established infection—with or without epidemiological importance—or just a trace of parasite–host contacts. Three examples of the application of nested PCR technique using rDNA sequence primers on different biological samples, are reported here: (a) five different bioptic samples were examined from 153 dogs of an endemic focus of canine leishmaniasis (CanL) (Salina Island, Sicily), giving a *Leishmania* prevalence of 82.8%, in contrast with 31.4% seroprevalence; (b) in a new CanL focus of Rome province; 60/127 *Phlebotomus perniciosus* (45.7,0%) were found positive by nested PCR, in contrast with 1.4% microscopically positive; and (c) leishmanial DNA was detected in a natural case of leishmaniasis in a novel mammal host, a domestic ferret (*Mustela putorius furo*). These results raise the following doubts, respectively: (a) Does the high “infection” prevalence detected in dogs correspond to the “infectiveness-to-the vector” prevalence? (b) Can the positive sandflies transmit infective promastigotes? and (c) Can an infected domestic ferret play a role as parasite reservoir?

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Unusual tubulins with deviant genetic code in the hypotrich ciliate *Euplotidium itoi*. MANUELA HARTMANN, WOLFGANG LUDWIG, KARL-HEINZ SCHLEIFER and GIULIO PETRONI*, Department for Microbiology, Technical University Munich, D-85380 Freising, Germany, *Department of Ethology, Ecology, Evolution, University of Pisa, I-56126 Pisa, Italy.

We characterized nine macronuclear tubulin gene minichromosomes in the hypotrich ciliate *Euplotidium itoi*: two α-tubulin genes, five β-tubulin genes, and two γ-tubulin genes. In particular, one peculiar β-tubulin gene was identified. It shows only 81.5% amino acid similarity to the other characterized β-tubulin types and an UAG stop codon in frame. Based on this result we started to analyse stop codon usage in *Euplotidium itoi* macronuclear tubulin genes and some unrelated macronuclear minichromosomes. Most of the ciliates independently evolved diverse non-canonical genetic codes. Differences to the standard code consist in the reassignment of one or two of the three standard stop codons to encode a particular amino acid. Class Spirotrichea represents a particular case comprising two groups of organisms that do not

share any stop codon. In oxytrichids, UAR is translated as glutamine and UGA is the single stop codon. This is in contrast to the genus *Euplotes*, close relatives of *Euplotidium*, in which UAR encodes a stop signal and UGA is translated as cysteine. Fourteen complete genes were characterized in *Euplotidium itoi*; in all cases UAA revealed as the single triplet used to code a stop. UGA was never and UAG rarely observed in frame. The UAG stop codon was observed in the peculiar type 3 β -tubulin gene, in two dynein partial sequences and in some other putative genes. We could confirm the active expression of three of these genes, the unusual β -tubulin, a putative lipase and a possible steroid-binding protein, by RT-PCR. Finally, a detailed analysis on the structure of the macronuclear tubulin genes, including nucleotide and amino acid differences among tubulin gene paralogs, polyadenylation sites, putative chromosome fragmentation sites, ATGC-content in coding and non-coding regions, telomere sequence and length will be presented.

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A multidisciplinary approach to describe protists: redescription of *Chattonidium setense* Villeneuve, 1937 (Ciliophora, Heterotricha). LETIZIA MODEO, GIULIO PETRONI, FRANCO VERNI and GIOVANNA ROSATI, Dipartimento di Etologia, Ecologia ed Evoluzione; Università di Pisa, I-56122 Pisa, Italy.

This study represents the first extended report on the heterotrichid ciliate *Chattonidium setense* since the original description by Villeneuve in 1937. This rare marine ciliate is in fact the only representative of the family Chattonidiidae, which is the eighth, often neglected, family of order Heterotrichida. As in our previous works, here we redescribe the species combining morphological observations (live, stained, scanning, and transmission electron microscope) with behavioral and molecular data as we are convinced that both taxonomic determination and molecular characterization are necessary for the clarification of phylogenetic relationships between a species and related genera. Live observation, Feulgen procedure and SEM analysis showed, besides a number of other interesting features, the presence of a characteristic aboral ciliated cavity of undetermined function, whose presence makes ciliate's identification unambiguous. TEM observation evidenced the following main features: well-divided ecto- and endoplasm; deep insertions of the rows of somatic dikinetids in cortex invaginations; paraglycogen granules as polysaccharide reserve; large bundles of microtubules and fibers delimiting the body region surrounding the aboral cavity; two kinds of subcortical extrusomes. Since the revision of heterotrichs ciliates by Villeneuve-Brachon (1940) many genera and families have been transferred to other classes on the basis of both new molecular data and reconsidered morphological characters. Our results support previous statements that *Chattonidium* has the general morphology typical of Heterotrichida and that the main similarities are with genus *Condylostoma*. 18S rRNA molecular data firmly confirm that *Chattonidium* belongs to class Heterotricha, order Heterotrichida (Lynn, D. H. 2003. The Ciliate Resource Archive. <http://www.uoguelph.ca/~ciliates> 2004.07.19) as an independent lineage. Preliminary molecular data also suggest an association between genera *Chattonidium* and *Condylostoma*. Finally, some behavioral and ecological notes on the ciliate are added.

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Searching for tubulin-like genes in free-living bacteria and symbionts of ciliates. Characterization of the genomic environment of tubulin genes from *Prostheco bacter*. MARTIN PILHO-

FER, GIOVANNA ROSATI*, WOLFGANG LUDWIG, KARL-HEINZ SCHLEIFER and GIULIO PETRONI*, Department for Microbiology, Technical University Munich, D-85380 Freising, Germany, *Department of Ethology, Ecology, Evolution; University of Pisa, I-56126 Pisa, Italy.

Tubulins are typical eucaryotic genes, which have recently been found in the free-living bacteria *Prostheco bacter* (*Verrucomicrobiales*). Nevertheless, microtubule-like structures were never observed in *Prostheco bacter*, whereas they were reported in epixenosomes which are symbiotic *Verrucomicrobiales* of the ciliate *Euplotidium*. In the present work, an extensive search for tubulin like genes was performed with several combinations of PCR-primers, designed according to different criteria. Although some combinations of these primers are able to amplify simultaneously eukaryotic α - and β -tubulin as well as *Prostheco bacter* tubulin genes, no bacterial-like tubulin genes were identified in other *Verrucomicrobiales* species. Because of host derived contaminations, even a subtractive PCR approach, using microtiter-plate-subtractive-hybridization, was used with the epixenosomes. The characterization of the genomic environment of the *Prostheco bacter* tubulin genes was performed by gene walking. The sequences obtained from *P. debontii* revealed a so-far undiscovered gene duplication. The bacterial tubulin operon looks conserved among *Prostheco bacter* species and includes an α -tubulin, a β -tubulin and a kinesin like protein. Upstream of two operons a repetitive sequence could be found which could be a hint for horizontal gene transfer. The analysis of the genomic environments of the tubulin genes provided open reading frames, which have homology to *Verrucomicrobium spinosum* genes (ongoing genome project data), whereas up to now the blast search did not reveal tubulin-like genes in this organism. Present results suggest, that the *Prostheco bacter* tubulin genes are possibly of recent origin, maybe acquired by horizontal gene transfer, or lost by other so far studied *Verrucomicrobiales*. The origin and nature of microtubular genetic determinants from epixenosomes remains open.

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Effect of euplotin C on swimming behavior in *Paramecium*. PAOLA RAMOINO, CESARE USAI*, GRAZIANO GUELLA** and FERNANDO DINI***, Dipartimento per lo Studio del Territorio e delle sue Risorse, Università degli Studi di Genova, I-16132 Genova, Italy, *Istituto di Biofisica, CNR, I-16100 Genova, Italy, **Dipartimento di Fisica, Laboratorio di Chimica Bioorganica, Università degli Studi di Trento, I-38100 Trento, Italy, ***Dipartimento di Etologia, Ecologia ed Evoluzione, Università degli Studi di Pisa, I-56126 Pisa, Italy.

The effect of euplotin C—a lipophilic toxic metabolite produced by the protist ciliate *Euplotes crassus*—on the swimming behavior was studied in a single-celled system represented by the ciliate *Paramecium primaurelia*. When in *P. primaurelia* internal Ca^{2+} concentration coupled to membrane depolarization increases, a reversal in the direction of ciliary beating and consequently in swimming direction occurs. The ciliary reversal is correlated to Ca^{2+} influx amount. In this study, evidence was given that continuous ciliary reversal (CCR) duration, induced by high external KCl concentrations, is longer in euplotin C-treated cells than in control cells. To test the hypothesis that euplotin C increases CCR duration by modulating a specific subtype of Ca^{2+} channel, selective Ca^{2+} channel blockers were used. Blocking L-type channels by nimodipine and verapamil, N- and Q-type channels by Ω -conotoxins, fractions GVIA and MVIIC, significantly reduced the CCR duration evoked by membrane depolarization, suggesting an involvement of these channels in ciliary

reversal in *Paramecium*. The effect of euplotin C on CCR duration persisted when Ω -conotoxin GVIA or Ω -conotoxin MVIIC were applied, conversely, it disappeared when L-type channel blockers were used. The magnitude of the block by nimodipine and verapamil in the presence of euplotin C was comparable with that observed in the absence of euplotin C, suggesting that the Ca^{2+} channels modulated by euplotin C were dihydropyridine-sensible calcium channels similar to L-type channels found in mammalian cells.

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Activity of a killer peptide on the growth and ultrastructure of leishmaniae. SARA SCUTERA, STEFANIA RAIMONDO, STEFANIA CONTI*, WALTER MAGLIANI*, LUCIANO POLONELLI* and DIANELLA SAVOIA, Department of Clinical and Biological Sciences, University of Torino, I-10043 Orbassano, Italy, *Department of Pathology and Laboratory Medicine (section of Microbiology), University of Parma, I-43100 Parma, Italy.

An engineered killer decapeptide (KP) has been recently synthesized based on the sequence of a recombinant, single-chain anti-idiotypic antibody acting as a functional internal image of a yeast killer toxin. This KP exerts a microbicidal activity by an interaction with β -glucan present in the surface of many microorganisms, in particular fungi. The activity of KP against the "in vitro" growth and vitality of promastigotes of two *Leishmania* species has been evaluated and compared with the activity due to a "scramble" irrelevant decapeptide (SP). The IC_{50} of KP corresponds to 7.2×10^{-5} mol/litre for *Leishmania infantum*, whereas both the inhibitory and leishmanicidal activity is superior on *Leishmania major*. Moreover, an increase of activity was observed by treating protozoa again with KP; SP is inactive. Structural damages of promastigotes were revealed by transmission electron microscopic studies; the treatment with KP induces a cell death via a non-apoptotic process. In particular, intracellular damage including cytoplasmic degeneration without disruption of the plasma membrane was noted. The activity of KP on leishmania promastigotes seems governed by ionic and/or electrostatic interactions with the glycoconjugates present on the surface of the parasite inducing an autophagic cell degeneration.

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Characterization of R-body genetic determinants in *Caedibacter caryophilus* a symbiont of *Paramecium caudatum*: preliminary results. MARTINA SCHRALLHAMMER, HANS-DIETER GÖRTZ*, WOLFGANG LUDWIG, KARL-HEINZ SCHLEIFER and GIULIO PETRONI**, Department for Microbiology, Technical University Munich, D-85350 Freising, Germany, *Biological Institute, University of Stuttgart, D-70569 Stuttgart, Germany, **Department of Ethology, Ecology, Evolution, University of Pisa, I-56126 Pisa, Italy.

Refractile inclusion bodies, called R-bodies were observed within the cells of some bacterial strains. They are protein ribbons, which are typically coiled into cylindrical structures. They are produced by members of the genus *Caedibacter*, gram-negative rod-shaped endosymbionts of paramecia and e.g. the free-living bacteria *Hydrogenophaga taeniospiralis* and *Acidovorax avenae*. The phylogenetic relationship even between the members of the genus *Caedibacter* is quite low: *C. taeniospiralis* belongs to the *Gammaproteobacteria* and is related to *Francisella tularensis*, *C. caryophilus* is affiliated with the *Alphaproteobacteria* and clusters with the obligate endosymbiont *Holospira*. In the case of *C. taeniospiralis* 51, the genetic determinants of R-body synthesis

are located on a plasmid, whereas in other strains like 7 and 562 it looks like phage particles are involved in their production. In the present study, we investigated *C. caryophilus*, endosymbiont of *Paramecium caudatum*. Separation of *C. caryophilus* cells was performed by PercollTM density gradient centrifugation. The isolated DNA was separated by pulsed-field gel electrophoresis and it was possible to visualize several bands referring to one or more extrachromosomal elements. A small gene library of these extrachromosomal elements was constructed and we already identified transposition-related genes; interestingly, similar genes were reported also on the plasmid of *C. taeniospiralis* 51.

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Effects of Xenobiotic Compounds on Cell Activities in *Euplotes crassus*. FRANCESCA TRIELLI, ANDREA AMAROLI, MARZIA OGNIBENE, CARLA FALUGI* and MARIA U. DELMONTE CORRADO, Dipartimento per lo Studio del Territorio e delle sue Risorse, Università di Genova, I-16132 Genova, Italy, *Dipartimento di Biologia; Università di Genova, I-16132 Genova, Italy.

It is now widely accepted that Protists are relevant bioassays to be exploited for the study of environmental modifications due to the presence of xenobiotic compounds. In this work, we evaluated the possibility of utilizing *Euplotes crassus*, an interstitial marine ciliate, for the pre-chemical screening of environmental sites, such as estuarine and coastal sediments. With this aim, we tested the sensitivity of *E. crassus* to exposure to three classes of pollutants: an organophosphate neurotoxic drug, basudin, largely used for pest control in agricultural sites, a toxic heavy metal, mercury (HgCl_2), and an aromatic polycyclic hydrocarbon, benzopyrene (BP). We found a dose-dependent effect of these compounds on cell viability at concentrations ranging from $1/10^2$ v/v to $1/10^7$ v/v for basudin, from 5 μM to 0.1 μM for HgCl_2 , and from 50 μM to 1 μM for BP. In particular, 100% mortality was caused by a 1-h exposure to $1/10^5$ v/v basudin, or 2 μM HgCl_2 , or 25 μM BP, and by a 24-h exposure to $1/10^6$ v/v basudin, 0.5 μM HgCl_2 , or 5 μM BP. A significant decrease in the daily mean fission rate ($P < 0.001$) was found after exposure to $1/10^7$ v/v basudin, or 0.25 μM HgCl_2 , or 1 μM BP. Moreover, as it is well known that the inhibition of acetylcholinesterase (AChE) activity represents a specific biomarker for neurotoxic drugs, we first detected this enzyme activity in *E. crassus*, using cytochemical, spectrophotometric, and electrophoretic methods; then, AChE activity was characterized by its sensitivity to specific AChE inhibitors and to variations in pH and temperature. Like AChE present in higher organisms, the AChE activity detected in *E. crassus* was inhibited by exposure to basudin. Conversely, exposure to HgCl_2 , or BP did not inhibit AChE activity, but caused a significant reduction in lysosomal membrane stability.

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Recent adaptation to symbiotic lifestyle: the *Polynucleobacter-Euplotes* system. CLAUDIA VANNINI, MARTIN HAHN*, GIULIO PETRONI, FRANCO VERNI, SAMUELE LUCCHESI and GIOVANNA ROSATI, Dipartimento di Etologia, Ecologia, Evoluzione, Università di Pisa, I-56126 Pisa, Italy, *Institute for Limnology, Austrian Academy of Sciences, A-5310 Mondsee, Austria.

Bacteria belonging to the genus *Polynucleobacter* were previously described as endosymbionts inhabiting the cytoplasm of freshwater ciliates of the genus *Euplotes* having a 9 type I cirrus pattern. They were all found to be necessary for the division process of their hosts. Here, we report the finding of a

Polynucleobacter bacterium as symbiont of *E. harpa*, a brackish species with ten frontoventral cirri. Identity of the symbionts was unambiguously determined by morphological description and by 16S rRNA gene sequence characterization. Furthermore, antibiotic treatments confirmed that these bacteria play the same functional role of the other *Polynucleobacter* symbionts; aposymbiotic *E. harpa* ceases to divide and eventually die. Phylogenetic analyses based on 18S rRNA gene show that *Euplotes* species harboring *Polynucleobacter* symbionts, included *E. harpa*, steadily cluster together. In order to clear whether this symbiosis was established as a single evolutionary event by a common ancestor of the ciliate hosts, or by multiple acquisition of different bacteria by the different *Euplotes* species, comparative phylogenetic analyses were performed within the so-called *Polynucleobacter*-

cluster, comprising bacterial symbionts as well as strictly related free-living bacteria. Due to a high degree of conservation of 16S rRNA gene sequences within this group, phylogenetic trees based on this gene did not resolve the matter. Surprisingly, even the characterization and the analysis of the 16-23S ITS sequences, usually exhibiting faster evolutionary rates, revealed a high similarity (>98%), thus hindering a stable and clear phylogenetic reconstruction. Data obtained cannot at present definitively confirm or reject any hypothesis on the origin of this symbiosis, revealing, at the same time, that we are dealing with a relatively young symbiotic system. As shown by the close phylogenetic relationship between free-living and symbiotic *Polynucleobacter*, this is an ideal model for studying development and adaptation to symbiotic lifestyle.